Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.

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Abstract

The rapid advancement of digital technologies transformed modern manufacturing by enabling organizations to improve production monitoring, operational efficiency, and data-driven decision-making. However, many food manufacturing companies continued to rely on manual and paper-based production recording systems, which often resulted in delayed reporting, inaccurate computations, data redundancy, and limited visibility of production performance. At AlDi Foods Inc., production personnel manually recorded raw material usage, extraction outputs, and packed yield information, making it difficult for management to monitor production activities in real time and evaluate production efficiency promptly. These challenges highlighted the need for a centralized digital production monitoring system capable of providing accurate, timely, and reliable production information.

This study aimed to develop a web-based system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* The proposed system was designed to digitize production recording by replacing manual documentation with a centralized platform that supported real-time monitoring of production activities. It was structured to automate the computation of yield percentage and extraction efficiency, classify production stability based on predefined organizational thresholds, generate dashboard visualizations, maintain historical production records, and produce automated weekly, monthly, quarterly, and yearly raw material utilization reports. Through these features, the system was expected to improve production visibility, reduce manual computation errors, enhance reporting accuracy, and support timely managerial decision-making.

The study employed a Developmental Research Design and utilized the Agile Development Model to guide the systematic analysis, design, development, testing, and evaluation of the proposed system. System requirements were gathered through interviews with production personnel, direct observation of existing production workflows, and a review of company production documents and records. The system was developed using HTML and CSS for the user interface, PHP for server-side application development, MySQL for centralized database management, and Chart.js for graphical visualization of production analytics. A three-tier architecture consisting of the Client Layer, Application Layer, and Database Layer was implemented to ensure efficient processing, secure data management, and reliable communication among system components.

The findings of the study were expected to demonstrate that the proposed system provided a practical, accessible, and cost-effective digital solution for improving production monitoring, real-time yield analytics, extraction efficiency monitoring, and evidence-based decision-making within AlDi Foods Inc., while contributing to the continuing digital transformation of food manufacturing operations.

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

In today's highly competitive and technology-driven environment, industries continuously seek innovative ways to improve operational efficiency, reduce waste, and enhance productivity. The manufacturing sector, particularly in food production, plays a critical role in meeting consumer demand while maintaining quality and cost-effectiveness. With the emergence of digital transformation, organizations are transitioning from traditional production methods toward more advanced, automated, and data-driven systems.

Digital Transformation introduces the integration of digital technologies, such as real-time analytics, cloud computing, and intelligent systems, into manufacturing processes. These technologies enable organizations to collect, process, and analyze data instantly, allowing for improved decision-making and enhanced operational control. A key advantage of this transformation is the ability to monitor production processes in real time, thereby minimizing inefficiencies and preventing costly errors.

Despite these advancements, many small and medium-sized enterprises (SMEs), including AlDi Foods Inc., continue to rely on manual, paper-based methods of recording production data. These traditional systems prove prone to human error, data inconsistencies, and delays in information processing. Consequently, businesses experience data latency and lack real-time visibility into production performance, which makes it difficult to detect yield losses, machine issues, and material wastage at an early stage.

In food production, particularly within juice manufacturing, yield serves as a critical performance indicator that represents the ratio of output (extracted juice) to input (raw materials). A high yield consistently indicates efficient resource utilization, whereas a low yield points to underlying inefficiencies in the extraction process.

Factors such as equipment condition, raw material quality, and operator performance significantly affect those yield outcomes. However, without a reliable monitoring system in place, identifying the impact of those variables remains difficult to analyze and optimize.

AlDi Foods Inc. relies on manual logging systems to track production data, which limits the company's ability to monitor extraction efficiency in real time. Production issues are often identified only after operations have been completed, resulting in delayed decision-making and missed opportunities for immediate corrective actions. To address these challenges, digitizing the production process through automated data collection and centralized software systems is essential for transforming raw data into actionable insights.

To solve these operational bottlenecks, this study proposes the development of *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* The system aims to replace manual recording methods with a centralized digital platform that provides real-time monitoring, analytics, and visualization of production data. Ultimately, the project seeks to improve efficiency, reduce waste, and enhance decision-making across the company's food production operations.

1.2 Statement of the Problem

The current production system of AlDi Foods Inc. relies heavily on manual, paper-based recording of production data. This approach presents several challenges that affect the efficiency, accuracy, and reliability of operations. Manual data recording consumes time, increases the possibility of human error, and limits the company's ability to monitor production performance in real time. As a result, production inefficiencies and extraction losses are often detected only after operations are completed, making it difficult to implement immediate corrective actions.

Additionally, the absence of a centralized digital monitoring system severely limits management's access to timely and accurate production information. Without this visibility, leadership faces significant hurdles when making critical, time-sensitive decisions, evaluating overall employee productivity, and maintaining high operational efficiency.

Furthermore, this lack of real-time analytics hinders the company's capability to identify recurring production patterns and monitor extraction performance on the fly. As a result, the facility is left vulnerable to unaddressed inefficiencies and preventable material waste during its core manufacturing processes.

Specifically, this study seeks to answer the following questions:

1. How does manual recording of production data affect accuracy, consistency, and reliability at AlDi Foods Inc.?
2. What are the limitations of the current system in terms of real-time monitoring and data accessibility?

3. How can real-time monitoring improve the tracking of raw material inputs and production outputs?
4. How can yield losses during the extraction process be detected at an earlier stage?
5. How can inefficiencies in the production process be measured and analyzed effectively?
6. How can a digital system enhance managerial decision-making through timely and accurate data?
7. What features should be included in a digital production monitoring system to address existing challenges?

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop a digital production monitoring system for AlDi Foods Inc. that provides real-time yield analytics and extraction efficiency monitoring. This system specifically focuses on improving the overall accuracy, efficiency, and accessibility of production data.

To achieve this, the system aims to completely replace the company's manual, paper-based recording processes. By implementing a centralized digital solution, the system enables seamless, real-time tracking of production inputs and outputs, particularly within their juice extraction operations.

Specifically, this study seeks to address the critical need for operational visibility by providing a highly structured and automated framework for monitoring vital production indicators. In traditional manufacturing setups, tracking metrics such as raw material usage, total juice output, overall yield percentage, and extraction efficiency often relies on fragmented, manual processes. By integrating an automated tracking system, this research aims to centralize these disparate data points into a single, cohesive stream of real-time intelligence. This transition not only streamlines the collection of core metrics but also establishes a standardized baseline for performance evaluation, ensuring that every drop of resource and ounce of output is accounted for accurately.

Beyond mere data collection, the implementation of this automated system is strategically designed to mitigate the inherent risks of manual oversight. By significantly minimizing human error in data recording and eliminating the regular delays typical of legacy reporting methods, the system ensures that information moves at the speed of production. Consequently, this drastically improves the consistency, timeliness, and reliability of the data. Armed with high-fidelity, up-to-the-minute production insights, stakeholders and floor managers are empowered to make proactive, well-informed operational decisions that optimize throughput and reduce waste.

In addition, the system is designed to support real-time visualization and analysis of production performance through a centralized dashboard. This digital solution allows production managers to immediately identify variations in yield and extraction efficiency, detect possible losses during processing, and evaluate overall machine performance more effectively.

Ultimately, the study intends to enhance production monitoring capabilities and support data-driven decision-making within AlDi Foods Inc. By providing a practical and user-friendly interface, the system empowers leadership to transform raw operational data into actionable insights that optimize daily workflows.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Analyze the existing production process and identify inefficiencies in data recording and monitoring.
2. Design a user-friendly digital system that replaces manual production logs.
3. Develop a real-time monitoring feature for tracking raw material inputs and production outputs.
4. Implement automated yield calculation and analytics functionalities.
5. Monitor extraction efficiency and identify performance trends during production.
6. Provide an interactive dashboard for real-time visualization of production data.
7. Store and manage production data securely using a centralized database.
8. Evaluate the system's performance in terms of functionality, usability, accuracy, efficiency, and reliability.
9. Reduce delays in production monitoring by providing instant access to operational data and analytics.
10. Improve the company's ability to identify extraction losses and production inefficiencies through automated monitoring and reporting features.

11. Support data-driven decision-making by generating accurate and timely production reports for management.
12. Enhance productivity and operational control through the integration of digital technologies and real-time monitoring tools.
13. Minimize manual recording errors and improve data consistency within production operations.
14. Provide a scalable foundation for future integration of IoT-based sensors and automated manufacturing technologies.

1.4 Scope and Delimitation

This study focuses on the design and development of a digital production monitoring system for AlDi Foods Inc. The system is intended to improve the tracking, monitoring, and analysis of production data, particularly regarding materials input, yield computation, and extraction efficiency. Ultimately, the proposed system aims to replace manual recording methods with a centralized digital platform capable of providing real-time monitoring and analytical functionalities.

The scope of the study encompasses the development of key system features, such as real-time production data input, automated yield calculation, and extraction efficiency monitoring. Additionally, the development effort incorporates interactive dashboard visualizations alongside centralized database management. These capabilities are designed to streamline data accessibility across operational shifts.

Through this digital interface, the system allows authorized users to monitor production performance, track raw material usage, and systematically evaluate production outputs. Furthermore, the study includes basic analytics and visualization tools that directly support ongoing production analysis and operational decision-making. The software is engineered to serve as a comprehensive visual hub for factory management.

The system is developed as a prototype specifically intended for use within the operational context of AlDi Foods Inc. During this phase, data is entered manually to effectively demonstrate the platform's real-time monitoring capabilities. To confirm its overall viability, the study also covers rigorous system testing and evaluation in terms of functionality, usability, efficiency, accuracy, and reliability.

However, the study is defined by several operational delimitations. The proposed system does not include full-scale industrial automation or advanced machine-learning capabilities, and integration with highly sophisticated Industrial Internet of Things (IIoT) infrastructures, robotics, or enterprise-level manufacturing systems is strictly beyond the scope of this research. Moreover, the study is limited only to monitoring production yield and extraction efficiency, excluding other business functions such as inventory management, sales, accounting, payroll, or supply chain management.

Finally, the system is tested within a controlled environment, meaning performance outcomes do not fully represent all real-world production conditions. Because the study focuses exclusively on the operational processes of AlDi Foods Inc., the findings cannot be fully generalized to other manufacturing companies with different production environments or operational requirements.

Consequently, financial analysis, large-scale deployment, and long-term implementation assessment are also excluded from the scope of this research.

1.5 Significance of the Study

This study is beneficial to the following:

AlDi Foods Inc.- The system will improve production efficiency, reduce waste, and enhance overall productivity and profitability through real-time monitoring and analytics.

Production Managers- Managers will have access to timely and accurate production data, enabling faster decision-making, improved monitoring, and immediate response to operational issues.

Employees- The system will simplify production data recording processes, reduce manual workload, minimize human error, and improve operational efficiency.

BSIT Students- The study serves as a practical application of knowledge in system development, database management, software engineering, and real-time analytics.Future ***Researchers***- This study can serve as a reference for future research related to digital production systems, real-time monitoring, Industry 4.0 technologies, and manufacturing analytics.

1.6 Definition of Terms

Analytics – The process of analyzing data to extract meaningful insights and support decision-making.

Centralized Database – A digital storage system where production data is organized and managed in a single location for easier access and monitoring.

Dashboard – A graphical user interface that displays real-time production information, analytics, and system reports.

Digitizing Production – The conversion of manual production records and monitoring processes into digital format using information technology.

Efficiency – A measure of how effectively resources are utilized to produce desired outputs.

Extraction **–** The process of obtaining juice or liquid product from raw materials during production.

Manufacturing Industry- The study may contribute to the advancement of digital transformation practices in manufacturing by demonstrating the benefits of real-time monitoring systems in improving efficiency and reducing production waste.

Monitoring**–** The continuous observation, tracking, and evaluation of production processes and operational performance.

Real-Time **–** Immediate processing, updating, and availability of data without significant delay.

Yield **–** The ratio of output produced compared to input materials, usually expressed as a percentage.

Automated Yield Calculation – The automated mathematical process of computing the ratio of usable output to raw material input without manual intervention.

Extraction Efficiency – A quantitative measurement comparing the actual amount of liquid extracted to the theoretical maximum yield possible from a given batch of raw materials.

Manual Recording Method – The traditional, paper-based practice of logging operational data, material usage, and production metrics by hand.

Operational Visibility – The degree to which real-time production activities, resource flows, and machine performances are accessible and transparent to managers and operators.

Production Monitoring System – A specialized digital application designed to continuously record, track, and evaluate production workflows, inputs, and outputs.

Prototype System – A preliminary, functional version of software developed to demonstrate core capabilities, validate system design, and test features in a controlled environment.

Raw Material Input – The baseline quantity of unprocessed agricultural goods or ingredients introduced into the extraction machinery at the start of a production run.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 Foreign Studies

A study by Verhoef et al. (2021) analyzed the impact of transitioning from legacy paper-based logs to centralized digital dashboards in manufacturing. The researchers highlighted that manual data silos create "blind spots" or information gaps, where production anomalies—such as waste hotspots or mechanical inefficiencies—are often only discovered long after a batch is completed. By implementing a digitized analytics system, the facility achieved a more responsive production environment, allowing managers to make data-driven decisions based on live efficiency metrics rather than historical, error-prone logs. This study serves as a foundational reference for your claim that digitizing the production floor is a prerequisite for Industry 4.0 and is essential for reducing financial losses caused by avoidable process inefficiencies.

A study by Makhmudah et al. (2025) explored the implementation of a digital production reporting system in a manufacturing environment. Using a qualitative case study approach, the researchers replaced manual reporting processes with a system integrated with sensors, automated data transmission, and dashboard visualization. The results showed improvements in data accuracy, operational efficiency, and decision-making. This study is relevant to the present research as it demonstrates the effectiveness of digital systems in improving production monitoring and data management.

A study by Mansour and Al-Hamdani (2024) examined the application of internal Key Performance Indicators (KPIs) in evaluating production efficiency within the food industry, with a specific focus on non-conforming products identified during manufacturing operations. The researchers proposed a structured performance measurement system centered on the internal parts per million (INTppm) indicator, which quantifies defective units detected within production orders relative to the total number of items produced. This approach allows organizations to assess production quality at the operational level rather than relying solely on end-product inspection, thereby enabling immediate, data-driven decision-making. By continuously tracking these metrics, the authors emphasized that food production performance depends heavily on the continuous monitoring of quality-related deviations to maintain process stability. In their case analysis involving food-related processing lines, Mansour and Al-Hamdani (2024) monitored various parameters across manual and automated operations, including quality control checks, labeling procedures, measurement systems, and line adjustments. The findings revealed that internal defect rates varied significantly depending on the production stage, with technical problems—such as mechanical errors, clamping inconsistencies, tool wear, and alignment issues—serving as primary drivers of elevated INTppm values.

Additionally, inadequate maintenance and delayed equipment replacements further exacerbated production instability. Consequently, the internal defect rate proved to be a critical indicator of both product quality and machine reliability. To mitigate these operational inefficiencies, Mansour and Al-Hamdani (2024) highlighted the necessity of corrective and preventive measures, such as enhancing automated process controls, executing random quality checks during shifts, and updating employee training programs. The researchers also recommended integrating skill evaluation checkpoints and fostering workforce communication to minimize human errors alongside technical updates. Cultivating this long-term data collection approach allows manufacturing organizations to transition from reactive problem-solving to proactive process optimization, shifting focus toward controllable internal variables like equipment conditions and operator behavior. Ultimately, Mansour and Al-Hamdani (2024) concluded that utilizing internal KPIs like INTppm offers a reliable methodology for optimizing food industry operations by linking real-time defect rates directly to strategic business outcomes like waste reduction and cost efficiency. This literature is highly relevant to the present study as it provides a robust structural foundation for utilizing internal performance indicators to target production inefficiencies. Furthermore, it directly supports the proposed framework of developing a data-driven, real-time tracking system for AlDi Foods Inc. to monitor extraction efficiency and establish long-term process optimization.

A comprehensive review conducted by Aftab Siddique, Ashish Gupta, Jason T. Sawyer, Tung-Shi Huang, and Amit Morey (2025), entitled " Big Data Analytics in the Food Industry: A State-of-the-Art Literature Review", examined the increasing influence of artificial intelligence (AI), machine learning (ML), and big data analytics on the modernization of food industry operations. Published in "npj Science of Food", the study explored numerous scholarly works related to digital transformation and data-driven technologies across food production, processing, quality assurance, supply chain management, and food safety. The researchers emphasized that the food industry is undergoing a significant technological shift as organizations increasingly adopt digital systems capable of collecting, storing, processing, and analyzing large volumes of operational data. As production processes grow more complex and consumer demands evolve, the deployment of advanced analytics has become a pivotal strategy for improving efficiency, productivity, and competitiveness. According to Siddique et al. (2025), modern food manufacturing facilities generate substantial amounts of data from various stages of production, including raw material handling, processing, packaging, storage, distribution, and quality control. Traditionally, much of this data remained underutilized due to legacy constraints in collection and analytical methods. However, advancements in digital technologies enable organizations to transform raw production data into actionable insights to support real-time decision-making and operational improvements.

The study explained that big data analytics provides organizations with the capacity to identify patterns, detect hidden inefficiencies, forecast outcomes, and monitor performance far more accurately than conventional, manual approaches. The review highlighted that AI and ML technologies have become essential tools for enhancing food production systems by processing large datasets to generate insights that are difficult to obtain through manual analysis. AI-based models learn from historical and real-time data, making it possible to predict trends, isolate operational bottlenecks, and optimize processing parameters. Through continuous data analysis, organizations can achieve tighter process control, minimize material waste, and enhance overall manufacturing efficiency. A major focus of the study involved the application of predictive analytics and artificial neural networks (ANNs) in food processing operations, particularly in activities such as drying, baking, fermentation, extrusion, freezing, and quality assessment. By utilizing historical operational data, these predictive systems can anticipate final product characteristics, estimate exact processing times, model complex multi-variable relationships, and dynamically identify optimal operating conditions. Furthermore, Siddique et al. (2025) underscored the importance of real-time monitoring systems over traditional monitoring methods, which heavily rely on manual recording procedures that are time-consuming and prone to human error. In contrast, digital monitoring systems continuously collect and analyze operational data, allowing managers and operators to observe production performance dynamically. This real-time monitoring provides immediate visibility into production loops and enables organizations to respond swiftly to operational anomalies. The authors further discussed the integration of computer vision systems and image-processing technologies to automatically inspect products during production. This automated visual intelligence reduces the need for manual inspection while increasing consistency and accuracy in food grading, defect detection, and quality evaluation. In addition to production monitoring and quality control, the review explored the role of big data analytics in food safety management, supply chain optimization, and resource sustainability. Advanced analytical systems can identify contamination risks, track disease outbreaks, and improve overall traceability to mitigate economic losses associated with product recalls. From an operational standpoint, AI-powered forecasting models analyze historical sales data and consumer behavior to predict future demand, enabling manufacturers to optimize inventory levels and schedule production efficiently. Siddique et al. (2025) also noted that data-driven systems support corporate sustainability goals by optimizing the use of raw materials, energy, and water, thereby reducing environmental impacts. Despite challenges such as data quality, system integration, and workforce readiness, the authors concluded that the future of the food industry depends on intelligent systems capable of transforming raw metrics into actionable knowledge. This extensive review is highly relevant to the present capstone project, "Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.", because it provides strong theoretical and empirical validation for the implementation of digital monitoring and analytics systems within food processing environments. The findings directly support the technical architecture of a platform capable of monitoring production yield and extraction efficiency in real time. By establishing a system for continuous data collection, management can move away from latency-heavy, manual logs to instantly identify process variations, evaluate extraction performance, and detect production losses. Consequently, the work of Siddique et al. (2025) serves as a foundational benchmark for the proposed system, demonstrating that real-time data integration is critical to modernizing waste management and achieving operational excellence.

A study by Rivadeneira et al. (2023), published in *Food Research*, examined the production of pectin from “saba” banana peel waste using ultrasound-assisted extraction and evaluated its performance as an emulsifying agent compared with commercial low-methoxyl pectin (LMP). The study emphasized the potential of agricultural waste valorization by converting banana peels into functional food ingredients suitable for emulsion-based systems. Findings revealed that the extracted pectin had lower equivalent weight and a reduced degree of esterification than commercial LMP, indicating differences in molecular structure that influence its functional behavior in food applications. Although UEP showed lower purity, it contained relatively higher amounts of protein and ash, which contributed positively to its interfacial and emulsifying properties.

Structural and physical analyses further confirmed these differences, where FTIR results indicated variations in esterified carboxyl groups between UEP and LMP. Microscopic observation showed that emulsions stabilized by UEP formed smaller and more evenly distributed oil droplets compared to those stabilized by commercial pectin. This droplet size reduction contributed to improved emulsion stability, as it enhanced interfacial coverage and minimized phase separation. The study also reported that changes in formulation factors such as pH level, oil concentration, and ionic strength significantly affected emulsion behavior, with UEP consistently showing stronger adaptability under varying conditions.

In terms of functional performance, emulsions containing UEP exhibited higher viscosity and better resistance to breakdown compared to those stabilized by LMP. The study also found that UEP influenced lipid digestion, as it reduced free fatty acid release during in vitro digestion, suggesting a potential role in controlling lipid bioaccessibility. Overall, the researchers concluded that ultrasound-extracted pectin from banana peel waste is a promising sustainable alternative to commercial pectin, offering improved emulsifying stability and functional benefits for food formulation and processing systems.

A study by Sabih, Farid, Ejaz, Husam, Khan, and Farooq (2023) investigated the use of computer vision and machine learning technologies to automate the measurement of raw material flow rates on a conveyor belt system used in soda ash manufacturing. The researchers recognized that many industrial processes still depend on manual monitoring and measurements, which can lead to errors, inefficiencies, and delays in decision-making. To address this challenge, they proposed an automated framework capable of estimating the flow rate of raw materials transported on a conveyor belt without requiring continuous human intervention.

The study was conducted in a soda ash manufacturing plant where limestone and coke are transported through a conveyor belt system. These materials are critical inputs in the production process because maintaining the correct ratio between limestone and coke is essential for generating carbon dioxide gas with the required purity level. Traditionally, operators manually conducted experiments and observations to determine the flow rate of materials and adjust conveyor settings accordingly. However, this approach was time-consuming and susceptible to inaccuracies. The researchers sought to improve this process through the implementation of an intelligent monitoring system.

To collect data, a high-resolution camera was installed above the conveyor belt to continuously capture video footage of the moving materials. The data collection process spanned twenty-four weeks and included videos recorded under various environmental conditions such as changes in lighting, dust levels, and weather conditions. Since the conveyor belt was located outdoors, these environmental factors introduced additional challenges that had to be addressed by the proposed system. The researchers designed their framework to be robust enough to operate effectively despite these changing conditions.

The methodology involved several stages of image processing and machine learning. First, a region of interest was extracted from each video frame so that only the conveyor belt area containing the raw materials would be analyzed. This step reduced computational requirements and eliminated unnecessary background information. The images were then preprocessed using sharpening and filtering techniques to improve image quality and reduce the effects of dust, noise, and poor lighting. Bilateral filtering was specifically selected because it reduced image noise while preserving important edge details needed for accurate material detection.

After preprocessing, the researchers focused on segmenting the raw materials from the conveyor belt background. Several segmentation approaches were evaluated, including deep learning-based methods, watershed segmentation, frame differencing, and background subtraction. However, because of limited labeled data and highly variable environmental conditions, some methods produced inconsistent results. To overcome these limitations, the researchers developed a voting-based segmentation approach that combined frame differencing and the BackgroundSubtractorCNT algorithm. By retaining only the pixels identified by both techniques, the system was able to isolate the most reliable regions representing the raw materials. This method improved segmentation accuracy and reduced the influence of environmental noise and lighting variations.

Once the materials were successfully segmented, the system extracted various image features to support flow-rate estimation. The researchers used moment-based features to describe the shape, size, position, and distribution of the segmented materials. Additional contour-based features were extracted to measure the area occupied by limestone and coke on the conveyor belt. K-means clustering was also applied to separate different image regions based on color information. These features were aggregated over specific time windows to accommodate the variable lengths of the video recordings and ensure consistent input dimensions for machine learning models.

To improve model performance and reduce computational complexity, the study employed feature selection techniques, particularly Recursive Feature Elimination with Cross-Validation (RFECV). This method identified the most significant features while eliminating less relevant variables. The selected features were then used to train and evaluate several machine learning algorithms, including Decision Tree, Random Forest, XGBoost, Gradient Boosting, Bagging Regressor, Bayesian Ridge, Gamma Regressor, RANSAC, and Theil-Sen Regressor. Both linear and non-linear models were assessed to determine the most effective approach for predicting material flow rates.

The researchers evaluated the performance of the models using several regression metrics, including Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). They also utilized leave-one-out validation and k-fold cross-validation to ensure the reliability of the results. The findings revealed that model performance was significantly influenced by the choice of window size, aggregation method, and feature selection strategy. While feature selection improved some non-linear models, certain linear models achieved better results when all extracted features were retained.

Among the evaluated algorithms, the Bagging Regressor demonstrated the best overall performance in terms of prediction accuracy. The final model was able to estimate material flow rates with a percentage error as low as seven percent, indicating a high
level of reliability. The study showed that combining computer vision techniques with machine learning algorithms can effectively automate flow-rate monitoring while minimizing the need for manual measurements. Furthermore, the proposed system successfully operated in challenging outdoor environments where variations in lighting, dust, and weather conditions could otherwise compromise accuracy.

The researchers concluded that the integration of computer vision and machine learning provides a practical and cost-effective solution for industrial process monitoring. By using only a standard RGB camera and advanced data analytics techniques, the system was able to deliver accurate real-time estimates of material flow rates. This capability supports more informed operational decisions, enhances process efficiency, and reduces the likelihood of human error in industrial settings.

A study by **Sharma et al. (2021)** examined the growing role of artificial intelligence (AI) and big data analytics (BDA) in transforming the food industry toward more sustainable, efficient, and intelligent operations. The researchers conducted a systematic literature review, gathering and analyzing previous studies from reputable academic databases, including Scopus, ScienceDirect, Wiley Online Library, Taylor & Francis, and Web of Science. Their objective was to investigate how AI and BDA technologies are being utilized across different stages of the food industry, from agricultural production and food processing to logistics, quality assurance, supply chain management, and distribution. The review emphasized that the increasing demand for food, combined with the need for higher product quality, food safety, and sustainable resource management, has encouraged food manufacturers to adopt digital technologies that can optimize production processes and improve operational performance.

The authors explained that artificial intelligence serves as a powerful computational technology capable of analyzing complex production data, recognizing patterns, making predictions, and supporting intelligent decision-making. AI encompasses various technologies such as machine learning, artificial neural networks (ANNs), computer vision, robotics, fuzzy logic, genetic algorithms, and expert systems. These technologies enable food manufacturers to automate repetitive tasks, monitor production processes, detect defects, forecast production outcomes, and improve quality control. Meanwhile, big data analytics complements AI by collecting, processing, and analyzing large volumes of structured and unstructured data generated throughout production operations. Through advanced data analysis, manufacturers can identify operational trends, predict equipment failures, optimize production schedules, reduce waste, and improve resource utilization. Together, AI and BDA create an intelligent production environment that supports real-time monitoring and continuous process improvement.

The review highlighted numerous applications of AI within food production systems. Machine learning algorithms are widely used for demand forecasting, production planning, yield prediction, and quality classification. Artificial neural networks assist manufacturers in modeling complex production processes and predicting product quality based on multiple process variables. Computer vision systems perform automated inspection by identifying defects, grading food products according to size, shape, color, and texture, and ensuring compliance with quality standards. Robotics contributes to automating labor-intensive operations such as food handling, sorting, packaging, and processing while maintaining product consistency and reducing human error. The authors also discussed intelligent systems such as electronic noses, hyperspectral imaging, infrared sensing, and machine vision technologies that improve food safety, contamination detection, and product evaluation with greater accuracy than traditional manual inspection methods. These technologies reduce processing time while maintaining high levels of precision and reliability.

Sharma et al. (2021) further explained that big data analytics strengthens decision-making by converting massive amounts of operational data into meaningful insights. The researchers described three primary categories of analytics used in food manufacturing. **Descriptive analytics** helps identify production patterns, operational trends, and performance indicators by analyzing historical production data. **Predictive analytics** estimates future outcomes such as production demand, equipment performance, product quality, weather impacts, customer behavior, and inventory requirements using statistical models and machine learning algorithms. **Prescriptive analytics** goes one step further by recommending the most effective actions based on optimization models and simulation techniques, allowing managers to make informed decisions regarding production scheduling, inventory management, logistics, and resource allocation. These capabilities enable manufacturers to respond more effectively to production challenges while minimizing operational risks and improving efficiency.

Another important contribution of the study is its discussion of real-time monitoring technologies integrated with AI and BDA. Modern food manufacturing facilities generate enormous amounts of production data through sensors, Internet of Things (IoT) devices, radio-frequency identification (RFID), automated equipment, and intelligent monitoring systems. These technologies continuously collect information regarding machine performance, production output, environmental conditions, inventory levels, and product quality. AI algorithms analyze these data streams in real time, enabling immediate detection of production abnormalities, equipment malfunctions, or quality deviations. Instead of relying on delayed manual reporting, manufacturers receive instant information that supports faster corrective actions and more effective production management. Real-time visibility also improves traceability throughout the production process, helping organizations maintain compliance with food safety regulations while reducing waste and operational costs.

The researchers also emphasized the growing role of AI and big data analytics across various sectors of the food industry. In dairy processing, AI models predict milk yield, monitor animal health, and estimate product shelf life. In meat processing, machine learning, computer vision, and electronic nose technologies assess meat freshness, identify spoilage, and automate quality grading. For fruits and vegetables, intelligent refrigeration systems monitor product aging and shelf life, while machine vision systems automate sorting and grading based on physical characteristics. AI applications have also been successfully implemented in bakery operations through intelligent oven control systems that optimize baking conditions and improve product consistency. Similarly, beverage manufacturers utilize AI technologies to evaluate flavor, aroma, appearance, and consumer preferences, while robotics assists in packaging and production automation. These examples demonstrate the versatility of AI technologies in addressing operational challenges across different food manufacturing environments.

Despite these advantages, Sharma et al. (2021) acknowledged several challenges associated with implementing AI and BDA in the food industry. The adoption of intelligent technologies requires substantial financial investment in digital infrastructure, sensors, automated equipment, and information systems. Organizations also need skilled personnel capable of operating AI-based technologies, interpreting analytical results, and maintaining digital production systems. Additionally, concerns regarding cybersecurity, data privacy, workforce adaptation, and organizational readiness remain significant barriers to successful implementation. However, the researchers argued that the long-term benefits—including improved operational efficiency, enhanced product quality, reduced costs, faster decision-making, better traceability, and increased production sustainability—far outweigh these implementation challenges. As digital technologies continue to evolve, their adoption is expected to become increasingly essential for maintaining competitiveness within the global food industry.

2.2 Local Studies

A study by Bagaforo and Albason (2024), presented at the 19th European Conference on Innovation and Entrepreneurship (ECIE), investigated the digital transformation journeys of Micro, Small, and Medium Enterprises (MSMEs) in Iloilo Province, identifying the specific enablers and barriers local businesses face when adopting new technologies. The research utilized a descriptive-correlational approach to evaluate how local infrastructure and workforce "readiness" influence the transition from traditional to digitized operations. Their findings highlighted that while digital tools significantly enhance local business competitiveness and operational efficiency, success is often hindered by technical literacy gaps and varying levels of infrastructural support. This study provides critical contextual evidence for AlDi Foods Inc. regarding the feasibility of implementing automated systems within the local economic landscape, directly supporting the Significance of the Study by illustrating how digital integration optimizes resource management and regional market positioning.

A study by De Leon et al. (2017), archived in the Central Philippine University (CPU) Institutional Repository, focused on the design and development of an online ordering system for Iloilo Supermart using the V-Model Software Development Life Cycle (SDLC). The researchers implemented a secure, centralized database and a user-friendly management dashboard to streamline operations for the Iloilo-based food retailer, effectively mitigating risks associated with physical data damage and manual record-keeping.

A study conducted by Palero et al. (2026), titled *“Green Extraction and Sustainability Assessment of Total Carotenoids from Carrot (Daucus carota) Peel Waste using Soybean Oil,”* explored a sustainable and environmentally responsible approach for recovering valuable bioactive compounds from agricultural waste materials. Published in the Davao Research Journal, the study primarily addressed the growing concern over the environmental and health hazards associated with conventional solvent-based extraction methods, particularly those involving organic solvents such as hexane. These traditional solvents, while efficient in extracting carotenoids, are widely criticized due to their toxicity, flammability, and environmental persistence. In response to these limitations, the researchers investigated soybean oil as a safer, biodegradable, and food-grade alternative solvent capable of efficiently extracting carotenoids from carrot peel waste while aligning with green chemistry and circular economy principles.

Carotenoids are naturally occurring pigments responsible for the orange, yellow, and red coloration in many fruits and vegetables, and they are widely recognized for their antioxidant, anti-inflammatory, and health-promoting properties. Because of their high commercial value and wide applicability in food, pharmaceutical, and cosmetic industries, there is increasing global interest in improving their extraction from natural sources, especially from agricultural by-products that are often discarded. Carrot peels, which are generated in large quantities during food processing, represent a significant yet underutilized source of carotenoids. The study emphasized that valorizing such waste materials not only contributes to resource efficiency but also reduces environmental pollution and supports sustainable production systems.

To achieve its objectives, the researchers employed a two-factor, three-level full factorial experimental design to systematically evaluate and optimize the extraction process. The independent variables considered were the liquid-to-solid (L/S) ratio and extraction temperature, both of which are known to significantly influence mass transfer and solute diffusion during extraction processes. The dependent variable was the total carotenoid content (TCC), expressed in micrograms of β-carotene per gram of dried carrot peel. The experimental procedure involved freeze-drying carrot peels to preserve bioactive compounds, followed by pulverization to increase surface area and enhance solvent penetration. The prepared samples were then mixed with soybean oil under controlled conditions and subjected to agitation using a shaking water bath. This setup ensured consistent contact between the solvent and plant material, facilitating efficient extraction of carotenoids into the oil phase.

Quantitative analysis of carotenoid content was performed using UV-Visible spectrophotometry at 450 nm, applying the Beer-Lambert law for concentration determination. The study also implemented rigorous statistical tools to evaluate the reliability and validity of the experimental model. These included analysis of variance (ANOVA), regression modeling, and diagnostic tests such as the Shapiro-Wilk test for normality, Durbin-Watson test for autocorrelation, Cook's distance for identifying influential outliers, and Variance Inflation Factor (VIF) for detecting multicollinearity. Additionally, model performance was assessed using R^2, adjusted R^2, and predicted R^2 values to ensure both accuracy and predictive capability.

The experimental results demonstrated that both the liquid-to-solid ratio and temperature significantly influenced carotenoid extraction efficiency, although their effects were not equal in magnitude. The highest observed total carotenoid content was 84.22 ± 5.56 μg β-carotene/g of dried peel, achieved at a liquid-to-solid ratio of 40 mL/g and a temperature of 45 °C. This result indicated that increasing the solvent proportion relative to the biomass significantly enhances extraction efficiency by improving concentration gradients and promoting better diffusion of carotenoids from the plant matrix into the oil solvent. In contrast, temperature exhibited a positive but comparatively weaker effect on extraction performance, suggesting that thermal enhancement plays a supporting role rather than a dominant one in the process.

Statistical analysis confirmed that the overall model was highly significant ($p < 0.0001$), with a strong coefficient of determination ($R^2 = 0.95$), indicating that the model effectively explained most of the variability in carotenoid recovery. The adjusted R^2 (0.94) and predicted R^2

(0.93) values further supported the robustness and predictive reliability of the model. However, a statistically significant lack of fit was also observed, suggesting that certain complex interactions or nonlinear behaviors within the system were not fully captured by the model. This indicates that while the model is strong in predictive capability, there remains room for refinement to improve its descriptive accuracy.

Through response surface methodology optimization, the study identified the most favorable extraction conditions as a liquid-to-solid ratio of 40 mL/g and a temperature of 30 °C. Under these conditions, the predicted carotenoid yield was 77.49 ± 13.13 μg β-carotene/g of dried peel. To validate the model, experimental runs were conducted under the same optimized conditions, yielding an actual average value of 76.04 ± 3.71 μg β-carotene/g. The small deviation between predicted and experimental values, reflected in a low relative error, confirmed the reliability and practical applicability of the developed model. This close agreement demonstrates that the optimization approach used in the study is effective for predicting carotenoid extraction performance under varying process conditions.

Further analysis revealed that the liquid-to-solid ratio was the most influential factor affecting carotenoid recovery. This can be attributed to its direct effect on mass transfer dynamics. A higher solvent volume increases the concentration gradient between the solid matrix and the liquid phase, thereby enhancing diffusion rates and preventing early saturation of the solvent. This allows more carotenoids to be released from the plant material into the soybean oil. While temperature also contributes by improving solubility and weakening cellular structures, its effect becomes limited once the system approaches equilibrium or when saturation conditions are reached. In oil-based extraction systems, this suggests that solvent quantity plays a more critical role than thermal energy in determining overall extraction efficiency.

The study also compared its findings with previous research involving other biomass sources such as pumpkin, peach, and carrot byproducts. These comparisons showed consistent trends, where higher solvent-to-solid ratios generally resulted in improved carotenoid recovery.

However, excessive temperature increases were often associated with reduced efficiency due to possible degradation of heat-sensitive carotenoid compounds. This reinforces the idea that moderate temperature conditions combined with optimized solvent ratios are ideal for maximizing yield in green extraction systems.

In addition to extraction efficiency, the study conducted a sustainability evaluation using the Path2Green assessment framework, which measures the environmental performance of extraction processes across 12 standardized principles. These principles include biomass sourcing, transport, pretreatment, solvent selection, scaling, purification, energy use, and waste management. The overall Path2Green score obtained was 0.593 out of a possible maximum of 1.00, indicating a moderately sustainable and environmentally friendly process.

The sustainability assessment revealed that the process performed strongly in several key areas. High scores were recorded for biomass utilization, as the study effectively repurposed agricultural waste rather than relying on freshly cultivated crops. Similarly, solvent selection scored highly due to the use of soybean oil, which is biodegradable, non-toxic, non-volatile, and safe for food-related applications. The process also demonstrated strong performance in scalability, purification, post-treatment, application potential, and waste management, as the extracted product could be directly utilized without requiring extensive purification or solvent removal.

However, certain limitations were identified in the sustainability analysis. Energy consumption received a low score due to the use of freeze-drying, grinding, and controlled temperature water bath systems, all of which require significant energy input. Pretreatment processes also contributed to reduced sustainability performance because mechanical processing steps were necessary to prepare the biomass for extraction. Additionally, yield efficiency was identified as an area for improvement, indicating that the extraction process was not fully exhaustive and that some carotenoids likely remained unextracted in the residual biomass. The study also noted the absence of a closed-loop extraction system, which limited its circular economy potential.

Despite these limitations, the study highlighted that soybean oil-based extraction remains a promising green alternative to conventional solvent systems. The integration of environmentally friendly solvents with waste valorization strategies demonstrates a strong alignment with sustainable development goals and green chemistry principles. Furthermore, the extracted carotenoid-rich oil can potentially be applied directly in food formulations, cosmetics, pharmaceuticals, and nutraceutical products without requiring further purification, thereby reducing processing costs and environmental burden.

In conclusion, Palero et al. (2026) successfully demonstrated that soybean oil is an effective and sustainable solvent for carotenoid extraction from carrot peel waste. The optimized extraction conditions provided high yields with strong model reliability, while the sustainability assessment confirmed its moderate environmental performance. Although improvements are still needed in energy efficiency, extraction completeness, and system integration, the study provides strong evidence that agricultural waste can be effectively transformed into high-value bioactive compounds using green extraction technologies. This research contributes significantly to the growing field of sustainable bioprocessing and supports the development of eco-friendly industrial practices that prioritize both efficiency and environmental responsibility.

A study by **Manalo (2021)** discussed the role of Digital Agriculture in supporting the modernization and industrialization goals of the Philippine agricultural sector through the adoption of Information and Communications Technologies (ICTs). The study highlighted how digital technologies have become important tools in improving agricultural productivity, efficiency, and access to information. Various ICT-based innovations have already been utilized in agriculture, including web-based disease diagnosis systems, crop monitoring sensors, mobile applications, and digital trading platforms. In the Philippines, DA-PhilRice developed several ICT-driven applications such as the Leaf Color Chart (LCC) app, e-binhi, e-damuhan, and AgriDoc, which provide farmers with timely and relevant agricultural information. These technologies were designed to support better crop management, improve decision-making, and enhance overall agricultural operations. The study emphasized that digital technologies can help farmers access critical information related to crop production, nutrient management, pest control, and market opportunities, enabling them to make informed decisions and improve farm performance. Through these innovations, Digital Agriculture has been recognized as a significant strategy in transforming traditional farming practices into more efficient, data-driven, and technology-enabled systems.

The study further examined the level of ICT adoption among Filipino farmers and identified several challenges that hinder the effective use of digital technologies. Based on the results of the 2016–2017 Rice-Based Farm Households Survey, approximately 93 percent of rice-farming households had access to mobile phones, smartphones, and internet services. Similarly, a large percentage of farmers expressed willingness to receive agricultural information through text messaging and internet-based platforms. Despite this high level of access, actual utilization of ICTs for farming activities remained relatively low. The survey revealed that only 31 percent of farmers used text messaging as a source of agricultural information, while only 21 percent actively searched for farming-related information online. These findings suggest that access to technology alone does not guarantee its effective use. The study argued that several factors contribute to this gap, including low levels of digital literacy, limited exposure to technology, lack of confidence in using digital tools, and ICT anxiety among farmers.

According to Manalo (2021), ICT anxiety is one of the major barriers to technology adoption in agriculture. ICT anxiety refers to the discomfort, fear, or uncertainty experienced by individuals when interacting with computers, mobile devices, or other digital technologies. This challenge is particularly common among older populations, including many Filipino farmers. Since a significant portion of the agricultural workforce in the Philippines consists of aging farmers, the adoption of new technologies often becomes difficult. The study explained that limited experience with digital tools and insufficient technical knowledge can discourage farmers from utilizing ICT-based solutions, even when such technologies are readily available. As a result, many farmers continue to rely on traditional methods of obtaining information, such as consulting fellow farmers, agricultural extension workers, and local community networks. This preference for interpersonal communication demonstrates that technological innovations must be accompanied by strategies that address behavioral and educational barriers to ensure successful implementation.

To address these challenges, the study introduced the concept of infomediaries as a means of bridging the gap between technology and farmers. Infomediaries are individuals who facilitate access to information by helping others use digital tools and communication technologies. The study described infomediaries as important agents in reducing the digital divide, particularly among populations that have limited technological skills. The digital divide refers to the disparity between individuals who can effectively access and utilize digital technologies and those who cannot. This divide is influenced by various factors, including educational attainment, socioeconomic status, technological literacy, and access to resources. By mobilizing infomediaries, agricultural organizations can help ensure that the benefits of digital technologies reach a wider population of farmers, including those who may be hesitant or unable to use ICTs independently.

The study highlighted the implementation of the Infomediary Campaign by DA-PhilRice from 2012 to 2017 as an innovative approach to promoting digital inclusion in agriculture. The campaign engaged high school students from rice-farming communities and encouraged them to serve as infomediaries for their parents and local farmers. These students accessed agricultural information through digital platforms and shared the information with farmers who lacked the skills or confidence to use ICT tools themselves. Through this initiative, young people became active participants in agricultural development while helping bridge the information gap within their communities. The campaign demonstrated how youth engagement can contribute to the dissemination of agricultural knowledge and encourage greater acceptance of digital technologies among farmers.

Results of the Infomediary Campaign showed positive outcomes in promoting information access and technology adoption. The study reported that more than 4,000 student-infomediaries submitted nearly 20,000 inquiries to the PhilRice Text Center between 2012 and 2016. These inquiries covered a wide range of agricultural topics, including rice varieties, pest management, nutrient management, crop establishment, and general farming practices. Students not only gathered information but also actively shared their knowledge with their parents, relatives, and other members of the farming community. The study observed that information dissemination often extended beyond the immediate recipients, creating a multiplier effect where agricultural knowledge was continuously shared among farmers. This process contributed to improved awareness of modern farming techniques and increased utilization of available information resources.

Furthermore, the study found evidence that the campaign positively influenced farmers' information-seeking behavior. Farmers who were exposed to the initiative became more likely to use text messaging services and digital communication platforms to obtain agricultural information. In some communities, the habit of consulting digital information sources continued even after the campaign ended. These findings suggest that targeted interventions and support systems can help overcome barriers to technology adoption and encourage the long-term use of ICT-based agricultural solutions. The study emphasized that successful digital transformation requires not only technological infrastructure but also educational programs, capacity-building initiatives, and community support mechanisms that enable users to maximize the benefits of digital tools.

The study concluded that Digital Agriculture can only achieve its full potential if efforts are made to ensure inclusivity and accessibility among all stakeholders. While digital technologies offer significant opportunities for improving productivity, information management, and decision-making, their effectiveness depends largely on the willingness and ability of users to adopt them. Initiatives such as the Infomediary Campaign demonstrate that human-centered approaches can play a critical role in facilitating technology adoption and reducing barriers associated with digital literacy and ICT anxiety. The study recommended expanding infomediary programs, strengthening ICT training initiatives, integrating agriculture-related digital skills into educational curricula, and providing continuous support to both farmers and agricultural extension workers.

This literature is relevant to the present study as it demonstrates the significant role of digital technologies in improving operational efficiency, information management, and data-driven decision-making. Similar to the proposed system for real-time yield analytics and extraction efficiency monitoring, the study highlights how digital solutions can enhance monitoring processes, facilitate timely access to information, and support more effective management practices. The findings support the development of digital production systems by emphasizing the value of technology in collecting, analyzing, and utilizing data to improve productivity and operational performance. Furthermore, the study reinforces the importance of user adoption and accessibility in ensuring the successful implementation of digital innovations within agricultural and food-processing industries

A study by **Barroga, Rola, Depositario, Digal, and Pabuayon (2019**) investigated the determinants of the extent of technological innovation adoption among micro, small, and medium food processing enterprises (MSMFEs) in the Davao Region, Philippines. The research focused on enterprises that were beneficiaries of the Department of Science and Technology–Small Enterprises Technology Upgrading Program (DOST-SETUP), a government initiative that provides technological support, upgraded equipment, and technical assistance to improve productivity and competitiveness in the food processing sector. The study aimed to determine not only whether MSMFEs adopt technological innovations but also the degree or extent to which these innovations are implemented within their operations.

The researchers utilized a census of 52 MSMFEs in Davao Region and employed both quantitative and qualitative methods to analyze the data. Specifically, an ordered logistic regression model was used to identify the factors influencing the level of technological innovation adoption. The study also used index construction to measure the extent of adoption across multiple technological interventions, including food safety training, Good Manufacturing Practices (GMP) assessment, plant layout improvement, packaging and labeling, laboratory testing, product development, energy audit, cleaner production technology (CPT) consultancy, MPEX consultancy, and technology training. These innovations represent a comprehensive package of technical and managerial upgrades introduced through the SETUP program.

Findings of the study showed that MSMFEs in the Davao Region generally exhibited varying levels of technological innovation adoption, ranging from non-adoption to full adoption. A small portion of the enterprises failed to adopt any of the technological innovations provided, while others demonstrated low to moderate adoption levels. However, the majority of MSMFEs were classified as highly adopters, indicating that most enterprises were able to integrate several SETUP innovations into their operations. Only a smaller segment of the sample achieved full adoption, meaning they implemented nearly all of the technological innovations offered by the program.

Among the specific innovations introduced by SETUP, food safety training emerged as the most widely adopted intervention, followed by GMP assessment and MPEX consultancy. These innovations were particularly important for MSMFEs because they directly address compliance with food safety standards and operational efficiency. On the other hand, innovations such as product development and cleaner production technology consultancy were among the least adopted. The study noted that more advanced or resource-demanding innovations tended to have lower adoption rates, especially among smaller enterprises with limited capacity and technical expertise.

The study further highlighted that the adoption of technological innovations does not only depend on whether enterprises accept or reject a particular technology, but also on the depth and breadth of implementation across various business functions. This means that even if MSMFEs adopt certain innovations, the level of integration into their production processes, management systems, and product development activities may differ significantly. Based on this, the researchers categorized MSMFEs into five groups according to their extent of adoption: did not adopt, less adopted, moderately adopted, highly adopted, and fully adopted.

In terms of distribution, most medium-sized enterprises were found to have lower levels of adoption compared to micro and small enterprises. Micro enterprises, despite their limited resources, showed a relatively higher tendency toward adopting multiple technological innovations. Small enterprises were mostly classified as moderate to high adopters, depending on their exposure to export markets and access to external support services. The study suggests that enterprise size alone does not fully determine innovation adoption, but interacts with other organizational and contextual factors.

One of the key findings of the study is that the combination of food safety training, GMP assessment, and MPEX consultancy was the most commonly adopted set of innovations. These three are closely interconnected, as MPEX consultancy often recommends improvements related to GMP compliance and food safety practices. This indicates that MSMFEs tend to adopt innovations in clusters rather than individually, especially when the technologies are complementary and collectively improve production efficiency and compliance with regulatory standards.

The study also revealed that full adoption of technological innovations was relatively limited, with only about 21% of MSMFEs achieving this level. Enterprises under this category were able to implement most or all of the SETUP-supported innovations, demonstrating strong organizational capacity and readiness for technological upgrading. Micro enterprises in this category adopted nearly all available innovations, while small enterprises typically adopted a slightly smaller set, often excluding more advanced innovations such as laboratory analysis or product development.

Highly adopted MSMFEs represented a significant portion of the sample. These enterprises generally implemented key foundational innovations such as GMP assessment and food safety training, along with selected productivity-enhancing interventions. However, they still lacked full integration of more advanced technological services. This suggests that while these enterprises are relatively progressive in adopting innovation, constraints such as cost, technical expertise, and organizational capacity still limit full adoption.

Moderately adopted MSMFEs showed partial integration of SETUP innovations, often focusing on basic operational improvements. These enterprises typically adopted essential interventions such as food safety training and plant layout improvements but did not fully engage in more complex innovations like laboratory analysis or product development. Their level of adoption indicates an intermediate stage of technological readiness, where awareness and initial implementation exist but full assimilation remains limited.

Less-adopted MSMFEs were characterized by minimal engagement with SETUP innovations, typically adopting only a few interventions such as technology training or food safety training. In this category, medium enterprises were more dominant, suggesting that some established firms may perceive less need for external technological assistance due to existing internal capabilities. This group highlights that adoption is not always higher among larger firms, especially when they already possess in-house technical expertise.

At the lowest level, the study identified a very small number of enterprises that did not adopt any technological innovations at all. These cases were linked to external shocks such as natural disasters and management issues, which disrupted operations and prevented the proper utilization of SETUP assistance. These findings emphasize that external environmental factors and governance issues can also significantly affect innovation adoption outcomes.

The study further examined the determinants of technological innovation adoption using an ordered logistic regression model. The results identified four statistically significant factors influencing the extent of adoption: sex of the owner, educational attainment, business scale, and type of market. These variables were found to consistently influence whether MSMFEs adopt innovations at a higher or lower level.

In terms of sex of the owner, the study found that male-owned enterprises were more likely to achieve higher levels of technological innovation adoption compared to female-owned enterprises. Although female entrepreneurs represented a larger proportion of MSMFE owners in the sample, male owners were more likely to engage in extensive adoption. This was attributed to differences in mobility, access to training opportunities, and exposure to external information networks, which may influence innovation-related decision-making.

Educational attainment was also found to have a positive relationship with the extent of adoption. MSMFE owners with higher educational levels were more likely to adopt a wider range of technological innovations. This is because education enhances the ability to understand technical information, evaluate innovation benefits, and manage the risks associated with technological change. As a result, more educated owners tend to be more open to upgrading their production systems and integrating new technologies into their operations.

Business scale, on the other hand, showed a negative relationship with adoption extent, indicating that micro enterprises were more likely to adopt innovations compared to small and medium enterprises. This finding suggests that smaller firms are more flexible and can implement changes more quickly due to simpler organizational structures and centralized decision-making. In contrast, larger firms may experience more complex internal processes that slow down innovation adoption.

The type of market was also identified as a significant determinant. MSMFEs that cater to both domestic and export markets were more likely to adopt technological innovations at higher levels. This is because export-oriented firms are required to comply with stricter international standards, which encourages them to adopt advanced technologies and quality control systems to remain competitive in global markets.

Other variables such as age, years of experience, organizational type, and sources of information were found to be statistically insignificant in influencing the extent of technological innovation adoption. This suggests that while these factors may have some descriptive influence, they do not significantly determine the level of adoption when analyzed together with stronger predictors such as education, gender, business scale, and market orientation.

Overall, the study concluded that technological innovation adoption among MSMFEs in the Davao Region is a multi-dimensional process influenced by both individual and organizational factors. The findings highlight that innovation adoption is not simply a matter of access to technology, but also depends on the characteristics of decision-makers and the market environment in which enterprises operate.

The study is relevant to the present research because it provides empirical evidence on how personal demographics and organizational characteristics influence the extent of technological innovation adoption in food processing enterprises. It also highlights the importance of government support programs such as DOST-SETUP in promoting innovation among MSMEs, particularly in developing regions.

A study by **Ang, Amongo, Paras Jr., Suministrado, and Peralta (2022),** entitled *Development of Banana Peduncle Juice Extractor for Ethanol and Fiber Production*, focused on the conceptualization, design, fabrication, and performance evaluation of a specialized mechanical system intended for the efficient processing of banana (Musa acuminata C.) peduncles. The research was motivated by the increasing concern regarding agricultural waste management in the Philippines, particularly within large-scale banana plantations where peduncles are typically discarded despite their significant potential as raw materials for bioethanol production and natural fiber utilization. The authors emphasized that although banana peduncles contain valuable lignocellulosic components and fermentable sugars, their utilization remains limited due to the lack of appropriate extraction technology that can simultaneously maximize juice recovery while preserving fiber integrity. Conventional extraction machines, originally designed for other agricultural commodities such as sugarcane, were found to be inefficient when applied to banana peduncles due to differences in material structure, density, and mechanical resistance. This gap in technology prompted the development of a dedicated juice extraction system tailored specifically for banana peduncle processing.

The developed machine was designed using engineering principles that prioritized functionality, manufacturability, safety, and cost-effectiveness. The researchers deliberately utilized locally available materials in order to ensure affordability and ease of replication, particularly for potential use in rural agricultural communities and small-scale processing facilities. The system employed a crushing roll mechanism as the primary extraction component, consisting of a main roll and multiple feeder rolls arranged in a configuration that enables controlled compression of the peduncle material. This configuration was intended to optimize juice release while minimizing excessive mechanical damage to the fiber structure, thereby allowing the biomass to be utilized for both ethanol fermentation and fiber recovery applications. The structural design also incorporated protective enclosures and adjustable components to enhance operational safety and accommodate varying operator conditions. In addition, the transmission system was engineered using a combination of engine-driven pulleys, gear reducers, and chain-sprocket assemblies, allowing controlled variation of rotational speed to evaluate performance under different operational settings.

To assess the effectiveness of the developed machine, the researchers employed a systematic experimental design using a two-factor, three-level full factorial approach. The two independent variables considered were roll rotational speed (50 rpm, 70 rpm, and 90 rpm) and crushing load (0.8 kN, 1.2 kN, and 1.6 kN). A total of 27 experimental runs were conducted to determine the influence of these variables on machine performance. Each trial utilized approximately 40 kilograms of freshly harvested banana peduncle samples, which were processed individually to ensure accurate measurement of performance indicators. The study followed standardized testing procedures based on established agricultural machinery evaluation protocols, specifically PAES 235:2008 for multi-crop juice extractors and PAES 229:2005 for fiber decorticators. These standards ensured that the evaluation process was consistent, reliable, and comparable to other agricultural machinery studies.

The performance of the machine was evaluated using multiple response variables, including input capacity, juice extraction rate, juice recovery, extraction losses, extraction efficiency, fuel consumption, fiber extraction rate, fiber recovery, and fiber tensile strength. In addition, the °Brix level of the extracted juice was measured to determine sugar concentration, which is an important factor in ethanol production. The inclusion of both mechanical and biochemical indicators allowed for a comprehensive assessment of the system, covering not only productivity but also product quality. The tensile strength of the extracted fiber was also analyzed using a universal testing machine following ASTM D3822-01 standards to determine whether mechanical processing had adversely affected fiber integrity.

Results of the study showed that both roll rotational speed and crushing load significantly influenced machine performance, although their effects varied depending on the specific output variable. Higher roll rotational speeds generally resulted in increased input capacity, juice extraction rate, and fiber extraction rate due to reduced processing time and faster material throughput within the crushing chamber. Conversely, lower rotational speeds tended to reduce throughput but improved certain recovery metrics due to longer retention time within the extraction system. Crushing load, on the other hand, exhibited a dual effect depending on the response variable. Lower crushing loads facilitated easier passage of peduncle material through the rolls, thereby increasing processing speed, while higher crushing loads enhanced the mechanical compression of the material, resulting in improved juice recovery and extraction efficiency.

The interaction between roll speed and crushing load revealed important trade-offs in system performance. For instance, although higher crushing loads improved juice recovery, they also increased resistance within the system, which reduced processing speed and increased fuel consumption. Similarly, higher rotational speeds improved throughput but contributed to higher extraction losses, as some juice was not fully recovered before the material exited the machine. These findings highlight the complexity of optimizing mechanical systems for biomass processing, where improvements in one performance aspect may negatively affect another. As such, the study emphasized the importance of multi-response optimization techniques to determine the most balanced operating conditions.

Using response surface methodology (RSM), the researchers performed a multi-objective optimization analysis to identify the most desirable combination of operating parameters. The optimization process considered multiple performance goals simultaneously, including maximizing input capacity, juice recovery, extraction efficiency, fiber extraction rate, and fiber recovery, while minimizing extraction losses and fuel consumption. Based on desirability function analysis, the optimal operating condition was identified at approximately 71.378 rpm and 0.8 kN crushing load. At this optimal setting, the machine achieved an input capacity of about 307.24 kg/h, a juice extraction rate of 82.92 kg/h, juice recovery of 26.08%, extraction efficiency of 18.46%, fiber extraction rate of 54.83 kg/h, fiber recovery of 42.29%, and fuel consumption of 0.391 L/h. These results demonstrate that the system can achieve a balanced level of efficiency while maintaining both juice and fiber output quality when operated under optimized conditions.

Further analysis revealed that neither roll rotational speed nor crushing load significantly affected the °Brix content of the extracted juice or the tensile strength of the recovered fiber. This indicates that while the mechanical parameters influence the quantity of juice and fiber produced, they do not significantly alter the chemical composition of the juice or the structural strength of the fiber. This finding is particularly important because it suggests that the quality of the end products remains stable regardless of operational variations, making the system more reliable for industrial applications where consistency is essential.

In terms of fermentation performance, the extracted juice was subjected to ethanol production analysis using Saccharomyces cerevisiae as the fermentation agent. The results showed that the fermented slurry contained approximately 10–11% ethanol by volume across all treatment combinations. Statistical analysis indicated no significant differences in ethanol content among the different mechanical settings, reinforcing the observation that sugar concentration, rather than mechanical extraction conditions, is the primary determinant of ethanol yield. This further supports the conclusion that the machine primarily influences extraction efficiency and throughput rather than biochemical conversion outcomes.

In conclusion, the study demonstrated that the developed banana peduncle juice extractor is an effective machine for converting agricultural waste into valuable products such as bioethanol feedstock and natural fiber. The system successfully addressed the limitations of conventional extraction machines by introducing a design specifically optimized for banana peduncle characteristics. The findings highlight the importance of mechanical optimization in improving productivity while maintaining product quality. Moreover, the study underscores the potential of banana peduncle as a sustainable raw material for bioenergy and fiber-based industries, contributing to waste reduction, renewable energy development, and value-added agricultural processing. The authors further recommended future improvements such as modifying roll geometry, improving transmission efficiency, experimenting with alternative crushing mechanisms, and testing additional banana varieties to enhance the versatility and efficiency of the system.

A s**tudy by Jiang and Po (2023**) investigated the influence of human resource practices on employee retention in the food manufacturing industry in the Philippines and examined the moderating role of the work environment. The researchers focused on five key human resource practices: training and development, employee participation in decision-making, compensation, job security, and performance appraisal. Grounded in Social Exchange Theory, the study proposed that employees are more likely to remain with an organization when they perceive that the organization provides adequate support, fair treatment, and opportunities for growth. In return, employees reciprocate through commitment, loyalty, and long-term retention.

The study employed a quantitative research design using a cross-sectional survey method. Data were collected from 206 employees working in various food manufacturing companies in the Philippines through both online and offline questionnaires. The researchers used a five-point Likert scale to measure employee perceptions regarding human resource practices, work environment, and employee retention. Statistical analyses, including regression and moderation analysis, were conducted using JAMOVI software to determine the relationships among the variables.

The findings revealed that employee participation in decision-making, compensation, job security, and performance appraisal significantly and positively influenced employee retention. Employees who were given opportunities to participate in organizational decisions tended to develop a stronger sense of belonging and commitment to their companies. Likewise, fair compensation and benefits were found to be major factors that encouraged employees to remain in their organizations. The results further indicated that job security contributed positively to retention because employees preferred stable employment arrangements that reduced uncertainty regarding their future careers. Performance appraisal was also shown to strengthen employee retention by providing feedback, recognition, and opportunities for professional improvement.

Among the identified human resource practices, compensation emerged as the strongest predictor of employee retention. The researchers found that employees were more likely to remain in organizations when they perceived that they were fairly compensated for their work and contributions. This finding emphasizes the importance of providing competitive salaries and benefits in order to attract and retain skilled employees in the food manufacturing sector. The study concluded that compensation plays a critical role in maintaining workforce stability and reducing employee turnover.

In contrast, the study found that training and development did not significantly influence employee retention. This result differed from several previous studies that suggested training opportunities increase employee loyalty and commitment. The researchers proposed that training programs may not automatically lead to retention unless employees perceive them as valuable, relevant, and beneficial to their career advancement. This finding highlights the need for organizations to carefully evaluate the effectiveness of their training and development initiatives and ensure that such programs align with employee needs and organizational objectives.

Another significant contribution of the study was its examination of the work environment as a moderating variable. The findings showed that the work environment strengthened the relationship between employee participation in decision-making, compensation, job security, performance appraisal, and employee retention. Employees working in supportive environments were more likely to respond positively to human resource practices and demonstrate stronger intentions to remain in the organization. The study emphasized that a positive work environment includes both physical and non-physical factors, such as workplace safety, adequate facilities, healthy interpersonal relationships, effective communication, and a supportive organizational culture.

The researchers further explained that improvements in the physical work environment, including proper ventilation, safe production equipment, reduced workplace noise, and clean facilities, can increase employee satisfaction and productivity. At the same time, non-physical aspects such as teamwork, leadership support, organizational structure, and employee relationships contribute significantly to employee motivation and retention. These findings suggest that organizations should not only focus on implementing effective human resource policies but also invest in creating a work environment that promotes employee well-being and engagement.

The study has important implications for food manufacturing companies in the Philippines. It suggests that organizations should prioritize compensation systems, employee involvement, job security measures, and performance evaluation mechanisms to improve retention rates. Furthermore, managers should recognize the importance of maintaining a positive work environment to maximize the effectiveness of these human resource practices. By doing so, organizations can reduce employee turnover, improve workforce stability, and enhance overall organizational performance.

This literature is relevant to the present study because it demonstrates how organizational practices, performance monitoring, and workplace conditions influence efficiency, productivity, and long-term organizational success in the food manufacturing industry. The findings support the development of digital systems that provide real-time monitoring, performance tracking, and data-driven decision-making capabilities. Similar to the proposed system for real-time yield analytics and extraction efficiency monitoring, the study highlights the importance of accurate information, continuous monitoring, and effective management practices in improving operational performance and achieving organizational goals.

2.3 Related Literature

A study by Somsen, Capelle, and Tramper (2004) introduced Production Yield Analysis (PYA), a systematic method used to evaluate raw material efficiency in food processing. The study developed a "Yield Index" to distinguish between necessary losses and avoidable losses, helping identify inefficiencies in production systems. The authors emphasized that relying solely on historical benchmarks may conceal actual waste, highlighting the importance of model-based analysis. This literature is relevant to the present study as it supports the development of a digital system capable of identifying and analyzing production losses in real time.

A study by **Charbonnier, Béranger, and Cognon (2005)** investigated the application of trend extraction and analysis techniques for complex system monitoring and decision support in industrial and medical environments. The researchers focused on developing an online monitoring methodology capable of analyzing continuously changing data from highly complex systems such as industrial automation processes, food manufacturing operations, and intensive care monitoring systems. According to the researchers, modern industrial environments generate large volumes of operational data that are difficult for human operators to interpret manually, especially in situations where process conditions rapidly change and immediate decisions are required. The study emphasized that traditional monitoring approaches often rely heavily on manual observation and operator expertise, which may result in delayed responses, reduced efficiency, and increased risk of operational errors during abnormal conditions.

To address these challenges, the researchers proposed a real-time trend extraction methodology designed to analyze univariate time-series data using semi-qualitative representations. The system utilized three primary trend primitives: **Increasing, Decreasing, and Steady**. These primitives were used to represent operational behavior over specific time intervals, allowing users to interpret process conditions more effectively. According to the study, converting numerical signals into symbolic trend representations helps reduce cognitive overload and improves operator understanding of complex operational situations.

The methodology involved several stages. First, continuously generated process data were segmented into linear intervals. These intervals were then classified into temporal patterns such as increasing trends, decreasing trends, stable conditions, transient variations, and step changes. After classification, the detected temporal patterns were transformed into semi-quantitative episodes describing the behavior of operational variables over time. The researchers explained that this approach enabled the monitoring system to identify abnormal process behavior without requiring highly complex mathematical models of the entire industrial system.

The study also explored the application of trend extraction for predictive maintenance and operational supervision in industrial systems. The researchers explained that highly automated production systems often operate under demanding conditions to maximize productivity and profitability, increasing the likelihood of equipment degradation and operational faults. By continuously monitoring operational variables and analyzing abnormal trend patterns, the proposed methodology could assist in identifying equipment deterioration before critical failures occurred. The researchers concluded that predictive monitoring capabilities support preventive maintenance strategies, reduce operational downtime, and improve overall equipment reliability.

A study by **Davies, Fried, and Gather (2003)** entitled “Robust Signal Extraction for On-line Monitoring Data” examined statistical methods for extracting meaningful information from continuously generated time-series data in real-time monitoring environments. The researchers focused on developing and evaluating robust techniques that can effectively separate true underlying signals from noise, outliers, and irregular fluctuations. The study was motivated by the increasing demand for reliable monitoring systems in critical environments such as intensive care units and industrial process control systems, where fast and accurate interpretation of data is essential for decision-making.

The researchers explained that modern monitoring systems generate large volumes of data due to continuous recording of physiological or operational variables. In healthcare settings, for example, patient data such as heart rate, blood pressure, and oxygen levels are recorded at very short intervals, producing complex datasets that must be analyzed immediately. Similarly, industrial systems also generate high-frequency operational data that reflect machine performance, process efficiency, and production status. However, these datasets are often affected by noise, measurement errors, and external disturbances, which make it difficult to identify meaningful patterns using traditional analysis methods.

According to the study, one of the major challenges in online monitoring is distinguishing clinically or operationally relevant changes from irrelevant noise and outliers. The researchers emphasized that false alarms in monitoring systems are a significant problem, particularly in intensive care environments where excessive alarms may lead to alarm fatigue among medical staff. This reduces trust in automated systems and may cause important warnings to be ignored. In industrial environments, inaccurate signal interpretation may also lead to unnecessary maintenance actions or failure to detect real system faults. Because of these issues, there is a strong need for robust statistical methods that can reliably extract true signals from noisy data in real time.

Another important contribution of the study is its discussion on the trade-off between robustness and computational efficiency. The researchers emphasized that while highly robust methods provide better resistance to data contamination, they may not always be suitable for real-time applications due to processing delays. Therefore, selecting an appropriate signal extraction method requires balancing accuracy, stability, and computational cost depending on the system requirements.

In conclusion, the study by Davies, Fried, and Gather (2003) provides valuable insights into the importance of robust statistical methods in real-time monitoring systems. Its emphasis on noise reduction, trend extraction, and reliable signal interpretation makes it highly applicable to modern industrial systems, particularly in the context of automated food production and digital manufacturing technologies.

A study conducted by **Charmine Sheena R. Saflor and Marvin I. Noroña (2021)** explored the implementation of Total Productive Maintenance (TPM) within the Philippine rice production sector. The research focused on addressing the increasing operational and maintenance challenges experienced by farmers and agricultural cooperatives due to the growing use of mechanized farming equipment. The authors emphasized that many rice production operators in the Philippines still rely on reactive maintenance practices, where machines are repaired only after breakdowns occur. This approach often results in frequent equipment failures, increased maintenance expenses, production delays, and reduced overall productivity. The study highlighted that inadequate maintenance management can shorten the lifespan of agricultural machinery and negatively affect the efficiency of rice production operations.

The study is relevant to the present research because it demonstrates the importance of monitoring equipment performance and maintenance activities to improve operational efficiency and reduce production losses. It supports the idea that implementing a digital and data-driven monitoring system can help organizations identify inefficiencies, manage maintenance schedules, and enhance productivity in real time. Similar to the proposed study on real-time yield analytics and extraction efficiency monitoring, the research emphasizes the value of systematic analysis and continuous monitoring in optimizing production processes, minimizing waste, and improving decision-making within industrial and agricultural operations.

A study conducted by **Desai Shivani Sarjerao (2020),** Design, Development and Performance Evaluation of Juice Extractor, presented the development of a hydraulically operated juice extraction machine intended for processing pomegranate arils and bottle gourd.

The experimental evaluation focused on three main independent variables: size reduction (shred size), applied hydraulic pressure, and filter bag pore size. These were tested in different combinations to determine their effects on several dependent performance indicators, namely juice yield, juice extraction efficiency, extraction losses, machine capacity, and sedimentation percentage. A series of experimental trials were conducted for both pomegranate arils and bottle gourd, with repeated measurements used to ensure reliability of results. Statistical analysis, including ANOVA, was applied to determine the significance of main effects and interaction effects among the variables.

In terms of juice extraction efficiency, the study reported maximum values of 94.30% for pomegranate and 80.27% for bottle gourd under optimal operating conditions. The results also showed a consistent decrease in extraction losses as pressure increased and shred size decreased, demonstrating improved material utilization and reduced wastage. Although most interaction effects between variables were statistically non-significant, individual factors such as pressure and filter bag pore size were found to have a significant impact on extraction performance.

Overall, the study emphasizes that mechanical design parameters and operating conditions play a critical role in determining the efficiency of juice extraction systems. It further suggests that optimizing these variables can significantly enhance juice recovery, reduce processing losses, and improve machine productivity. This research is relevant to the present study as it provides an engineering-based framework for evaluating extraction systems and supports the development of optimized, efficiency-driven food processing technologies in agricultural production systems.

A study by **Bartoszewicz and Wdowicz (2019)** presented an automated reporting approach that integrates the SAP ERP Production Planning (PP) module with Microsoft Excel and Visual Basic for Applications (VBA). The researchers observed that while SAP ERP provides extensive support for production planning and operations, its standard reporting features may not always satisfy the analytical and reporting needs of production managers. To address this limitation, they developed the **Follow the Plan (FTP) Report**, an application that automatically retrieves production data from SAP ERP, processes the information, and generates reports comparing planned production activities with their actual execution. This automation reduces the need for manual data handling while improving the speed and consistency of production reporting.

The proposed system gathers production information from the **COOISPI** and **MB51** transactions within SAP ERP. These transactions contain data on production orders, work centers, planned schedules, completed orders, and material movements. After extracting the required information, the application organizes and analyzes the data in Microsoft Excel, allowing users to determine whether production orders were completed as planned. The system also identifies production deviations, such as orders that were not started, orders produced beyond planned quantities, and production results that fall outside the accepted tolerance. In addition, users can record standardized root causes, including raw material shortages, equipment failures, low machine performance, quality issues, and inaccurate system data, to support further investigation and continuous process improvement.

The researchers found that automating the reporting process significantly improved operational efficiency. Preparing production compliance reports manually previously required approximately two hours because users had to retrieve, organize, and validate data from multiple SAP transactions. With the FTP Report application, the same process could be completed in about five minutes, while also reducing the possibility of human error. The study concluded that integrating ERP systems with automated reporting tools enhances production monitoring, supports faster managerial decision-making, and provides more reliable information for evaluating production performance. This literature is relevant to the present study because it demonstrates how digital technologies can automate production monitoring and reporting, supporting real-time analysis and more effective evaluation of production efficiency, which aligns with the objectives of developing a real-time yield analytics and extraction efficiency monitoring system for AlDi Foods Inc.

2.4 Synthesis of the Review Literature

The reviewed foreign and local literature consistently demonstrates that production systems in agriculture and food manufacturing are undergoing a significant transformation from traditional manual operations toward digitally enabled, data-driven, and automated environments. This transformation is largely driven by rapid advancements in Information and Communications Technology (ICT), artificial intelligence (AI), machine learning, big data analytics, and real-time monitoring systems. Collectively, these technologies enhance operational efficiency, improve production accuracy, and support faster, evidence-based decision-making. Across the studies reviewed, there is a clear consensus that digital integration is no longer optional but a necessary component for modern, competitive, and efficient production systems, particularly in industries involving food processing and extraction operations.

One of the most consistent findings across the literature is the growing importance of real-time monitoring systems and centralized digital platforms in production environments. Studies such as Verhoef et al. (2021), Makhmudah et al. (2025), and Bartoszewicz and Wdowicz (2019) emphasize that traditional paper-based recording systems and manual reporting processes are highly inefficient due to delays, data inconsistencies, and lack of immediate accessibility. These limitations prevent production managers from identifying inefficiencies, equipment malfunctions, and yield losses at the moment they occur. In contrast, digital dashboards, ERP-integrated systems, and automated reporting tools provide real-time visibility of production activities, allowing faster response to operational issues. This shift highlights that real-time data accessibility is a critical requirement for effective production monitoring systems such as the proposed yield and extraction efficiency monitoring system for AlDi Foods Inc.

Another major finding is the increasing role of data analytics, artificial intelligence, and intelligent computing systems in optimizing production processes. Studies by Siddique et al. (2025) and Sharma et al. (2021) show that AI-powered systems are capable of processing large and complex datasets to identify operational patterns, predict production outcomes, and optimize resource allocation. These systems support predictive, descriptive, and prescriptive analytics, enabling manufacturers to minimize waste, improve productivity, and enhance decision-making accuracy. Furthermore, technologies such as machine learning and computer vision are increasingly used to automate inspection, monitor production flow, and detect irregularities, reducing reliance on manual observation and improving operational reliability in industrial environments.

The literature also highlights the critical importance of production efficiency metrics, yield analysis, and performance evaluation systems in modern manufacturing. Mansour and Al-Hamdani (2024) introduced internal key performance indicators (KPIs) as tools for identifying production inefficiencies and defects, while Somsen et al. (2004) emphasized Production Yield Analysis (PYA) as a structured method for distinguishing between avoidable and unavoidable production losses. Similarly, Dora et al. (2020) found that most food losses occur during the production stage due to machine inefficiencies, human errors, poor process control, and inadequate operational management. These findings strongly indicate that continuous monitoring of yield and efficiency is essential in minimizing production losses, improving resource utilization, and ensuring consistent operational performance—core objectives of real-time production monitoring systems.

In food extraction and processing systems, multiple studies such as Desai (2020), Mphahlele et al. (2016), Aviara et al. (2020), and Ang et al. (2022) demonstrate that mechanical and operational parameters significantly influence extraction yield, product quality, and production efficiency. Variables such as pressure, rotational speed, feed condition, and extraction technique directly affect both the quantity and quality of output. In addition, Mphahlele et al. (2016) further explain that extraction methods not only influence yield but also affect chemical composition, antioxidant content, aroma compounds, and overall product quality. These findings reinforce the idea that monitoring extraction processes is necessary not only for improving efficiency but also for ensuring product consistency and quality control in food manufacturing systems.

The reviewed literature also shows a growing adoption of automation technologies and intelligent monitoring systems in industrial environments. Sabih et al. (2023) demonstrated that computer vision and machine learning can effectively automate material flow monitoring, improving accuracy while reducing manual intervention. Similarly, ERP-based systems studied by Bartoszewicz and Wdowicz (2019) show that automated reporting systems significantly enhance production tracking efficiency, reduce human error, and support faster managerial decision-making. These studies collectively indicate a global shift toward integrated digital production systems that enable real-time monitoring, automated computation, and improved operational transparency.

Local studies further confirm the relevance and urgency of digital transformation in the Philippine context, particularly in agriculture and food manufacturing industries. Bagaforo and Albason (2024) and Barroga et al. (2019) found that while MSMEs benefit from technological adoption, they continue to face challenges such as limited technical expertise, financial constraints, and inadequate infrastructure. De Leon et al. (2017) emphasized that centralized digital systems significantly improve data reliability, production tracking, and operational efficiency in local enterprises. Meanwhile, Manalo (2021) highlighted that although ICT access is widespread among Filipino farmers, actual utilization remains low due to digital literacy gaps, ICT anxiety, and lack of technical support. These findings suggest that the success of digital systems depends not only on availability but also on usability, accessibility, and user capability.

Collectively, the literature establishes that digital transformation, real-time monitoring, and data-driven analytics are essential components of modern production systems. This is particularly significant in food processing and extraction industries where efficiency, yield optimization, and waste reduction directly affect productivity and profitability. The findings strongly support the development of integrated systems that provide real-time data capture, automated yield computation, and extraction efficiency monitoring—such as the proposed system for AlDi Foods Inc.—to enhance operational performance and support data-driven decision-making in production environments.

Research Gaps Identified

Despite the extensive body of literature on digital transformation, production optimization, and intelligent manufacturing systems, several important gaps remain evident, particularly when applied to food processing and juice extraction operations.

First, there is a noticeable lack of research and system development focused specifically on real-time monitoring systems designed for juice extraction and similar food processing operations. Most existing studies concentrate on broader industrial manufacturing environments, agricultural ICT applications, or general automation systems. While these studies provide valuable insights into production optimization, they do not sufficiently address the unique requirements of extraction-based processes, where continuous measurement of input materials, juice yield, and extraction efficiency is critical. As a result, there is limited exploration of systems tailored to monitor and evaluate performance in real-time within juice production environments such as AlDi Foods Inc.

Second, the current literature often treats key production functions—such as production monitoring, yield analysis, quality control, and reporting—as separate and disconnected systems. This fragmented approach creates inefficiencies in data handling and delays in decision-making, as information must be processed across multiple platforms or manual procedures. There is a clear absence of a unified and integrated system that consolidates these functions into a single real-time platform capable of automatically computing, analyzing, and presenting production performance indicators in a cohesive manner.

Third, many of the existing advanced solutions in production monitoring rely heavily on high-cost and technically complex technologies, including IoT-based sensor networks, machine learning algorithms, cloud computing infrastructures, and industrial automation systems. While these technologies are highly effective in large-scale manufacturing environments, they are often impractical for small and medium-sized enterprises (SMEs) due to financial constraints, lack of technical expertise, and limited infrastructure support. This creates a technological gap where SMEs in the food processing sector are unable to fully benefit from advanced production monitoring innovations.

Fourth, although local studies in the Philippines confirm the advantages of digital transformation in agriculture and food manufacturing, there is still a lack of localized, real-time production monitoring systems specifically developed for juice extraction industries. Most available systems or studies focus on agricultural production, MSME business processes, or general ICT adoption rather than actual operational monitoring within food processing plants. This gap is particularly evident in companies such as AlDi Foods Inc., where production tracking is still largely dependent on manual or semi-digital recording systems.

Fifth, while modern production environments generate large volumes of operational data, there remains a significant gap in transforming raw production data into real-time, actionable insights. Many systems are capable of collecting data but lack intelligent mechanisms for immediate interpretation, such as automated yield computation, extraction efficiency evaluation, and instant performance feedback. This limitation reduces the usefulness of collected data and prevents production managers from making timely, data-driven decisions during ongoing operations.

Lastly, there is still limited research addressing the systematic transition from traditional manual recording systems to fully digital, real-time production monitoring systems, particularly in food processing industries. Many facilities continue to rely on paper-based logs, spreadsheets, or semi-manual recording methods, which are prone to human error, data loss, and delayed reporting. The literature lacks practical frameworks and implemented systems that guide SMEs in shifting toward fully integrated digital production monitoring environments without requiring high-cost industrial infrastructure.

Overall, these gaps highlight the need for a cost-effective, integrated, and real-time production monitoring system that is specifically designed for juice extraction processes. The proposed study, *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."*, directly addresses these gaps by developing a centralized system that enables real-time tracking of production input, output, yield computation, and extraction efficiency analysis, thereby supporting improved operational efficiency and data-driven decision-making in food processing operations.

How the Current Project Addresses the Gaps

The proposed study, *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.",* directly addresses the research gaps identified in the reviewed literature through the development of a centralized, digital, and real-time production monitoring system specifically designed for juice extraction operations. The system aims to replace traditional paper-based production recording with a more efficient digital platform that enables accurate data collection, automated computation, and timely access to production information. By providing production managers with organized and up-to-date operational data, the system supports more effective production management and data-driven decision-making.

First, the proposed system specifically addresses the lack of real-time monitoring solutions developed for juice extraction and food processing operations. While many existing studies focus on general manufacturing automation, agricultural information systems, or industrial monitoring applications, few concentrate on the unique production requirements of juice extraction. The proposed system is designed to monitor critical production variables such as raw material input, extracted juice output, packed yield, yield percentage, and extraction efficiency. These indicators are essential in evaluating production performance and identifying areas where operational improvements can be made. By focusing on extraction-specific performance measures, the study contributes practical knowledge to food processing research while addressing the operational needs of AlDi Foods Inc.

Second, the proposed system addresses the fragmented nature of existing production monitoring approaches by integrating multiple production management functions into a single centralized platform. Instead of maintaining separate records for production monitoring, yield computation, efficiency analysis, and report generation, the system consolidates these processes into one digital environment. Production data entered into the system are automatically processed to generate yield percentages, extraction efficiency values, and production summaries. This integration reduces redundant work, minimizes inconsistencies caused by manual computation, and provides production managers with a comprehensive view of operational performance through a single interface.

Third, the proposed study offers a cost-effective and practical digital solution that is suitable for small and medium-sized food manufacturing enterprises. Unlike many advanced production monitoring systems discussed in the literature that require IoT devices, sophisticated sensors, machine learning algorithms, or expensive industrial automation equipment, the proposed system emphasizes centralized data management, automated calculations, and real-time reporting using readily available digital technologies. This approach makes the system more feasible for organizations with limited financial and technical resources while still providing significant improvements in production monitoring and operational efficiency.

Fourth, the proposed system improves the utilization of production data by transforming manually recorded operational information into real-time analytical insights. Rather than simply storing production records for documentation purposes, the system automatically computes key production indicators and presents them through organized reports and dashboards. This allows production managers to monitor production progress continuously, identify abnormal production trends, detect extraction losses, and evaluate operational performance while production activities are ongoing. As a result, corrective actions can be implemented more quickly, reducing delays associated with end-of-day or end-of-batch manual reporting.

Fifth, the study contributes to the digital transformation of food manufacturing operations within the Philippine context. Although previous local studies have demonstrated the importance of digital technologies in agriculture and manufacturing, relatively few have developed practical production monitoring systems specifically intended for food processing companies. By implementing a real-time yield analytics and extraction efficiency monitoring system at AlDi Foods Inc., the study provides a localized example of how digital technologies can improve production management, support operational transparency, and enhance overall manufacturing efficiency. The proposed system may also serve as a reference model for other small and medium-sized food processing enterprises that seek to modernize their production monitoring practices.

Lastly, the proposed system converts raw production data into meaningful and actionable information that supports continuous process improvement. Through automated computation of yield percentages, extraction efficiency, raw material utilization, and production output, the system enables managers to evaluate production performance objectively using measurable indicators. These real-time analytics facilitate evidence-based decision-making, improve resource utilization, reduce production losses, and enhance overall operational efficiency. By providing timely and accurate production information, the system supports continuous monitoring and process optimization, ultimately contributing to improved productivity, better product consistency, and more effective management of juice extraction operations at AlDi Foods Inc.

Overall, the proposed study responds directly to the limitations identified in the reviewed literature by developing an integrated, practical, and real-time production monitoring system specifically tailored to the operational requirements of juice extraction. Through digitalization, automated yield analytics, extraction efficiency computation, and centralized reporting, the system supports the modernization of production management practices while promoting greater operational efficiency, improved resource utilization, and informed decision-making in food processing environments.

A study by Mphahlele, Fawole, Mokwena, and Opara (2016) investigated the influence of different extraction methods on the chemical composition, bioactive compounds, volatile compounds, antioxidant properties, and overall quality of pomegranate (*Punica granatum* L. cv. Wonderful) juice. The researchers emphasized that the quality of fruit juice is not determined solely by the characteristics of the fruit itself but is also greatly affected by the method used during juice extraction. Since consumers increasingly prefer fruit juices that retain their natural nutritional value and health-promoting compounds, identifying extraction methods that maximize juice yield while preserving important chemical and nutritional properties has become an important area of research. The study was conducted to determine how different extraction techniques influence the final composition and quality of pomegranate juice and to provide scientific evidence that could assist food processors in selecting the most appropriate extraction method.

To achieve this objective, the researchers collected commercially harvested pomegranate fruits of the cultivar 'Wonderful' from South Africa and processed them using four different extraction methods. These methods included extracting juice from the arils without crushing the seeds, blending the arils together with the seeds, pressing the whole fruit, and extracting juice from halved fruits using a commercial hand press. The resulting juice samples were analyzed for several quality parameters, including total soluble solids, titratable acidity, pH, juice yield, sugar composition, organic acid content, phenolic compounds, flavonoids, anthocyanins, antioxidant activity, and volatile organic compounds responsible for aroma. Statistical analyses, including analysis of variance and principal component analysis, were also performed to determine whether significant differences existed among the extraction methods and to identify which technique produced the most desirable juice characteristics.

The results demonstrated that the extraction method significantly influenced many of the physicochemical properties of pomegranate juice. Although the total soluble solids remained relatively similar across all extraction methods, noticeable differences were observed in acidity, pH, juice yield, and color. Juice extracted from halved fruits exhibited higher titratable acidity and lower pH, indicating a more acidic product. The researchers explained that lower pH levels may contribute to better microbial stability because acidic environments inhibit the growth of many spoilage microorganisms. Differences were also observed in juice yield, with extraction methods producing varying amounts of juice depending on how effectively the fruit tissues were disrupted during processing. Juice extracted from halved fruits produced the highest juice yield, while pressing whole fruits resulted in the lowest yield because some juice remained trapped inside the fruit tissues during extraction. These findings demonstrated that extraction efficiency is directly affected by the extraction technique employed during processing.

The researchers also examined the sugar and organic acid composition of the juice produced by each extraction method. Glucose and fructose were identified as the major sugars present in all samples, while citric acid was found to be the predominant organic acid. Although sugar concentrations showed only minor differences among the extraction methods, variations in organic acid content were more pronounced. Juice obtained from the arils contained relatively higher citric acid levels, whereas blending the arils with the seeds produced lower concentrations because the seed components diluted the juice composition. The study emphasized that organic acids play an important role in determining the taste, flavor, and stability of fruit juice, making their accurate evaluation essential for maintaining product quality.

In addition to evaluating basic chemical properties, the researchers investigated the effects of extraction methods on bioactive compounds that contribute to the nutritional value of pomegranate juice. Several flavonoids, including catechin, epicatechin, and rutin, together with gallic acid, were identified in all juice samples. Catechin was the most abundant flavonoid regardless of the extraction method, while gallic acid was the predominant phenolic acid. However, the concentration of these compounds varied depending on the extraction technique. Blending the arils together with the seeds produced higher catechin concentrations because phenolic compounds were also extracted from the seeds. On the other hand, juice extracted from halved fruits contained higher concentrations of several beneficial phenolic compounds because the pressing process allowed compounds from the peel and internal membranes to become incorporated into the juice. These findings demonstrate that the fruit fractions included during extraction significantly influence the nutritional composition of the final product.

The study further analyzed anthocyanins, which are natural pigments responsible for the characteristic red color of pomegranate juice. Eight individual anthocyanin compounds were identified, with cyanidin-3,5-diglucoside representing the dominant pigment in all juice samples. Although the same anthocyanin compounds were detected across all extraction methods, their concentrations differed depending on the extraction technique. Juice extracted from whole fruits exhibited relatively higher total anthocyanin content than several of the other extraction methods.

According to the researchers, extraction conditions such as acidity and pressing pressure may influence anthocyanin stability and concentration, ultimately affecting the visual appearance and consumer acceptability of the juice.

The antioxidant properties of the juice were also evaluated using radical scavenging activity and ferric reducing antioxidant power assays. The findings revealed that juice extracted from halved fruits consistently exhibited the highest antioxidant activity among all extraction methods. This increase was closely associated with higher concentrations of total phenolic compounds, tannins, and vitamin C present in the juice. Statistical analysis showed a positive relationship between phenolic content and antioxidant activity, indicating that phenolic compounds are major contributors to the antioxidant capacity of pomegranate juice. These results suggest that extraction methods capable of incorporating beneficial compounds from additional fruit tissues may improve the nutritional value and functional properties of the final product.

Furthermore, the researchers investigated the volatile organic compounds responsible for the aroma and flavor of pomegranate juice. Ten volatile compounds belonging to different chemical groups, including esters, ketones, alcohols, and terpenes, were identified. Ethyl acetate and 3-octanone were the dominant volatile compounds detected in all juice samples. However, their relative concentrations differed according to the extraction method, demonstrating that processing conditions influence aroma development as well as chemical composition. Juice produced by blending arils together with seeds contained the highest overall concentration of volatile compounds, suggesting that including additional fruit tissues during extraction may enhance aroma intensity. Since aroma is one of the primary quality attributes evaluated by consumers, these findings further highlight the importance of selecting appropriate extraction techniques during juice processing.

To better understand the relationships among the measured quality parameters, the researchers performed principal component analysis. The statistical analysis successfully distinguished the juice samples according to their extraction methods based on differences in chemical composition, antioxidant activity, bioactive compounds, sugars, organic acids, and volatile compounds. This finding confirmed that each extraction technique produced a unique quality profile. The researchers concluded that extraction methods influence multiple quality characteristics simultaneously rather than affecting only a single property, emphasizing the importance of considering overall product quality when selecting processing techniques.

Overall, the study concluded that extraction method is a critical factor influencing juice yield, chemical composition, nutritional value, antioxidant capacity, aroma, and overall product quality. The findings demonstrated that selecting an appropriate extraction technique can improve production efficiency while preserving valuable bioactive compounds that contribute to the health benefits and sensory characteristics of pomegranate juice. The study also showed that monitoring extraction conditions allows food manufacturers to optimize both product quality and processing performance.

This literature is relevant to the present study because it demonstrates that extraction methods directly affect production yield, extraction efficiency, and product quality. The findings support the importance of continuously monitoring production parameters to identify variations that influence output and resource utilization. Likewise, the proposed system, *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.*, aims to provide real-time monitoring and analysis of production yield and extraction efficiency, enabling manufacturers to evaluate processing performance, minimize production losses, improve operational efficiency, and make data-driven decisions that enhance overall productivity and product quality.

A study by **Aviara, Lawal, Nyam, and Bamisaye (2020)** focused on the design, construction, and performance evaluation of a multi-fruit juice extractor intended for small- and medium-scale fruit processing operations. The researchers developed the machine to overcome the limitations of traditional juice extraction methods, which are often labor-intensive, time-consuming, unhygienic, and inefficient. The extractor was designed to operate using the combined principles of compression and shear force through a screw conveyor mechanism that gradually squeezes juice from fruits as they move through the extraction chamber. The machine was constructed using locally available materials to ensure affordability, ease of maintenance, and suitability for local fruit juice processors.

To evaluate the performance of the machine, the researchers conducted experimental tests using pineapple, orange, and watermelon in both peeled and unpeeled conditions. The study measured three primary performance indicators: percentage juice yield, extraction efficiency, and extraction loss. Each fruit sample underwent repeated testing to ensure the reliability of the results, and the collected data were analyzed using Analysis of Variance (ANOVA) to determine the effects of fruit type and peel condition on machine performance. The researchers found that both the type of fruit and whether it was peeled significantly influenced the juice extraction results, while variations in feed rate had little effect on overall machine performance.

The findings revealed that peeled fruits generally produced higher juice yields and extraction efficiencies than unpeeled fruits because the removal of the peel reduced resistance during the extraction process. Watermelon recorded the highest juice yield at approximately 89.5% for peeled samples, while peeled pineapple achieved the highest extraction efficiency of about 96.9%. The study also found relatively low extraction losses across all tested fruits, indicating that the machine effectively recovered a large proportion of the available juice. These results demonstrated that the extractor could consistently produce high-quality output while minimizing waste during juice processing.

Furthermore, the researchers concluded that the machine was simple to operate, economically feasible, and capable of supporting small-scale fruit juice production using locally sourced materials. Its efficient design provides an alternative to conventional manual extraction methods by improving productivity, reducing processing time, and increasing juice recovery. The study emphasized that the machine can contribute to reducing post-harvest losses of tropical fruits while promoting value-added processing within local communities.

This literature is relevant to the present study because it highlights the importance of evaluating production performance through measurable indicators such as juice yield, extraction efficiency, and extraction loss. These performance metrics demonstrate how production processes can be monitored to identify inefficiencies and improve operational outcomes. Likewise, the proposed IoT-based production management system aims to digitally record production data, monitor raw material utilization and packed yield, and generate real-time production reports. By providing accurate and organized production information, the system can assist manufacturers in analyzing production efficiency, minimizing losses, and supporting data-driven decision-making throughout the juice production process.

A study by **Dora et al. (2020)** examined the role of sustainable operations in minimizing food loss within the Belgian food processing industry. The researchers recognized that although food waste at the consumer level has been extensively studied, there is limited research focusing on food losses that occur during industrial food processing. To address this gap, they investigated food loss across different production stages in food processing companies to determine where losses most frequently occur and what operational factors contribute to them. The study aimed to identify food loss hotspots and evaluate how sustainable operational practices can help reduce waste and improve production efficiency.

To conduct the research, the authors employed a multiple case study approach involving 47 food processing companies from various sectors in Belgium, including fruit and vegetable processing, dairy, beverages, bakery, chocolate, grain milling, sauces, ready meals, and meat processing. Primary data were collected through a semi-structured questionnaire, company interviews, on-site observations, and documentary analysis. The questionnaire gathered information about company operations, production capacity, food loss at different stages of processing, and the perceived causes of these losses. The collected quantitative data were analyzed using Microsoft Excel and SPSS statistical software, while qualitative observations were used to support and validate the findings.

The findings revealed that food loss accounted for an average of 2.1% of total production among the participating companies. Among the three production stages examined—before production, during production, and after production—the production stage, particularly the processing phase, was identified as the most significant hotspot for food loss. Statistical analysis showed that increases in operational problems during processing were directly associated with higher food loss. The researchers identified several major causes of food loss, including human errors, interrupted production, transportation problems, machine inefficiencies, product changes during production, product defects, packaging errors, poor inventory management, and buyer contract issues. Companies involved in fruit and vegetable processing experienced the highest percentage of food loss, while chocolate and ready-meal manufacturers incurred the greatest financial losses. The study also found that smaller food processing companies generally experienced higher levels of food loss than larger companies.

Based on these findings, the researchers emphasized that sustainable operational practices play a critical role in reducing food loss throughout food processing operations. They recommended improving production planning, inventory management, equipment maintenance, employee training, and quality control procedures to minimize operational inefficiencies. The study also highlighted the importance of implementing management approaches such as Lean Manufacturing, Lean Six Sigma, effective scheduling, and the use of Key Performance Indicators (KPIs) to continuously monitor and reduce food loss. Furthermore, the researchers stressed that systematic measurement of food loss enables companies to identify problem areas, develop targeted improvement strategies, and enhance overall production efficiency. These findings demonstrate that effective operational management not only reduces food waste but also improves productivity, sustainability, and cost efficiency within the food processing industry.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study employs a Developmental Research Design as the primary research methodology to guide the planning, design, development, implementation, testing, and evaluation of the system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* Developmental research is widely used in information technology and software engineering because it focuses on designing and developing innovative technological solutions that address existing organizational problems while ensuring that the developed product meets user requirements and operational objectives.

The primary purpose of this research is to develop a digital production monitoring platform that replaces the company's existing manual, paper-based production recording and monitoring process. The existing production monitoring procedure requires production personnel to manually record raw material usage, extraction output, packed yield, and production data using logbooks and spreadsheets. This traditional method increases the possibility of recording errors, delays report generation, produces inconsistent data, and makes it difficult to track production performance over time. These challenges hinder management from obtaining accurate and timely information needed for effective operational decision-making.

To address these issues, the proposed system automates the recording, monitoring, computation, and reporting of production data through a centralized digital platform. The system enables production personnel to record production information electronically while automatically calculating yield percentages, extraction efficiency, production losses, and production stability indicators. By digitizing these processes, the organization reduces manual computation, improves data accuracy, and enhances the overall efficiency of production monitoring.

Developmental research follows a systematic process that begins with identifying the existing problems within the organization. To accomplish this, the researchers conduct data-gathering activities such as interviews with production supervisors and management personnel, direct observation of the existing production workflow, and a review of existing production records and company documents. These activities provide a comprehensive understanding of the organization's operational processes, existing challenges, and information requirements. The collected information serves as the foundation for identifying system requirements and designing appropriate technological solutions.

Following the requirements-gathering phase, the researchers proceed with system analysis and design. During this stage, the functional and non-functional requirements of the proposed system are identified and documented. Process models, database structures, system architecture, user interface layouts, and workflow diagrams are prepared to ensure that every component of the system aligns with the operational processes of AlDi Foods Inc. The design phase also aims to create a user-friendly interface that enables production personnel and management to efficiently perform their respective tasks with minimal technical difficulty.

The development phase involves the actual construction of the proposed production monitoring system based on the approved system specifications. The researchers use various software development tools, programming languages, database management systems, and web technologies to implement the required functionalities. The core modules of the system include user management, production data entry, raw material monitoring, extraction monitoring, packed yield recording, automated computation of yield percentages, extraction efficiency analysis, production stability monitoring, dashboard visualization, and report generation.

A significant characteristic of developmental research is its iterative nature. Throughout the software development process, the researchers continuously conduct testing, validation, debugging, and refinement of individual system components. Feedback from the company representatives and intended users is incorporated during each stage of development to ensure that the proposed system effectively addresses operational requirements. This iterative approach allows improvements to be implemented before the final deployment of the system, thereby increasing its usability, reliability, and overall quality.

The proposed system specifically monitors critical production variables that directly influence manufacturing efficiency. These include the quantity of raw materials used during production, extracted product output, packed yield records, computed yield percentages, extraction efficiency indicators, production losses, and production stability status. The system automatically performs mathematical computations based on production inputs, eliminating the need for manual calculations and minimizing computational errors.

In addition to real-time production monitoring, the proposed system generates comprehensive analytical reports that summarize production performance over different reporting periods. Reports include daily, weekly, monthly, quarterly, and yearly summaries of raw material utilization, extraction performance, production yield trends, efficiency measurements, and production stability analyses. These reports enable management to evaluate operational performance, identify production inefficiencies, monitor resource utilization, and support evidence-based decision-making for continuous process improvement.

The proposed platform also features a centralized database that securely stores historical production records. This capability allows authorized personnel to retrieve previous production information, compare production performance across different periods, identify long-term operational trends, and generate historical analytics that are previously difficult to obtain using manual records. Data security measures, user authentication, and controlled system access are likewise incorporated to ensure the confidentiality and integrity of production information.

Furthermore, the effectiveness of the developed system is evaluated through system testing and user acceptance evaluation. Functional testing verifies whether every module performs according to the specified requirements, while usability evaluation determines whether the intended users can efficiently operate the system. User feedback regarding functionality, usability, reliability, accuracy, efficiency, and overall satisfaction is collected and analyzed to assess the quality and acceptability of the proposed solution.

Through the application of the Developmental Research Design, this study produces a functional, reliable, and user-centered digital production monitoring platform that addresses the operational needs of AlDi Foods Inc. The developed system improves production monitoring efficiency, reduces manual recording errors, automates production analytics, provides accurate real-time information, enhances reporting capabilities, optimizes raw material utilization, supports managerial decision-making, and contributes to the company's continuous improvement initiatives and digital transformation efforts.

3.2 System Development Methodology

The study utilizes the Agile Development Model as the primary software development methodology in designing, developing, testing, and implementing the proposed system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* Agile is an iterative and flexible software development methodology that emphasizes collaboration, continuous improvement, customer involvement, rapid delivery of working software, and adaptability to changing requirements. Compared to traditional software development approaches, Agile allows developers to respond quickly to feedback and make incremental improvements throughout the development process, resulting in a more reliable and user-centered information system.

The selection of the Agile Development Model is appropriate for this study because the proposed production monitoring system consists of several interconnected modules that can be developed and evaluated incrementally. Since production requirements evolve during system development, Agile provides the flexibility needed to modify features, improve functionality, and accommodate user suggestions without significantly affecting the overall project timeline. Continuous communication with the management and production personnel of AlDi Foods Inc. ensures that the developed system remains aligned with the organization's operational requirements and production monitoring objectives.

Unlike the traditional Waterfall Model, which follows a strictly sequential process, Agile divides the project into smaller and manageable development cycles called sprints. Each sprint focuses on a specific set of system functionalities that are planned, designed, developed, tested, evaluated, and refined before moving to the succeeding iteration. At the end of every sprint, a functional component of the system is produced, allowing stakeholders to review its performance and recommend improvements. This iterative approach minimizes development risks, facilitates early detection of errors, and promotes continuous enhancement of the proposed system.

The Agile Development Model adopted in this study consists of the following phases:

Requirement Analysis

The first phase of the Agile methodology is the Requirement Analysis phase. This stage focuses on understanding the existing production monitoring process practiced by AlDi Foods Inc. The researchers gather comprehensive information regarding the company's operational workflow through various data collection techniques, including interviews with company management, production supervisors, and production personnel. Direct observation of daily production activities is also conducted to understand how production data is recorded, monitored, and reported. In addition, existing production forms, manual logbooks, spreadsheets, and production reports are reviewed to identify the strengths and limitations of the existing system.

The primary objective of this phase is to determine both the functional and non-functional requirements of the proposed digital production monitoring platform. Functional requirements include identifying necessary features such as production data entry, automated yield computation, extraction efficiency monitoring, production stability interpretation, dashboard visualization, and report generation. Non-functional requirements include system usability, security, reliability, performance, accessibility, and maintainability.

Special emphasis is placed on identifying operational challenges associated with the manual process, including delayed report preparation, repetitive manual computations, inconsistent data recording, difficulty in retrieving historical records, and limited visibility of real-time production performance. The information gathered during this phase serves as the foundation for designing a system that effectively addresses these operational issues while supporting the company's digital transformation initiatives.

System Design

After completing the requirements analysis, the researchers proceed to the System Design phase. This phase involves translating the identified requirements into detailed technical specifications that serve as the blueprint for developing the proposed system.

The system architecture is designed to support centralized production data management, secure user authentication, automated processing, and real-time analytics. Database design activities involve creating relational database structures capable of storing and managing production information efficiently. Database tables include user accounts, production batches, raw material records, extraction records, packed yield data, computed yield percentages, extraction efficiency results, production stability classifications, historical production records, and generated reports.

The user interface is carefully designed to provide an intuitive, responsive, and user-friendly experience for production personnel and management. Dashboard layouts are organized to display production information clearly through tables, summary cards, charts, and graphical indicators, enabling users to monitor production performance efficiently.

System process flow diagrams, use case diagrams, activity diagrams, and navigation structures are also prepared during this phase to define how users interact with the system and how information flows between various modules. Automated workflows are incorporated to support real-time monitoring, automatic computations, report generation, historical data retrieval, and dashboard visualization.

The overall design prioritizes simplicity, accuracy, efficiency, scalability, and ease of maintenance to ensure that the proposed platform remains effective for future operational requirements.

Development

During the Development phase, the researchers construct the proposed production monitoring system based on the approved system design. The development process follows Agile principles by implementing system modules incrementally, allowing each component to be tested and refined before it is integrated into the complete system.

The web-based application is developed using HTML and CSS for the front-end user interface to provide an accessible and responsive design. PHP is utilized as the server-side programming language responsible for implementing the application's business logic, processing production data, managing user authentication, and performing automated computations. MySQL serves as the centralized relational database management system that securely stores all production-related information. Chart.js is integrated into the system to generate interactive charts, graphs, and dashboard visualizations that enable users to analyze production trends and performance effectively.

The development activities focus on implementing the following major modules:

- User Authentication and Access Management Module
- Production Management Module

- Raw Material Recording Module
- Extraction Monitoring Module
- Packed Yield Monitoring Module
- Real-Time Yield Analytics Module
- Extraction Efficiency Monitoring Module
- Production Stability Interpretation Module
- Historical Data Management Module
- Dashboard and Data Visualization Module
- Automated Report Generation Module

The User Authentication Module ensures that only authorized personnel can access the system based on their assigned user roles and permissions.

The Production Management Module manages production batches and organizes production records according to production schedules.

The Raw Material Recording Module enables production personnel to digitally record the quantity and type of raw materials utilized during production.

The Extraction Monitoring Module records extracted product quantities while automatically linking extraction outputs to their corresponding production batches.

The Packed Yield Monitoring Module monitors the quantity of finished products obtained after packaging and serves as the basis for yield computation.

The Real-Time Yield Analytics Module automatically calculates production yield percentages immediately after production data is entered, eliminating manual computation and providing management with up-to-date production performance indicators.

The Extraction Efficiency Monitoring Module computes extraction efficiency by comparing raw material inputs with extraction outputs. The generated efficiency values allow management to evaluate how effectively raw materials are converted into usable products.

The Production Stability Interpretation Module automatically interprets production performance by comparing calculated efficiency values against predefined organizational thresholds. Based on these standards, production batches are classified as either Stable or Not Stable, enabling management to quickly identify production inconsistencies and operational concerns.

The Historical Data Management Module securely archives production records, allowing authorized users to retrieve historical information for trend analysis, comparison of production periods, and long-term operational evaluation.

The Dashboard and Data Visualization Module presents production information through interactive charts, summary indicators, graphs, and analytical dashboards that provide management with a comprehensive overview of production activities.

Finally, the Automated Report Generation Module generates production reports categorized into daily, weekly, monthly, quarterly, and yearly summaries. These reports include raw material utilization, extraction output, packed yield, yield percentages, extraction efficiency, production stability results, and historical production performance. The generated reports support production planning, resource allocation, operational monitoring, and managerial decision-making.

Testing

Following system development, comprehensive testing procedures are conducted to ensure that all modules perform according to the specified requirements. Testing is an essential component of the Agile methodology because it allows developers to identify defects early and continuously improve system quality throughout development.

The testing process includes several levels of evaluation.

Unit Testing verifies the functionality of individual modules independently to ensure that each feature performs accurately according to its intended purpose.

Integration Testing evaluates the interaction between interconnected modules to confirm that information is transferred correctly throughout the system.

System Testing assesses the complete application as a fully integrated system. This phase ensures that all functionalities operate together without conflicts and that the overall performance meets the project's objectives.

Finally, User Acceptance Testing (UAT) is conducted with selected personnel from AlDi Foods Inc. End users evaluate the system's functionality, usability, accuracy, efficiency, reliability, and overall effectiveness in supporting production monitoring activities. Their feedback is documented and incorporated into subsequent refinements of the system.

Particular attention is devoted to validating the accuracy of automated yield computations, extraction efficiency calculations, production stability interpretations, dashboard analytics, historical record retrieval, and automated report generation. This ensures that the system produces reliable information for operational and managerial decision-making.

Deployment (Prototype)

Upon successful completion of system testing and refinement, the proposed production monitoring platform is deployed as a prototype within a controlled environment at AlDi Foods Inc. Prototype deployment allows the organization to evaluate the functionality and practicality of the developed system under actual operational conditions without immediately replacing the existing production monitoring procedures.

During this phase, production personnel and management use the prototype to perform actual production recording, monitor real-time production activities, generate analytical dashboards, calculate yield percentages, evaluate extraction efficiency, classify production stability, and produce automated reports. This practical implementation enables stakeholders to assess how effectively the proposed system addresses the operational limitations of the manual process.

The deployment phase also provides an opportunity to collect user feedback regarding system usability, functionality, interface design, report accuracy, processing speed, and overall user experience. Suggestions gathered from stakeholders are carefully analyzed and used to identify additional improvements, optimize existing features, and refine system performance before any future full-scale implementation.

By following the Agile Development Model throughout the project lifecycle, the researchers aim to produce a reliable, efficient, and user-centered production monitoring platform that supports the operational objectives of AlDi Foods Inc. The iterative nature of Agile ensures continuous system enhancement while enabling the organization to benefit from improved production monitoring, accurate real-time analytics, efficient resource utilization, and informed managerial decision-making.

3.3 System Architecture

The proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.* adopts a three-tier system architecture consisting of the Client Layer, Application Layer, and Database Layer. This architectural design provides a structured framework that separates the user interface, application processing, and data management components of the system. By separating these layers, the proposed platform becomes more organized, maintainable, scalable, and secure while ensuring efficient communication among system components.

The three-tier architecture allows each layer to perform specialized responsibilities independently while continuously exchanging information with the other layers. This separation improves system reliability, simplifies maintenance, facilitates future enhancements, and enables efficient processing of production information. It also minimizes system complexity by organizing operational tasks according to their specific functions.

Within the proposed production monitoring platform, users interact only with the Client Layer. The Client Layer communicates with the Application Layer, which processes requests, performs computations, and retrieves or stores data within the Database Layer. Once processing is completed, the Application Layer returns the processed information to the Client Layer, allowing users to immediately view updated production analytics, reports, and dashboard visualizations.

Client Layer

The Client Layer serves as the presentation layer and acts as the primary interface between users and the proposed production monitoring system. It provides an accessible web-based environment where authorized users interact with the application using computers or other compatible devices connected to the organization's local network or internet connection.

The primary users of the system include production personnel, supervisors, and system administrators. Each authorized user receives secure login credentials that determine the functions accessible according to their assigned role. Through this interface, production personnel perform daily production recording activities efficiently, while supervisors and management personnel monitor operational performance using analytical dashboards and generated reports.

The Client Layer enables users to digitally record all production-related information required throughout the manufacturing process. These records include production batch information, quantities of raw materials utilized, extraction output measurements, packed yield records, production dates, operator information, and other relevant production details. Replacing manual recording with electronic data entry minimizes clerical errors, eliminates redundant paperwork, and ensures that production records are stored in a consistent format.

After production information is submitted, users receive updated production results through an interactive dashboard. The dashboard presents key performance indicators using summary cards, tables, graphical charts, and visual analytics. Users monitor production activities, compare production batches, and evaluate operational performance without manually computing production statistics.

The Client Layer also provides real-time visualization of important production indicators such as:

- Raw material consumption
- Extraction output
- Packed yield
- Yield percentage
- Extraction efficiency
- Production stability status
- Historical production trends
- Periodic production summaries

One of the major advantages of the dashboard is its ability to automatically display the operational status of each production batch. Once production information is processed, the dashboard immediately indicates whether production performance is classified as Stable or Not Stable, allowing production personnel and supervisors to identify operational concerns without waiting for manual report preparation.

To improve usability, the user interface is designed following principles of simplicity, consistency, responsiveness, and accessibility. Navigation menus are organized logically to minimize user confusion, while input forms are structured to reduce encoding errors and improve efficiency. Dashboard components are arranged to provide clear visualization of production performance while requiring minimal technical expertise from users.

By providing immediate access to production information, analytical dashboards, and automated reports, the Client Layer significantly improves operational visibility and enables management to make faster and more informed production decisions.

Application Layer

The Application Layer serves as the central processing component and core engine of the proposed production monitoring platform. It functions as the intermediary between the Client Layer and the Database Layer by receiving production information from users, processing the submitted data according to predefined business rules, performing automated analytical computations, and returning meaningful information to the user interface.

All major operational logic of the proposed system is implemented within this layer. Every production transaction initiated by users undergoes validation to ensure that required information is complete, accurate, and logically consistent before being stored in the database. Validation procedures help minimize data entry errors and improve the overall quality of production records.

One of the primary responsibilities of the Application Layer is performing automated production calculations. Instead of requiring production personnel to manually compute operational statistics, the system automatically calculates important production indicators immediately after production data is entered.

These automated computations include:

- **Yield percentage calculation**
- **Extraction efficiency calculation**

- **Packed yield computation**
- **Raw material utilization analysis**
- **Production loss monitoring**
- **Historical production comparison**

The automated computation process significantly reduces computational errors, eliminates repetitive manual calculations, and provides management with accurate production information in real time.

Another important function of the Application Layer is the implementation of the Production Stability Interpretation Module. This module continuously evaluates production performance using predefined organizational standards and operational thresholds established by AlDi Foods Inc. The system compares computed yield percentages and extraction efficiency values against these predefined criteria to determine whether production performance meets acceptable operational standards.

Based on the computed analytical results, the system automatically classifies each production batch into one of two operational categories:

- **Stable**
- **Not Stable**

This automated interpretation assists management in quickly identifying production inconsistencies, abnormal production behavior, excessive production losses, and potential operational issues that require immediate corrective action. Instead of manually reviewing numerous production records, supervisors determine production status immediately through dashboard indicators generated by the system. The Application Layer also manages the Real-Time Yield Analytics Module, which continuously updates production performance indicators whenever new production information is recorded. This enables management to monitor production progress as manufacturing activities occur rather than waiting until the completion of production reports.

In addition, the Application Layer supports the Extraction Efficiency Monitoring Module, which evaluates how efficiently raw materials are converted into extracted products. By continuously monitoring extraction efficiency, management identifies production batches with unusually low extraction performance and investigates possible operational causes such as equipment performance, processing methods, or raw material quality. The system also incorporates an Automated Report Generation Module, which prepares production reports without requiring manual encoding or spreadsheet calculations. Using production information stored in the database, the Application Layer generates comprehensive reports that summarize operational performance according to various reporting periods.

Generated reports include:

- Daily production summaries
- Weekly raw material utilization reports
- Monthly production performance reports
- Quarterly extraction efficiency reports
- Yearly production analytics
- Historical production trend reports

These reports enable management to monitor operational efficiency, evaluate resource utilization, identify production trends, and support long-term strategic planning.

The Application Layer further manages user authentication, role-based access control, session management, historical data retrieval, dashboard visualization, and communication between the user interface and the database. Centralizing these responsibilities ensures consistency, improves processing efficiency, enhances system reliability, and maintains data integrity throughout the entire application.

Database Layer

The Database Layer functions as the centralized repository of all information generated and processed by the proposed production monitoring platform. This layer is responsible for securely storing, organizing, maintaining, retrieving, and managing production records throughout the system's operation. The proposed system utilizes a relational database to ensure that production information is properly structured and easily accessible. By maintaining a centralized database, the organization eliminates dependence on manual record books, spreadsheets, and scattered paper documents, thereby improving data consistency, security, and accessibility.

The database stores multiple categories of operational information, including:

- User roles and permissions
- Production batch records
- Raw material input records
- Extraction output records
- Packed yield information

- Yield percentage calculations
- Extraction efficiency results
- Production stability classifications
- Historical production records
- Dashboard summary information
- Generated production reports
- System activity logs

Every production transaction submitted by authorized users is securely stored within the database and becomes immediately available for future retrieval and analysis. Historical records remain preserved, enabling management to compare production performance across different operational periods and identify long-term production trends.

The centralized database significantly improves information management by eliminating problems commonly associated with paper-based record keeping, including misplaced documents, duplicate records, inconsistent information, delayed retrieval, and accidental data loss. To maintain data security and integrity, access to stored information is controlled through authenticated user accounts and predefined user permissions. Only authorized personnel are permitted to view, edit, or manage specific production records according to their assigned responsibilities.

Furthermore, the Database Layer serves as the primary source of information used by the reporting and analytics modules. Every dashboard visualization, graphical chart, analytical report, production summary, yield computation, extraction efficiency analysis, and production stability interpretation is generated using structured data retrieved from the centralized database.

Because all production information is stored within a single integrated repository, the organization can efficiently retrieve historical production records, monitor operational performance, evaluate production efficiency, and generate comprehensive management reports whenever needed.

System Data Flow

The overall operation of the proposed production monitoring platform follows a continuous flow of information among the three architectural layers. The process begins when authorized production personnel log into the system through the Client Layer using valid authentication credentials. Once access is granted, the user records production information such as production batch details, raw material quantities, extraction outputs, packed yield measurements, and other required production data.

The entered information is transmitted to the Application Layer, where the system first validates the submitted data to ensure completeness and accuracy. After successful validation, the Application Layer automatically performs production-related computations, including yield percentage calculation, extraction efficiency analysis, raw material utilization computation, and production stability interpretation. Once processing is completed, all production information and computed analytical results are securely stored within the Database Layer. The database immediately updates historical production records, ensuring that newly entered information becomes available for future retrieval, reporting, and trend analysis. Whenever users access the dashboard or request analytical reports, the Application Layer retrieves the required production information from the Database Layer, processes the requested data, generates graphical visualizations and analytical summaries, and returns the processed results to the Client Layer.

The Client Layer then displays updated dashboards containing real-time production statistics, extraction efficiency indicators, yield percentages, production stability classifications, historical production records, and periodic raw material utilization reports. Because the information is processed automatically, users receive immediate feedback regarding production performance without performing manual calculations. This continuous exchange of information among the Client Layer, Application Layer, and Database Layer enables the proposed system to provide accurate real-time monitoring, centralized information management, automated analytical processing, efficient report generation, and timely decision support. Through this architecture, AlDi Foods Inc. improves operational efficiency, reduces manual recording errors, enhances production visibility, and supports data-driven decision-making across its manufacturing operations.

3.4 Tools and Technologies Used

The development of the proposed system, *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.,"* requires the integration of various software technologies and development tools that work together to create an efficient, reliable, and user-friendly web-based production monitoring platform. The selection of these technologies plays a significant role in ensuring that the system accurately records production data, performs automated analytical computations, generates real-time dashboards, and produces comprehensive production reports that support the operational needs of the organization.

The proposed system is designed to replace the existing manual production monitoring process with a centralized digital platform. To achieve this objective, the researchers select technologies that are widely recognized for their stability, compatibility, flexibility, and ease of implementation. These technologies enable the system to automate production monitoring activities, reduce manual computations, improve data accuracy, and provide management with timely information for decision-making.

Each technology is carefully evaluated based on several criteria, including compatibility with other development tools, processing efficiency, security, maintainability, accessibility, scalability, and suitability for web-based information systems. Since the proposed platform consists of multiple interconnected modules—including production recording, yield analytics, extraction efficiency monitoring, production stability interpretation, dashboard visualization, and report generation—the selected technologies need to work seamlessly together to ensure smooth system operation.

Another important consideration in selecting these technologies is their ability to support centralized information management. Rather than storing production records in paper logbooks or disconnected spreadsheet files, the proposed system consolidates all production information into a centralized database that can be accessed by authorized personnel. This approach improves information consistency, minimizes duplication of records, enhances data security, and enables faster retrieval of historical production information whenever required.

The selected development tools also support the implementation of real-time monitoring features. As production personnel record operational data into the system, the application immediately processes the information, performs the required analytical computations, updates production dashboards, and stores records in the centralized database. This automation significantly reduces delays associated with manual report preparation while allowing management to monitor production performance more effectively.

Furthermore, the chosen technologies are widely used in both academic and professional software development environments. Their extensive documentation, active developer communities, and long-term support make them practical choices for developing an information system that can be maintained, improved, and expanded in the future as organizational requirements evolve.

Frontend Technologies

HTML (HyperText Markup Language)

HTML (HyperText Markup Language) serves as the primary markup language used to construct the structural components of the proposed web-based production monitoring system. It provides the basic framework upon which all web pages, forms, navigation menus, tables, dashboards, and other interface components are built. HTML defines how information is organized and displayed, ensuring that users can interact efficiently with the different modules of the application. Within the proposed system, HTML is used to develop user interfaces for production recording, raw material encoding, extraction monitoring, packed yield recording, production dashboards, historical record viewing, report generation, and administrative functions. Production personnel use HTML-based forms to enter production information, while supervisors and management personnel access analytical dashboards and production reports through organized web pages.

The flexibility of HTML allows the proposed application to maintain compatibility across modern web browsers, ensuring that authorized users can access the system without installing specialized software. Because the platform is web-based, users can conveniently perform production monitoring activities through standard computers connected to the organization's network.

HTML also supports integration with CSS, PHP, JavaScript libraries, and database-driven applications, making it an appropriate foundation for developing an interactive production monitoring system that meets the operational requirements of AlDi Foods Inc.

CSS (Cascading Style Sheets)

CSS (Cascading Style Sheets) is utilized to improve the visual appearance, usability, responsiveness, and overall user experience of the proposed system. While HTML defines the structural layout of the application, CSS controls the presentation of system components by specifying colors, fonts, spacing, positioning, menus, buttons, tables, dashboards, and graphical layouts.

The implementation of CSS enables the researchers to design a clean, organized, and professional user interface that minimizes user confusion and improves operational efficiency. Proper visual organization is particularly important because production personnel frequently interact with production records, dashboard summaries, and analytical reports during daily operations.

CSS is also used to implement responsive design techniques that allow the application to adapt to different screen sizes and display resolutions. Whether the system is accessed using desktop computers or laptops within the organization, the interface remains consistent, readable, and easy to navigate.

In addition, CSS enhances the dashboard by improving the presentation of production statistics, graphical indicators, report summaries, and navigation menus. These improvements enable users to quickly identify important production information and support faster operational decision-making.

Backend Technology

PHP (Hypertext Preprocessor)

PHP (Hypertext Preprocessor) serves as the primary server-side programming language responsible for implementing the core functionality and business logic of the proposed production monitoring platform. PHP processes requests submitted by users, communicates with the database, performs automated computations, validates production information, and generates dynamic web pages.

As the central processing component of the system, PHP performs numerous operational tasks, including:

- User authentication and login verification
- User account and access control management
- Production batch processing
- Raw material recording
- Extraction monitoring
- Packed yield recording
- Yield percentage calculation
- Extraction efficiency computation

- Production stability interpretation
- Historical record retrieval
- Dashboard data processing
- Automated report generation

One of PHP's most important responsibilities within the proposed system is performing automated analytical computations immediately after production data is entered. Rather than requiring production personnel to manually calculate yield percentages and extraction efficiency values, PHP executes these calculations automatically, reducing computational errors while improving processing speed.

PHP also manages communication between the user interface and the MySQL database, ensuring that production records are stored securely and retrieved accurately whenever requested by users. Because PHP integrates efficiently with MySQL, it is well suited for developing database-driven information systems such as the proposed production monitoring platform.

Analytics Processing Technology

Chart.js

Chart.js is utilized as the primary data visualization library for displaying production analytics and operational reports. The library provides interactive graphical representations of production data, enabling users to understand performance trends quickly and effectively.

The system uses Chart.js to visualize:

- Current Yield Percentage

- Extraction Efficiency Performance
- Stable and Not Stable Production Trends
- Raw Material Consumption Analysis
- Production Output Trends
- Weekly Usage Reports
- Monthly Usage Reports
- Quarterly Usage Reports
- Yearly Usage Reports

By converting raw numerical information into visual charts and graphs, Chart.js improves data interpretation and supports data-driven decision-making within AlDi Foods Inc.

Database Technology

MySQL

MySQL serves as the centralized relational database management system for the proposed production monitoring platform. It is responsible for securely storing, organizing, retrieving, updating, and managing all production-related information generated throughout system operation.

The centralized database serves as the foundation of the entire application by ensuring that production records remain accurate, organized, and readily accessible to authorized users. Unlike traditional paper-based recording methods, MySQL enables efficient information management while minimizing duplication, inconsistency, and loss of production data.

The database stores various categories of operational information, including:

- User Accounts
- User Roles and Permissions
- Production Batches
- Raw Material Inputs
- Extraction Outputs
- Packed Yield Records
- Yield Percentage Results
- Extraction Efficiency Results
- Production Stability Status
- Historical Production Records
- Dashboard Summary Data
- Generated Reports

Historical production information stored within MySQL enables management to perform long-term production analysis, compare operational performance across different reporting periods, identify production trends, and generate comprehensive management reports. The database also supports automated report generation and dashboard visualization by providing structured production information that can be processed efficiently by PHP and displayed through Chart.js.

Development Environment

Visual Studio Code

Visual Studio Code (VS Code) is utilized as the primary Integrated Development Environment (IDE) during the development of the proposed production monitoring platform. VS Code provides a lightweight yet powerful environment that supports both frontend and backend development, enabling researchers to efficiently create, edit, test, and maintain the application's source code.

Several built-in and extensible features contribute to improving development productivity, including:

- Syntax highlighting
- Intelligent code completion
- Error detection
- Debugging support
- Extension marketplace
- Integrated terminal
- Version control integration
- File management tools

These capabilities enable the researchers to organize project files efficiently, detect programming errors early, simplify debugging activities, and maintain code quality throughout the development lifecycle.

Local Development Server

XAMPP

XAMPP is used as the local development server environment during system development and testing. It provides the necessary components required for executing PHP applications and managing MySQL databases within a local machine.

XAMPP includes:

- Apache Web Server
- MySQL Database Server
- PHP Interpreter
- phpMyAdmin Database Management Tool

The use of XAMPP allows the researchers to develop, test, and validate system functionalities before deployment.

Technology Integration

The proposed system operates through the integration of these technologies. HTML and CSS provide the user interface through which production personnel interact with the system. PHP processes incoming production data and performs calculations related to yield analytics and extraction efficiency monitoring. MySQL stores production records and historical information, while Chart.js converts processed data into graphical dashboards and analytical reports. Visual Studio Code and XAMPP serve as the development and testing environments that support the creation and validation of the entire application.

Together, these technologies provide the technical foundation necessary to implement a digital production monitoring platform capable of supporting real-time yield analytics, extraction efficiency monitoring, production stability interpretation, and periodic raw material usage reporting for AlDi Foods Inc.

3.5 Data Gathering Techniques

To obtain the necessary information for the design, development, and implementation of the proposed system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.,"* the researchers employ a combination of carefully selected data-gathering techniques. Gathering accurate and comprehensive information is an essential phase of the system development process because it provides the researchers with a clear understanding of the organization's existing production monitoring practices, operational requirements, and the challenges associated with its current manual procedures.

Rather than relying on a single source of information, the study adopts a multi-method approach that combines different data-gathering techniques to obtain information from various operational perspectives. This approach improves the reliability and completeness of the collected data by allowing the researchers to compare information obtained from different sources and verify its consistency. The integration of multiple techniques also enables the researchers to develop a more comprehensive understanding of how production information flows throughout the manufacturing process and how different personnel perform their respective responsibilities.

The primary objective of the data-gathering process is to examine the current production monitoring procedures implemented by AlDi Foods Inc. The researchers analyze how production data is generated, recorded, processed, monitored, and reported from the initial receipt of raw materials up to the completion of the production cycle. Particular attention is given to the recording of raw material inputs, extraction outputs, packed yield records, production computations, and report preparation. Understanding these procedures allows the researchers to identify areas where digital technologies can improve operational efficiency and reduce dependence on manual recording practices.

In addition, the data-gathering activities are intended to identify operational challenges that affect the efficiency of the company's current production monitoring process. These challenges include delayed report preparation, repetitive manual computations, inconsistent production records, difficulty retrieving historical information, data duplication, limited visibility of production performance, and human errors during recording and calculation. Identifying these issues enables the researchers to establish the functional and non-functional requirements of the proposed digital production monitoring platform.

The information collected during this phase serves as the foundation for designing the proposed system's database structure, user interface, system workflows, automated analytical computations, dashboard visualizations, and report generation features. Every major component of the proposed system is developed based on actual operational requirements gathered from the organization to ensure that the platform effectively supports daily production activities and management decision-making.

Furthermore, the data-gathering process assists the researchers in determining the information required for implementing the system's core functionalities, including digital production recording, automated yield percentage calculation, extraction efficiency monitoring, production stability interpretation, centralized historical data management, dashboard visualization, and periodic production reporting. By understanding how information is handled within the organization, the researchers develop a system that accurately reflects existing operational practices while improving efficiency through automation.

The use of multiple data-gathering techniques also strengthens the credibility of the study by allowing information collected through one method to be verified using another. Observations of actual production activities are compared with interview responses, while existing production documents are reviewed to validate operational procedures and reporting practices. This process helps ensure that the proposed system is based on factual operational information rather than assumptions.

Ultimately, the data-gathering phase serves as the foundation of the entire system development process. The findings obtained during this stage guide the succeeding phases of system analysis, design, development, testing, implementation, and evaluation. Through this comprehensive approach, the researchers aim to develop a digital production monitoring platform that effectively addresses the operational needs of AlDi Foods Inc. while improving production efficiency, reporting accuracy, resource utilization, and managerial decision-making.

Interviews

The interview technique is conducted with selected personnel of AlDi Foods Inc., including production supervisors, production personnel, and other authorized representatives involved in the production monitoring process. The purpose of the interviews is to obtain first-hand information regarding the organization's current production workflow, existing monitoring procedures, reporting practices, operational challenges, and expectations for the proposed system.

Semi-structured interviews are used to allow respondents to describe their daily responsibilities, explain the current production recording procedures, identify difficulties encountered in manual monitoring, and provide suggestions for improving the overall production monitoring process. The researchers also gather information regarding the production data currently being recorded, the frequency of report preparation, methods of calculating yield and extraction efficiency, and the information needed by management for operational decision-making.

The responses collected through the interviews provide valuable insights into user requirements and serve as an important basis for identifying the necessary features and functionalities of the proposed production monitoring platform.

Observation

The researchers conduct direct observation of the production monitoring activities within AlDi Foods Inc. to obtain a clear understanding of how production information flows throughout the manufacturing process. During this activity, the researchers observe the procedures used in recording raw material inputs, monitoring extraction outputs, documenting packed yields, calculating production results, and preparing production reports.

Observation enables the researchers to examine actual operational practices, identify workflow inefficiencies, and understand how production personnel interact with existing recording procedures. This technique also allows the researchers to verify whether the information obtained during interviews accurately reflects actual production activities.

Through direct observation, the researchers can identify repetitive manual tasks, unnecessary documentation procedures, delays in information processing, and opportunities where automation can improve production monitoring efficiency. The observations also assist in designing system workflows that closely match the organization's existing operational processes while reducing unnecessary manual activities.

Document Review

The researchers conduct a comprehensive review of the existing documents, records, and production reports utilized by AlDi Foods Inc. as part of the data-gathering process. Document review is an essential technique because it provides objective, factual, and historical information regarding the organization's production monitoring practices. Unlike interviews and observations, which primarily rely on current activities and stakeholder responses, document review enables the researchers to examine actual production records that are used in daily operations. These documents serve as reliable references for understanding how production information is currently collected, organized, processed, and reported within the organization.

The primary objective of the document review is to analyze the existing production monitoring system and identify the information requirements needed for the development of the proposed digital production monitoring platform. By examining the company's operational records, the researchers can determine the types of production data that must be captured, stored, processed, and presented within the new system. This activity also allows the researchers to understand the format, structure, and frequency of existing reports, ensuring that the proposed system supports the organization's current reporting requirements while improving efficiency through automation.

The researchers review various production-related documents currently maintained by AlDi Foods Inc., including but not limited to:

- Production Logs
- Daily Production Reports
- Raw Material Usage Records
- Extraction Output Records
- Packed Yield Reports
- Inventory Records
- Historical Production Data
- Existing Monitoring Forms

Each of these documents contains valuable information regarding different stages of the production process. Production logs and daily production reports provide detailed records of manufacturing activities conducted during each production cycle, while raw material usage records document the quantity of materials consumed during processing. Extraction output records and packed yield reports contain information necessary for monitoring production efficiency, evaluating yield performance, and analyzing extraction results. Inventory records provide additional information regarding the availability and utilization of raw materials, while historical production records enable the researchers to examine production performance over extended periods.

A major focus of the document review is the examination of the company's existing production monitoring forms. These forms serve as the primary source of production information under the current manual monitoring system and illustrate how production personnel record operational data throughout the manufacturing process. By carefully reviewing these forms, the researchers gain a comprehensive understanding of how production information is organized, categorized, and documented. This analysis enables the researchers to identify the sequence of production activities, determine the relationships among various production records, and understand how information flows from raw material preparation through extraction, packaging, and final reporting.

The researchers also analyze the layout, content, and structure of the existing documents to determine the specific data elements required for the proposed system. Particular attention is given to identifying the production variables that need to be recorded during each production cycle, including raw material quantities, extraction outputs, packed yields, production dates, batch information, operator details, and other operational data necessary for monitoring production performance. Identifying these essential data elements ensures that the proposed system captures complete and accurate information required for automated computations and report generation.

Furthermore, document review assists the researchers in identifying the methods used to calculate production performance indicators, particularly yield percentages and extraction efficiency measurements. By examining existing reports and production summaries, the researchers understand how these values are computed, documented, and interpreted by production personnel and management. This information serves as the basis for designing automated computation modules that accurately replicate and improve the organization's existing production monitoring procedures.

Historical production records are also analyzed to identify long-term production trends, recurring operational patterns, and variations in production performance over time. Reviewing historical data enables the researchers to establish baseline production performance, identify seasonal or operational fluctuations, evaluate raw material utilization, and examine extraction efficiency across different production periods. These historical records provide valuable insights that support the design of the system's historical data management, trend analysis, and dashboard visualization features.

In addition, document review enables the researchers to identify weaknesses associated with the existing manual documentation process. The analysis reveals issues such as duplicate data entry, inconsistent record formats, incomplete production information, delayed report preparation, difficulty retrieving historical records, and the possibility of computational errors resulting from manual calculations. Identifying these limitations allows the researchers to design system features that address these operational challenges through centralized data management, automated computations, standardized record formats, and digital report generation.

The findings obtained through the document review play a significant role in the development of the proposed system's database structure. The researchers use the identified production data elements to design database tables, establish relationships among production records, and organize information in a manner that supports efficient storage, retrieval, and analysis. A well-designed database ensures that production information remains accurate, consistent, secure, and readily accessible for operational monitoring and management decision-making.

Moreover, the information gathered through document review supports the development of the system's reporting and analytics modules. Existing production reports serve as references for designing automated weekly, monthly, quarterly, and yearly production summaries. Similarly, the production data identified during document analysis is utilized to develop dashboard visualizations that display yield percentages, extraction efficiency, production stability status, raw material utilization, and historical production trends in an organized and meaningful manner.

The document review also serves as a validation mechanism for information obtained through interviews and direct observation. By comparing documented production records with the responses of production personnel and observations conducted within the facility, the researchers verify the consistency and accuracy of the collected information. This process strengthens the reliability of the study and ensures that the proposed production monitoring system accurately reflects the actual operational practices of AlDi Foods Inc.

Ultimately, the document review provides a solid foundation for the successful design and development of the proposed digital production monitoring platform. The information obtained through this technique guides the identification of system requirements, database design, workflow development, dashboard visualization, automated yield analytics, extraction efficiency monitoring, production stability interpretation, historical data management, and report generation. By thoroughly examining the organization's existing production documents, the researchers ensure that the proposed system is fully aligned with the operational needs of AlDi Foods Inc. and effectively supports the organization's transition from manual production monitoring to a centralized, data-driven digital information system.

Summary of Data Gathering Process

The data gathering process serves as one of the most critical stages in the development of the proposed system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* This phase provides the researchers with a comprehensive understanding of the organization's existing production monitoring procedures, operational workflows, and information management practices. By collecting accurate and reliable information from multiple sources, the researchers establish a strong foundation for designing a system that effectively addresses the operational needs of the company.

To obtain a complete and accurate understanding of the existing production environment, the researchers employ a combination of interviews, direct observation, and document review. These complementary data-gathering techniques enable the researchers to collect information from different perspectives, allowing the findings from one method to be verified through the others. This multi-method approach increases the credibility, reliability, and completeness of the collected information while providing a more detailed picture of the organization's production monitoring process.

The interviews provide firsthand information from production managers, supervisors, operators, and other personnel directly involved in manufacturing operations. Through these discussions, the researchers gain valuable insights into the production monitoring procedures, reporting practices, operational challenges, and user expectations for the proposed digital production monitoring system. The information obtained from these interviews helps identify user requirements and determine the features necessary to improve production efficiency, reporting accuracy, and decision-making.

Direct observation complements the interview findings by allowing the researchers to examine actual production activities as they occur within the facility. Observing the production environment enables the researchers to understand how production information is recorded, processed, and utilized throughout each stage of manufacturing. This technique also assists in identifying workflow inefficiencies, repetitive manual tasks, delays in report preparation, and opportunities where automation can improve operational performance without disrupting existing production processes.

The document review further strengthens the data-gathering process by providing objective and historical information regarding the organization's production activities. Existing production logs, raw material usage records, extraction reports, packed yield reports, inventory records, historical production data, and monitoring forms are examined to determine the information being collected and the reporting formats utilized by the organization. These documents help the researchers identify the specific production data elements required for the proposed system and ensure that the application's database structure and reporting modules accurately reflect the organization's operational requirements.

By integrating the information obtained through interviews, direct observation, and document review, the researchers develop a comprehensive understanding of the complete production monitoring workflow of AlDi Foods Inc. This process enables the researchers to trace the movement of production information from the recording of raw material inputs through extraction, packed yield documentation, production analysis, and final report generation. Understanding this complete flow of information is essential in designing a digital system that supports existing operational practices while improving efficiency through automation and centralized information management.

The collected data also enables the researchers to identify operational challenges associated with the organization's existing manual production monitoring process. These challenges include delayed report preparation, repetitive manual computations, inconsistent record-keeping, duplication of production data, difficulty retrieving historical records, limited real-time production visibility, and the possibility of human error during production recording and calculation. Identifying these operational concerns allows the researchers to establish clear functional and non-functional requirements for the proposed system.

Furthermore, the findings gathered during this phase serve as the basis for designing the system architecture, database structure, user interface, workflow processes, and analytical modules of the proposed application. The information obtained from the production environment guides the development of essential system functionalities, including digital production recording, automated yield percentage computation, extraction efficiency monitoring, production stability interpretation, centralized historical data management, interactive dashboard visualization, and automated generation of weekly, monthly, quarterly, and yearly production reports.

The data gathered also supports the development of the system's automated analytical capabilities. By understanding how production data is collected and analyzed, the researchers design computation processes that accurately calculate yield percentages, evaluate extraction efficiency, monitor raw material utilization, and classify production performance as either *Stable* or *Not Stable* based on the organization's predefined operational standards. These automated processes minimize manual computations while providing management with accurate and timely production information for decision-making.

In addition, the information collected during this phase assists in designing a centralized database capable of securely storing production records, historical information, user accounts, production batches, extraction results, packed yields, and generated reports. A well-structured database improves information consistency, eliminates duplicate records, facilitates rapid retrieval of historical production information, and supports long-term production analysis.

The comprehensive data-gathering process also contributes to the successful implementation of real-time dashboard visualization. The researchers identify the production indicators that management regularly monitors, enabling the proposed system to display relevant production metrics through charts, tables, graphical summaries, and performance indicators. These dashboards provide immediate visibility into production performance, allowing supervisors and managers to monitor yield percentages, extraction efficiency, production stability, and raw material utilization as production activities occur.

Ultimately, the information collected through interviews, direct observation, and document review serves as the primary foundation for the successful development of the proposed digital production monitoring platform. Every stage of the system development process—including requirements analysis, system design, software development, database implementation, testing, deployment, and evaluation—relies on the findings obtained during data gathering. By ensuring that the proposed system is designed based on actual operational requirements rather than assumptions, the researchers develop a reliable, efficient, and user-centered application that supports the digital transformation initiatives of AlDi Foods Inc.

Through this comprehensive and systematic data-gathering process, the proposed system is capable of replacing manual production monitoring procedures with a centralized digital platform that improves data accuracy, enhances real-time operational visibility, automates yield analytics and extraction efficiency monitoring, simplifies report generation, and provides management with reliable information for timely and informed decision-making. The successful completion of this phase ensures that the proposed system is closely aligned with the operational objectives and production monitoring requirements of AlDi Foods Inc., thereby contributing to improved productivity, resource utilization, and overall manufacturing performance.

3.6 Testing Procedures

The proposed system undergoes a series of structured testing procedures before its official deployment at AlDi Foods Inc. This rigorous evaluation phase is designed to ensure that all developed functionalities operate correctly, efficiently, and reliably under real-world conditions. By systematically walking through each feature in a controlled environment, the project team ensures that the platform is robust, stable, and safe to integrate into daily operations.

These thorough testing procedures are essential to validate the system's capability to handle complex, data-driven manufacturing workflows. Specifically, the technical team tests the platform's ability to support real-time yield analytics and extraction efficiency monitoring. Ensuring these features work correctly means that the data flowing from the production floor is captured accurately and processed without unnecessary delay.

Furthermore, the testing process focuses heavily on the system's analytical capabilities, such as automated production stability interpretation and periodic raw material usage reporting. The software is required to evaluate production trends automatically and generate precise and dependable summaries of resource consumption. Testing these components ensures that management receives accurate insights that help optimize resources and reduce waste.

Ultimately, the primary aim of this comprehensive testing process is to identify hidden software errors, eliminate system bugs, and verify overall performance. By matching the system's actual behavior against its original design specifications, the team ensures that the developed platform meets the functional and non-functional requirements of the study. This final validation ensures a high-quality digital production monitoring system for AlDi Foods Inc.

Unit Testing

Unit testing is conducted to evaluate individual components or modules of the system independently. Each function is tested to ensure that it performs its intended task accurately before being integrated into the full system.

The following modules will be subjected to unit testing:

- User authentication and login module
- Raw material input module
- Extraction output and packed yield input module
- Yield percentage computation module
- Extraction efficiency calculation module
- Production stability interpretation module (Stable / Not Stable classification)
- Dashboard data retrieval module

- Report generation module (weekly, monthly, quarterly, yearly)

During unit testing, the researchers verify the correctness of essential calculations, focusing specifically on the accuracy of yield percentage computations and extraction efficiency results. By isolating these mathematical modules, the team confirms that raw sensor inputs and operational data reliably translate into precise metric outputs.

Additionally, the stability classification logic is thoroughly evaluated to ensure that production status is correctly determined based on predefined thresholds. This step ensures that system alerts and performance indicators accurately reflect real-time conditions, preventing false alarms during active monitoring.

System Testing

System testing is performed to evaluate the complete and integrated system as a whole. This stage ensures that all modules work together properly and that data flows correctly across the Client, Application, and Database layers.

The system is tested under simulated production scenarios that reflect the actual workflow of AlDi Foods Inc., including:

- Input of raw material quantities during production
- Recording of extraction and packed yield outputs
- Real-time computation of yield percentage
- Continuous monitoring of extraction efficiency

- Automatic classification of production stability
- Generation of dashboard analytics and visual reports
- Creation of periodic raw material usage reports

System testing ensures that all components interact seamlessly across the underlying architecture. By validating end-to-end data pipelines and API integrations, the team confirms that every module communicates reliably under varying workload conditions.

As a result, real-time updates are reflected accurately on the dashboard without data loss, delay, or inconsistency. This rigorous verification gives operators immediate visibility into active floor operations, ensuring complete trust in the visual reporting interface.

User Acceptance Testing (UAT)

User Acceptance Testing is conducted with selected personnel from AlDi Foods Inc., including production managers, supervisors, and production staff. This phase evaluates the system based on actual user experience and operational suitability.

The respondents assess whether the system meets the organization's requirements in terms of:

- Ease of use of the interface
- Accuracy of yield and efficiency calculations
- Clarity of dashboard visualizations
- Effectiveness of stability interpretation (Stable / Not Stable)
- Usefulness of periodic reports (weekly, monthly, quarterly, yearly)

- Overall system performance in real production monitoring scenarios

Feedback gathered during UAT is used to identify necessary improvements and refinements to ensure that the system effectively supports production monitoring, decision-making, and operational efficiency. Through these testing procedures, the system is validated as a reliable digital solution capable of replacing manual production monitoring methods and providing real-time analytics for improved production management at AlDi Foods Inc.

3.7 Evaluation Criteria

The proposed system undergoes a formal evaluation to determine its overall effectiveness, performance, and suitability for the operational needs of AlDi Foods Inc. This assessment is designed to verify that the digital platform can reliably support the demands of a fast-paced manufacturing environment. By gathering targeted feedback from key stakeholders, the evaluation aims to demonstrate that the system is ready for full-scale implementation.

To ensure a rigorous and standardized assessment, the evaluation is conducted using a Likert scale questionnaire carefully adapted from the ISO 25010 Software Quality Model. Utilizing this internationally recognized standard ensures that the system is not only judged based on opinion but is systematically measured against established software quality characteristics such as usability, reliability, and functional suitability. This framework provides a benchmarked and professional foundation for the entire feedback process.

A primary focus of this evaluation is determining whether the new system successfully replaces the company's legacy manual production monitoring processes. Moving away from traditional paper-and-pen logging or siloed spreadsheets represents a major shift for the production floor.

The survey closely examines how well the technology digitizes day-to-day operations and whether it creates a more streamlined and error-free workflow for the staff.

In addition to evaluating daily operations, the assessment evaluates user adoption and the overall usability of the digital interface. Gathering structured input from both frontline operators and supervisory personnel ensures that the software caters effectively to all user groups. This helps verify that the platform minimizes operational friction and allows staff to transition smoothly away from traditional tracking methods.

Furthermore, the evaluation measures the system's capacity to handle advanced, data-driven manufacturing tasks. Specifically, it assesses how effectively the platform supports real-time yield analytics, extraction efficiency monitoring, and automated stability interpretation. The results confirm that these automated capabilities provide operational visibility that far surpasses previous manual methods.

Ultimately, the process validates the system's ability to generate accurate periodic reports for raw material usage and key performance metrics. By delivering reliable, automated analytics, the evaluation confirms that management gains the precise insights needed to optimize resources, minimize operational waste, and support long-term production goals at AlDi Foods Inc.

Functionality

Functionality refers to the degree to which the system performs its intended tasks accurately and completely. This includes the system's ability to correctly process production inputs, compute yield percentages, and generate extraction efficiency results.

The system is evaluated based on its capability to:

- Accurately record raw material inputs and production outputs
- Compute real-time yield percentage without errors
- Calculate extraction efficiency correctly based on input-output relationships
- Automatically interpret production status as Stable or Not Stable based on defined thresholds
- Generate accurate production reports, including weekly, monthly, quarterly, and yearly summaries
- Provide real-time dashboard analytics for production monitoring

This criterion ensures that the system fulfills all required functionalities for digitized production monitoring. By integrating core operational metrics into a centralized digital architecture, the platform fully automates tracking across every stage of the manufacturing process.

As a result, management eliminates the need for manual logging and fragmented paper records. This complete digital coverage provides a reliable, end-to-end framework that captures every critical data point needed to optimize shop-floor performance.

Usability

Usability measures how easy and efficient the system is for users to operate. It evaluates the user interface design, navigation structure, and overall user experience of the system.

The system is assessed based on:

- Ease of navigating the dashboard and system modules
- Clarity of data presentation in charts, tables, and reports
- Simplicity of entering production data such as raw materials and output yields
- Accessibility of real-time analytics and reports
- User-friendliness for production staff and managers with minimal technical background

This ensures that the system can be effectively used by AlDi Foods Inc. personnel without extensive training. By prioritizing an intuitive, user-friendly interface, the team minimizes technical barriers and simplifies daily operational workflows.

As a result, staff across various departments can comfortably integrate the software into their routine tasks right away. This seamless adoption significantly reduces onboarding time and allows the organization to maintain high productivity without productivity losses during the transition.

Accuracy

Accuracy refers to the correctness and precision of the system's data processing and output generation. This is a critical criterion because the system focuses on production analytics and efficiency monitoring.

The system was evaluated based on:

- Correct computation of yield percentage values
- Accurate extraction efficiency calculations
- Precise classification of production status (Stable or Not Stable)
- Consistency of reported data across different time periods

- Accuracy of raw material usage reports and historical records

This ensures that all analytical outputs reflect the actual production performance of the organization. By grounding the metrics directly in real-time operational data, the team maintains strict data integrity across all reporting channels. This direct alignment eliminates discrepancies between estimated projections and true yield. Consequently, leadership gains a dependable foundation for making strategic operational decisions.

Efficiency

Efficiency measures the system's ability to process data and deliver results with minimal delay and optimal resource usage. Since the system supports real-time monitoring, efficiency is a critical requirement.

The system was evaluated based on:

- Speed of data processing from input to dashboard display
- Real-time updating of production metrics and analytics
- Fast generation of reports (weekly, monthly, quarterly, yearly)
- Minimal system lag during simultaneous data input and monitoring
- Effective handling of multiple production records without performance degradation

This ensures that the system delivers real-time operational data right when supervisors need it most. By providing immediate visibility into floor activities, management can instantly spot bottlenecks and keep workflows moving smoothly.

Consequently, teams can adjust production parameters on the fly to prevent costly downtime. This proactive response capability helps the organization maintain peak productivity while consistently meeting tight output deadlines.

Reliability

Reliability refers to the system's ability to operate consistently and correctly over time without failure. It ensures that the system remains stable during continuous use in a production environment.

The system will be evaluated based on:

- Stability of system operations during continuous data input
- Consistency of real-time analytics and dashboard updates
- Ability to recover or handle errors without data loss
- Proper storage and retrieval of historical production records
- Dependability of system outputs under normal operational conditions

This criterion ensures that the system can be trusted for daily production monitoring and long-term operational use. By delivering consistently accurate data and maintaining stable uptime, the platform establishes strong operational reliability across every shift.

As a result, management can confidently rely on these metrics to manage day-to-day output and guide strategic planning. This long-term dependability prevents data fatigue and ensures the system remains a core asset as business needs evolve.

Summary of Evaluation Approach

To evaluate the overall impact of the new system, selected users from AlDi Foods Inc., including frontline production staff and upper management, complete a comprehensive Likert scale questionnaire. This structured survey captures standardized feedback regarding user experience, system functionality, and daily operational changes. By involving both the staff on the factory floor and the decision-makers in leadership, the evaluation ensures a well-rounded perspective on how the system performs across different levels of the organization.

The data gathered from these evaluation results serves as the primary metric to determine the overall effectiveness of the platform. Specifically, the feedback measures how successfully the digital platform replaces long-standing manual processes across the facility. Transitioning away from paper-based tracking represents a major operational shift, and this assessment pinpoints how smoothly the team adapts to the digital transition.

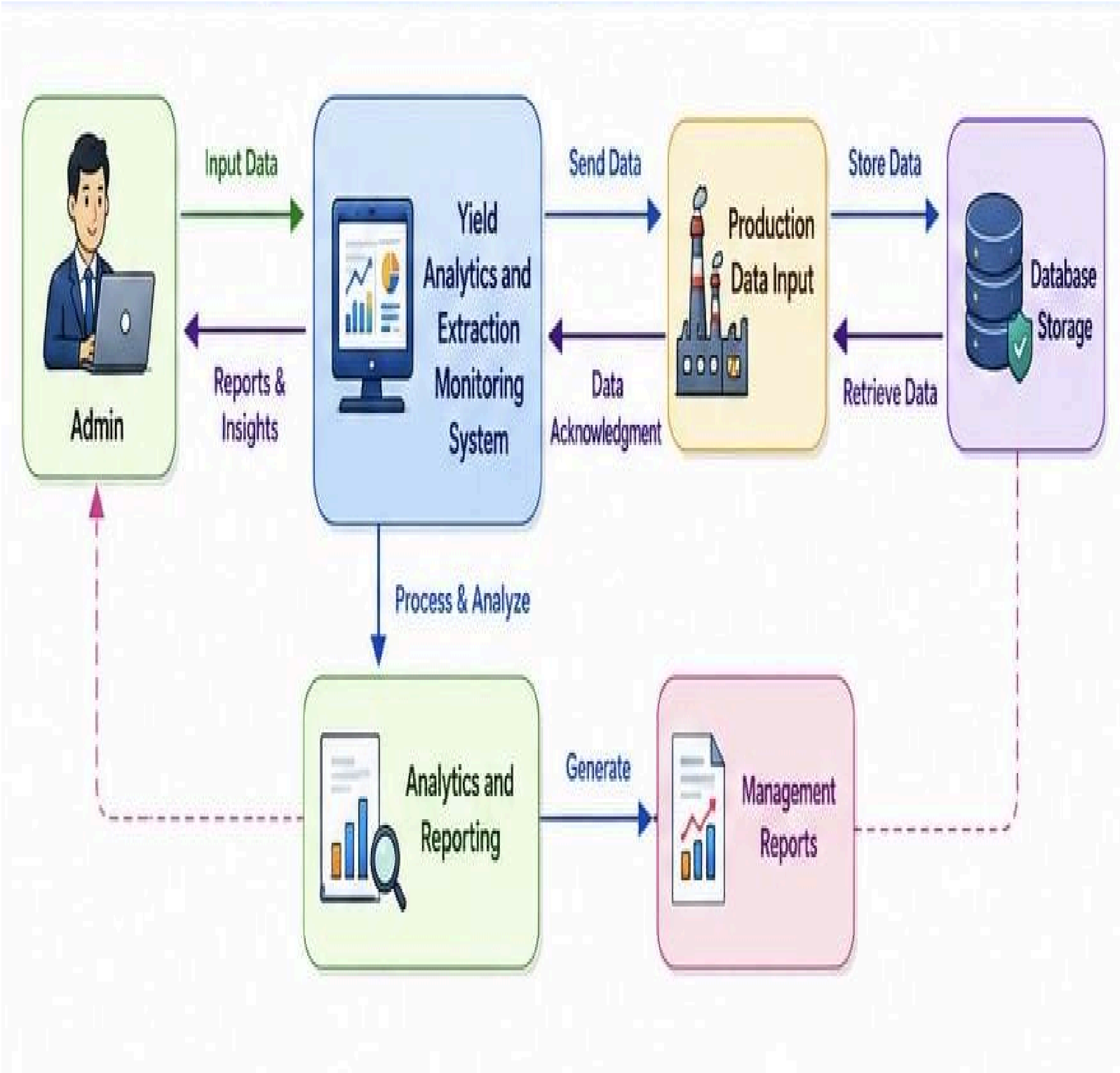
Furthermore, the evaluation heavily focuses on the system's ability to support real-time digital production monitoring throughout daily operations. Frontline data entry and automated tracking seamlessly translate into live updates that reflect current floor conditions. The outcome of this assessment confirms whether the technology consistently delivers the accurate, up-to-the-minute data stream required for modern manufacturing.

In addition to real-time tracking, the survey examines how well the system improves collaboration between different operational tiers. By capturing metrics on data accessibility, the evaluation demonstrates how communication between floor workers and management improves once manual bottlenecks are eliminated. This ensures that operational feedback is converted into actionable insights for continuous floor improvements.

Ultimately, the final results validate whether the system successfully achieves its core operational goals across all parameters. A positive evaluation demonstrates that the platform improves production visibility, enhances management decision-making, and reduces manual data-entry errors. Most importantly, it confirms that the system effectively supports the precise and efficient management of critical yield and extraction processes.

Overall, the comprehensive evaluation process confirms that the technological transition yields significant improvements in workflow efficiency, operational control, and data accuracy. By successfully replacing manual record-keeping with an automated solution, AlDi Foods Inc. establishes a solid foundation for sustained, data-driven production management.

DFD (Data Flow Diagram)



ERD (Entity Relationship Diagram)


ADMINS
email
id
username
PACK_YIELDS
batch_id (FK)
id
total_packs
produced
yield_total_kg
user_id (FK)
records
N
1
USERS
email
full_name
username
id
yields
N
1
MATERIAL_EXTRACTIONS
extracted_by
batch_id (FK)
id
material_name
kg_extracted
produces
N
1
PRODUCTION_BATCHES
product_name
production_
datetime
batch_number
created_by
has
1
N
DAILY_ANALYTICS
SNAPSHOTS
efficiency_
percentage
total_input_kg
total_output_kg
snapshot_date
id
BATCH_MATERIALS
original_kg
batch_id (FK)
id
material_name

Users

The Users entity represents production personnel who interact with the system. Users are responsible for creating production batches, encoding raw material data, and recording packed yield outputs. Users play a central role in the system because they provide the primary production data used for real-time analytics and efficiency monitoring.

Relationships:

- A User could create multiple Production Batches.
- A User could record multiple Pack Yield entries under different batches.

Production Batches

The Production Batch entity represents a single production run or cycle within AlDi Foods Inc. Each batch contains detailed information about the production process, including raw materials used and outputs produced. This entity serves as the main container of production data used for yield computation and extraction efficiency analysis.

Relationships:

- A Production Batch belonged to one User.
- A Production Batch contained multiple Batch Materials records. A
- Production Batch could generate multiple Pack Yield records.

Raw Materials

The Raw Materials entity stores information about raw materials used in each production batch. This includes the quantity of raw inputs such as fruits or raw juice materials used during the extraction process. This entity is essential for calculating production efficiency and yield percentage.

Relationships:

- Batch Materials belonged to a specific Production Batch.
- Each Production Batch could have multiple material entries depending on production requirements.

Extraction Records

The Extraction Records entity captures intermediate output data obtained during the processing or extraction phase of a production batch. It logs extracted quantities, timestamps, and operator remarks to track step-by-step progress before final packaging. This entity helps monitor inline operational yield and spot losses occurring mid-process.

Relationships:

- Multiple Extraction Records can be logged for one Production Batch.

Pack Yields

The Pack Yield entity represents the final output of the production process. It records the amount of finished products (e.g., bottled or packed juice) produced from a specific production batch. This data is crucial for real-time yield analytics and extraction efficiency computation.

Relationships:

- Pack Yield belonged to a Production Batch. A Production Batch could produce multiple
- Pack Yield records depending on product variations or packaging types.

Yield Analytics

The Analytics entity is a standalone system module responsible for processing production data into meaningful insights. It handles real-time computation and visualization of production performance. This entity is central to the system's objective of providing digital production intelligence.

Functions of Analytics:

- Computed real-time yield percentage
- Analyzed extraction efficiency performance
- Generated production trends and visual reports
- Classified production status (Stable / Not Stable)
- Generated periodic reports (weekly, monthly, quarterly, yearly)

Relationships:

- Analytics was independent but used data from Production Batch, Batch Materials, and Pack Yield.

Efficiency Records

The Efficiency Records entity tracks performance efficiency rates for individual production runs. It evaluates expected output standard rates against actual packed output to identify operational bottlenecks and measure worker/machine throughput.

Relationships:

- An Efficiency Record belongs to one Production Batch.
- Efficiency Records data can be compiled into multiple Reports.

Production Stability

The Production Stability entity logs batch consistency metrics and quality status updates throughout the production cycle. It ensures that outputs meet safety, stability, and quality standards before final distribution.

Relationships:

- A Production Stability record belongs to one Production Batch.
- Production Stability data can be compiled into multiple Reports.

Reports

The Reports entity aggregates operational, yield, efficiency, and stability metrics into formal documents. It tracks report types and generation dates to support management oversight, compliance audits, and long-term decision-making.

Relationships:

- A Report compiles analytical data originating from a Production Batch (including Yield Analytics, Efficiency Records, and Production Stability).

Summary of Relationships

Users

- **Creates** multiple Production Batches.
- **Records** multiple Pack Yield entries across various batches.

Production Batches

- **Belongs to** one User.
- **Contains** multiple Raw Materials (Batch Materials) entries.
- **Generates** multiple Extraction Records.
- **Produces** multiple Pack Yield records (for different variations or packaging types).
- **Has** one Efficiency Record.
- **Has** one Production Stability record.

Raw Materials (Batch Materials)

- **Belongs to** one specific Production Batch.

Extraction Records

- **Belongs to** one specific Production Batch (logs mid-process intermediate step data).

Pack Yields

- **Belongs to** one Production Batch.
- **Recorded by** a User.

Yield Analytics

- **Operates independently** while drawing source data directly from Production Batches, Batch Materials, and Pack Yields.

Efficiency Records

- **Belongs to** one Production Batch.
- **Compiles into** multiple Reports.

Production Stability

- **Belongs to** one Production Batch.
- **Compiles into** multiple Reports.

Reports

- **Aggregates** analytical data originating from Production Batches, including Yield Analytics, Efficiency Records, and Production Stability metrics.

System Role in Production Monitoring

This ERD structure supports the main goal of the system, which is to digitize production monitoring at AlDi Foods Inc. It enables:

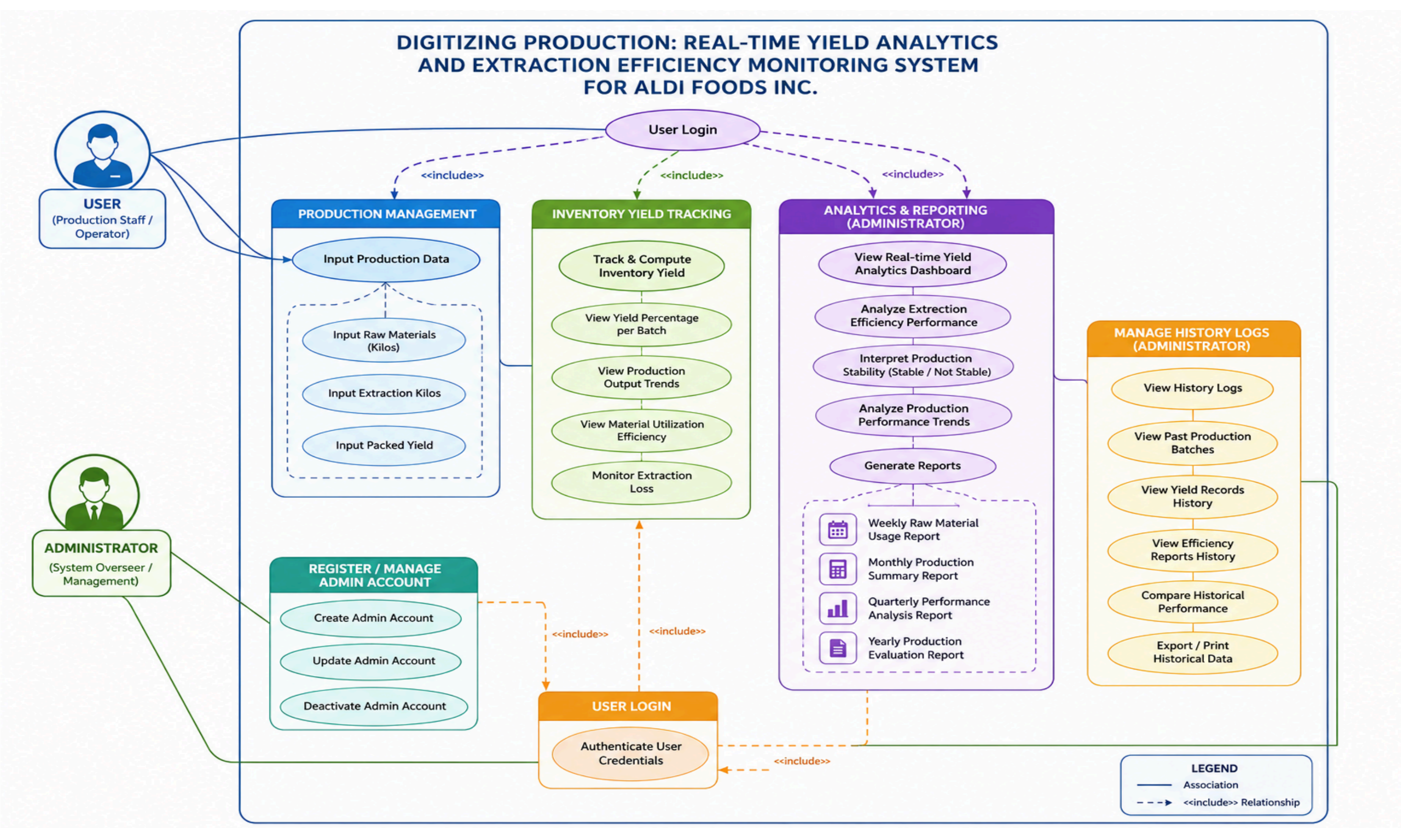
- **Real-time tracking of raw material usage** by logging specific inputs, quantities, and units mapped directly to each active production batch.
- **Automated yield computation** through the continuous integration of raw material inputs and packed yield outputs to determine yield percentages.
- **Extraction efficiency monitoring** by capturing mid-process operational logs, timestamps, and output volumes during intermediate extraction stages.
- **Production stability classification** by logging batch consistency metrics to automatically categorize quality and operational status.
- **Centralized reporting and analytics** by consolidating yield metrics, efficiency records, and stability status into downloadable management reports.

By structuring the system in this way, the platform ensured accurate, efficient, and real-time monitoring of production activities, replacing manual record-keeping processes and improving decision-making capabilities.

Use Case Diagram

The Use Case Diagram illustrates the interactions between the system and its primary actors within the Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc. It clearly defines the functional requirements of the platform and demonstrates how each actor engages with the system to facilitate production monitoring, data analytics, and operational reporting.

The system replaces manual production tracking by establishing a centralized digital platform. This transformation enables the real-time computation of yield metrics, extraction efficiency analysis, production stability interpretations, and automated report generation across the entire manufacturing workflow.



Actors

The system consisted of two primary actors:

User

The User represents production staff or operators responsible for encoding production data into the system. Users are directly involved in recording real-time production information such as raw material usage, extraction output, and packed yield data. The User plays a critical role in ensuring that production data is accurate, complete, and updated in real time, which is essential for generating reliable analytics and efficiency reports.

Administrator

The Administrator is responsible for overseeing the entire system. This actor manages user accounts, monitors production performance, reviews analytics dashboards, and generates reports for management decision-making. The Administrator has full access to system data and is responsible for evaluating production performance trends, monitoring efficiency levels, and ensuring data integrity across all modules.

Register / Manage Admin Account

This use case allows the Administrator to create, update, and manage admin accounts within the system. It ensures that only authorized personnel have access to sensitive production analytics and system configuration features.

User Login

Both Users and Administrators have to authenticate through the login system to access their respective functionalities. This ensures system security and controlled access to production data and analytics.

Production Management

This is the core module of the system where Users input all production-related data. It supports real-time monitoring and serves as the foundation for yield analytics and extraction efficiency computation.

Sub-functions included:

Input Raw Materials (Kilos)

Users encode the quantity of raw materials used in production batches. This data is essential for computing yield and efficiency.

Input Extraction Kilos

Users record the amount of juice or extracted output obtained from the raw materials during production.

Input Packed Yield

Users enter the final packaged product output, which is used to evaluate production losses and efficiency.

Inventory Yield Tracking

This use case allows the system to automatically track and compute production yield based on raw material input and output data. It provides real-time updates on:

- Yield percentage per batch
- Production output trends
- Material utilization efficiency
- Extraction loss monitoring

This feature supports continuous monitoring of production performance and helps identify inefficiencies early.

Analytics & Reporting (Administrator)

This use case enables the Administrator to access and analyze system-generated production data. It includes real-time dashboards and structured reports that support decision-making.

Key functionalities included:

- Real-time yield analytics visualization
- Extraction efficiency performance analysis
- Production stability interpretation (Stable / Not Stable)
- Trend analysis of production performance
- Generation of periodic reports:
    - Weekly raw material usage report
    - Monthly production summary report
    - Quarterly performance analysis
    - Yearly production evaluation report

This module transforms raw production data into meaningful insights for management and operational improvement.

Manage History Logs

This use case **allows** the Administrator to view and manage historical production records stored in the system database. It **provides** access to past production batches, yield records, and efficiency reports.

This supports:

- Audit trail of production activities
- Historical comparison of yield performance
- Detection of recurring inefficiencies
- Long-term production analysis and reporting

System Interaction Summary

The system operates through a structured and seamless interaction between Users and the Administrator, establishing a cohesive workflow across the entire operational environment. By defining clear roles and responsibilities, this framework ensures that data flows efficiently from initial entry to high-level management review.

Frontline Users are primarily responsible for encoding real-time production data directly at the operational level. By capturing critical metrics such as raw material usage, extraction output, and final packaged yield right as they occur, these operators provide the essential groundwork for accurate data collection.

Once these inputs are recorded, the system systematically processes the raw operational data into actionable intelligence. It automatically calculates real-time yield percentages, measures extraction efficiency, and categorizes overall production stability, effectively replacing the need for manual calculations.

The Administrator then utilizes these automated outputs to oversee plant-wide operations, monitor ongoing performance trends, and generate detailed analytical reports. Access to these clear, data-driven insights allows management to identify operational bottlenecks quickly and make informed, strategic decisions to optimize output.

Through this structured interaction, the platform successfully achieves its primary objective of digitizing production processes at AlDi Foods Inc. By integrating real-time monitoring, automated analytics, and streamlined reporting, the system significantly improves operational efficiency and transforms daily manufacturing workflows.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULT, AND DISCUSSION

4.2 Overview of the Developed System

The Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc. is a web-based application developed to modernize and improve the company's existing production monitoring process. The proposed system replaces the traditional manual and paper-based recording of production information with a centralized digital platform that enables administrators to record, process, analyze, and retrieve production data efficiently. By utilizing digital technologies, the system addresses the limitations of manual production monitoring while providing accurate and timely information that supports operational efficiency and management decision-making.

The development of the proposed system is based on the findings presented in Chapters 1, 2, and 3 of the study. The researchers identify several operational challenges in the existing production monitoring process, including delayed recording of production information, inaccurate manual computations, duplication of records, difficulty in retrieving historical production data, and limited visibility of production performance. These problems affect the company's ability to monitor production yield, evaluate extraction efficiency, and make timely production decisions. Consequently, the researchers design and develop a web-based production monitoring system that automates these processes while maintaining a centralized database for efficient data management.

The proposed application provides an integrated platform that allows administrators to encode production batch information, automatically compute production yield and extraction efficiency, monitor production performance through graphical dashboards, and generate production reports. Unlike the existing manual process, the developed system performs automatic calculations once production information is entered, significantly reducing computational errors and minimizing the time required to prepare production summaries. This automation enables production personnel and management to obtain accurate information immediately after data entry, thereby improving production monitoring and operational control.

The system is developed using PHP as the server-side programming language, HTML, CSS, and JavaScript for designing the user interface, Chart.js for graphical data visualization, and MySQL as the centralized database management system. These technologies are selected because they are reliable, cost-effective, widely supported, and suitable for developing web-based information systems for small and medium-sized enterprises (SMEs) such as AlDi Foods Inc. The application is deployed through the XAMPP development environment, allowing administrators to access the system through a standard web browser within the organization's local network.

One of the primary features of the proposed system is the Production Monitoring Module, which enables administrators to encode production batch information such as production date, raw material quantity, operator information, and production output. These production records serve as the foundation for all succeeding computations and analyses performed by the system. By maintaining complete and organized production records, the application provides a reliable source of information for evaluating production activities and monitoring operational performance.

Another important feature of the system is the Yield Analytics Module, which automatically computes the production yield percentage using the recorded production input and output values. The computed yield percentage allows administrators to evaluate how efficiently raw materials are converted into finished products. Since the computation is performed automatically by the system, the possibility of manual computational errors is significantly reduced. The resulting values are immediately displayed on the dashboard through charts and summary tables, enabling administrators to evaluate production performance without performing manual calculations.

Complementing the yield analysis is the Extraction Efficiency Monitoring Module, which evaluates the effectiveness of the juice extraction process by computing the extraction efficiency percentage for every production batch. This module allows production personnel and management to continuously monitor extraction performance and determine whether the extraction process operates efficiently. The generated information assists management in identifying production trends, recognizing variations in extraction performance, and implementing corrective actions whenever necessary to improve operational efficiency.

The proposed system also includes a Reporting Module that automatically generates production reports based on the stored production information. These reports summarize production activities, production yield, extraction efficiency, and other relevant production data within a selected period. The reporting function minimizes the time required to prepare production documents and enables administrators to retrieve historical production reports whenever needed. Through centralized report management, management personnel can compare production performance across multiple production batches and evaluate long-term production trends.

To ensure data integrity and accessibility, all production information is stored in a centralized MySQL database. Centralized storage eliminates redundant records, improves data consistency, and allows authorized administrators to retrieve historical production information quickly. Since all production records are maintained within a single database, the system provides better organization of production information while reducing the possibility of data loss associated with paper-based record keeping.

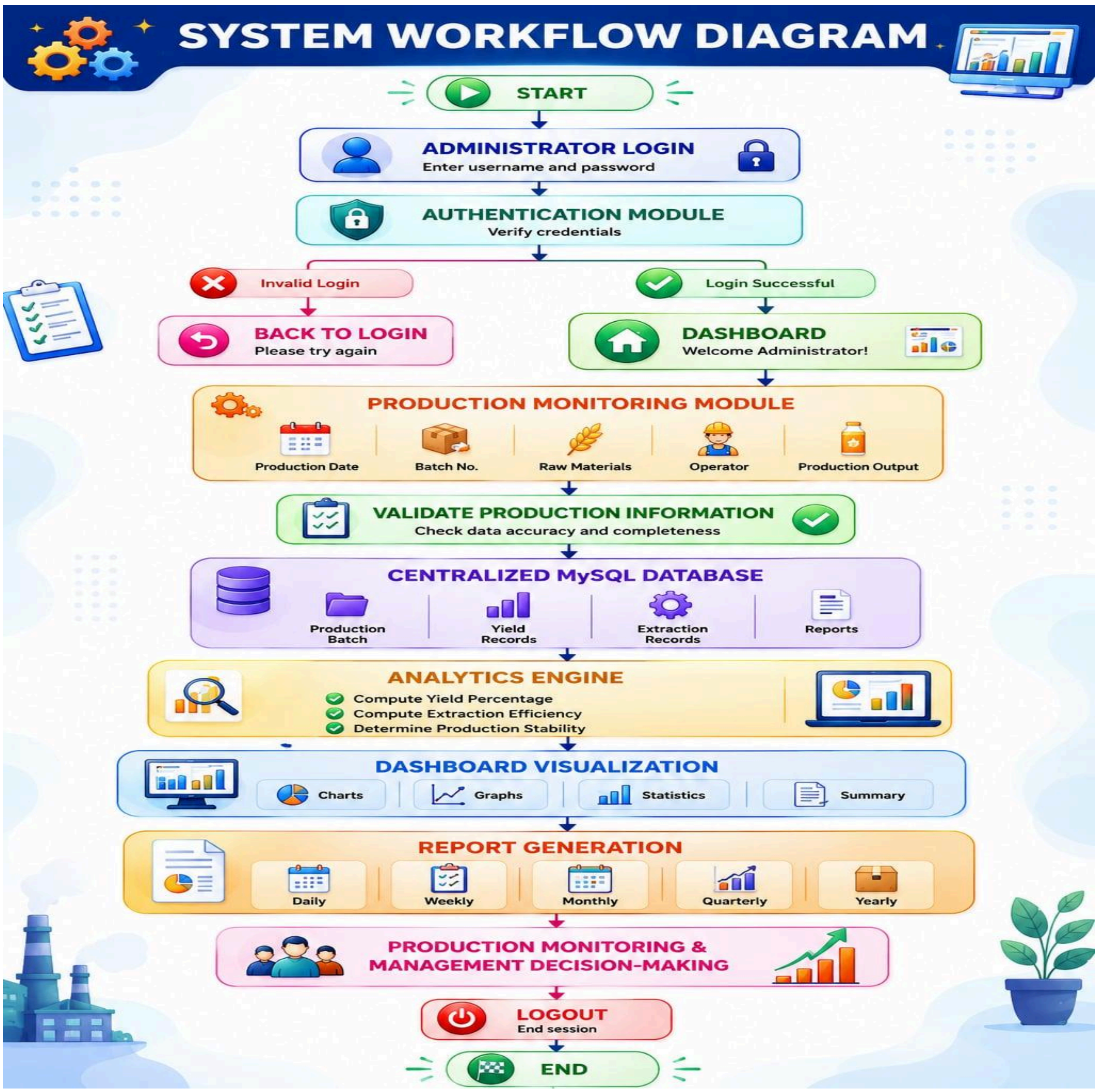
Security is also incorporated into the proposed application through the Administrator Module, which requires authorized users to log in before accessing the system. User authentication restricts unauthorized access to production information and ensures that only designated administrators can encode production records, modify existing information, monitor production performance, and generate reports. This security mechanism protects the integrity and confidentiality of the company's production data.

The integration of these functional modules provides AlDi Foods Inc. with a comprehensive production monitoring solution capable of supporting daily production operations. By replacing manual recording procedures with an automated web-based application, the proposed system improves production monitoring, enhances data accuracy, reduces processing time, minimizes computational errors, and supports data-driven decision-making. Furthermore, the centralized storage of production information provides management with reliable historical records that can be used to evaluate operational performance, monitor production consistency, and identify opportunities for continuous process improvement.

Overall, the developed system demonstrates how digital technologies can be applied within the food manufacturing industry to improve production monitoring and operational efficiency.

Through automated computations, centralized database management, graphical visualization, and report generation, the proposed application provides a practical and cost-effective solution that addresses the production monitoring requirements of AlDi Foods Inc. while supporting the objectives of the study.

Figure 4.1 illustrates the **System Workflow Diagram** of the system.



The workflow begins with the Administrator logging in and passing authentication to access the dashboard, where production details such as batch numbers, raw materials, and output are encoded and validated before being stored in the centralized MySQL database. Once stored, the system processes this data through an analytics engine to calculate yield and efficiency, presents visual insights on the dashboard, and generates periodic reports that empower management decision-making prior to session logout.

4.3 System Development Methodology

The development of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* follows the Agile Software Development Methodology, an iterative and incremental approach to software development that emphasizes flexibility, continuous collaboration, and regular evaluation throughout the project lifecycle. Agile is selected because it enables the researchers to divide the project into manageable phases while allowing continuous refinement of the system based on user feedback and evaluation. Unlike traditional software development methodologies, Agile encourages frequent testing, evaluation, and enhancement of each completed module before proceeding to the next development stage. This approach ensures that the proposed system remains aligned with the operational requirements of AlDi Foods Inc. and effectively addresses the identified problems in the existing production monitoring process.

The adoption of the Agile methodology also supports the developmental nature of this research. Since the objective of the study is to design, develop, implement, and evaluate a web-based production monitoring system, Agile provides an organized framework for developing the application through successive iterations. Every completed feature is reviewed, tested, and improved before additional functionalities are implemented. This continuous cycle of planning, development, testing, and feedback enables the researchers to identify and resolve system issues early in the development process, thereby improving software quality, minimizing development risks, and ensuring that the final application satisfies the functional and non-functional requirements identified during the requirement analysis phase.

The Agile development process used in this study consists of seven major phases, namely Planning, Requirement Analysis, System Design, Development, Testing, Deployment, and Review and Feedback. Each phase contributes to the successful completion of the proposed system and ensures that every module functions efficiently before the application is deployed.

Planning

The development process begins with the Planning phase, where the researchers establish the overall direction of the project. During this stage, the objectives, scope, limitations, expected outputs, project timeline, and development resources are identified and documented. The researchers carefully examine the existing production monitoring practices of AlDi Foods Inc. to understand how production information is recorded, processed, and reported.

Through interviews with production personnel and direct observation of the company's production operations, the researchers identify several challenges associated with the existing manual recording process. These include delayed submission of production reports, inaccurate manual computations, inconsistent recording of production data, duplication of information, difficulty in retrieving historical production records, and limited visibility of production performance. These findings serve as the foundation for defining the functional scope of the proposed web-based system.

During the planning stage, the researchers also select the technologies that are used to develop the application. The system is designed using PHP for server-side programming, HTML, CSS, and JavaScript for developing the user interface, Chart.js for graphical visualization of production data, MySQL as the centralized database management system, and XAMPP as the local web server environment. These technologies are selected because they are reliable, open-source, cost-effective, and appropriate for small and medium-sized manufacturing enterprises.

Requirement Analysis

After completing the planning activities, the researchers proceed to the Requirement Analysis phase. The primary objective of this stage is to identify all functional and non-functional requirements needed to successfully develop the proposed system.

The researchers gather information through interviews with production supervisors and personnel, direct observation of the juice extraction process, and examination of existing production records and manual log sheets. These activities enable the researchers to understand the complete production workflow and determine the information required for production monitoring.

Based on the gathered data, the researchers identify several essential functional requirements of the system. These include secure administrator login, recording of production batch information, automatic computation of production yield, automatic computation of extraction efficiency, dashboard visualization of production performance, centralized database management, historical data retrieval, and automated generation of production reports.

In addition to the functional requirements, several non-functional requirements are also identified. The system needs to provide a user-friendly interface, ensure reliable system performance, maintain data accuracy and integrity, support secure administrator access, and provide efficient retrieval of production information. These requirements serve as the basis for designing the system architecture and implementing the application.

System Design

After identifying the system requirements, the researchers proceed to the System Design phase. During this stage, the overall structure of the application is carefully planned before actual coding begins. The researchers prepare the system architecture, database structure, user interface layouts, process flow diagrams, and database relationships to serve as blueprints during software development.

The System Workflow Diagram illustrates the Agile development process followed throughout the project. The System Architecture Diagram defines the interaction between the administrator and the major system components responsible for production monitoring, analytics, reporting, and database management. The Entity Relationship Diagram (ERD) describes the relationships among the five primary database entities, namely Admin, Production Batch, Yield Record, Extraction Record, and Reports. The Data Flow Diagram (DFD Level 0) illustrates the overall movement of production data from the administrator to the database and ultimately to report generation.

The graphical user interface is also designed during this phase. The interface emphasizes simplicity, consistency, and ease of navigation so that administrators can efficiently record production information, monitor production performance, generate reports, and access historical production records with minimal training.

Development

The Development phase involves translating the completed system design into a fully functional web-based application. The researchers implement the application using PHP as the primary server-side programming language, while HTML, CSS, and JavaScript are utilized for developing responsive and interactive web pages. Chart.js is integrated into the application to provide graphical visualization of production yield and extraction efficiency, while MySQL is used to store and manage production records within a centralized database.

The development process follows Agile principles by implementing each module incrementally. The Administrator Module is developed first to provide secure user authentication and system access. This is followed by the Production Monitoring Module, which enables administrators to encode production batch information such as production date, raw material quantity, operator information, and production output.

The Analytics Module is then developed to automatically compute production yield and extraction efficiency using the encoded production information. The module also generates graphical charts and summary tables that allow administrators to monitor production performance more effectively. Finally, the Reporting Module is implemented to generate production reports and summaries that assist management in evaluating production operations and making informed decisions.

Continuous integration and module verification are performed throughout development to ensure that newly implemented features operate correctly with previously completed modules.

Testing

Following the completion of system development, the application undergoes a comprehensive Testing phase to verify that all modules operate according to the specified requirements. The primary objective of testing is to ensure that the application functions correctly, produces accurate computations, maintains data integrity, and satisfies the operational requirements of AlDi Foods Inc.

The researchers initially perform Unit Testing, where each module is tested individually to verify its functionality. The login module, production monitoring module, analytics module, reporting module, and database operations are evaluated independently before integration.

After verifying individual modules, Integration Testing is conducted to ensure that all modules communicate correctly with one another. This phase confirms that production information entered into the system is successfully stored in the database, automatically processed by the analytics module, and correctly displayed in the reports and dashboard.

The researchers also conduct System Testing to evaluate the performance of the complete application under normal operating conditions. Finally, User Acceptance Testing (UAT) is performed with selected personnel from AlDi Foods Inc. to determine whether the developed system meets the company's operational requirements and expectations. Any identified software defects are corrected before the application is deployed.

Deployment

Upon successful completion of testing, the project enters the Deployment phase. During this stage, the developed application is installed and configured within the operational environment of AlDi Foods Inc. The researchers configure the Apache web server using XAMPP, initialize the MySQL database, create administrator accounts, and import the required database tables.

The intended users are oriented on the proper use of the application, including procedures for logging into the system, encoding production information, monitoring production performance, generating reports, and retrieving historical production records. Sample production batches are also encoded to verify that the deployed application functions correctly within the production environment.

Deployment marks the transition of the proposed system from the development environment to actual operational use.

Review and Feedback

The final phase of the Agile methodology involves Review and Feedback. After deployment, the developed application is evaluated by its intended users to determine its overall effectiveness, usability, reliability, and functionality. Administrators and production personnel provide comments and recommendations regarding the operation of the application, user interface, report generation, dashboard visualization, and overall system performance. The researchers document all observations and suggestions for future enhancement.

Consistent with Agile principles, the collected feedback serves as the foundation for future development iterations. The continuous evaluation and improvement of the application ensure that the system remains responsive to the operational needs of AlDi Foods Inc. while providing reliable production monitoring, accurate yield analytics, efficient extraction efficiency monitoring, and effective production report generation.

Figure 4.2 presents the interaction between the Administrator and the Web-Based System, highlighting the major functional modules of the proposed application.

As *illustrated in Figure 4.2,* the Administrator serves as the primary user of the web-based application and is responsible for managing all production monitoring activities. After successfully logging into the system, the administrator interacts with the Web-Based System, which serves as the central platform for recording, processing, storing, and presenting production information.

The application is composed of three major functional modules: the Production Monitoring Module, the Analytics Module, and the Reporting Module. The Production Monitoring Module enables administrators to encode production batch information, including the production date, raw material quantity, operator information, and production output. These production records are stored securely in the centralized MySQL database, ensuring data consistency and efficient retrieval of historical production information.

The Analytics Module retrieves production data from the database and automatically computes the production yield percentage and extraction efficiency for each production batch. The computed results are displayed through graphical dashboards, charts, and summary tables, allowing administrators to monitor production performance, evaluate extraction efficiency, identify operational trends, and assess overall production stability.

The Reporting Module generates comprehensive production reports based on the processed information stored in the database. These reports provide summarized production information, yield analytics, extraction efficiency results, and other relevant operational data that support production monitoring, managerial evaluation, and data-driven decision-making.

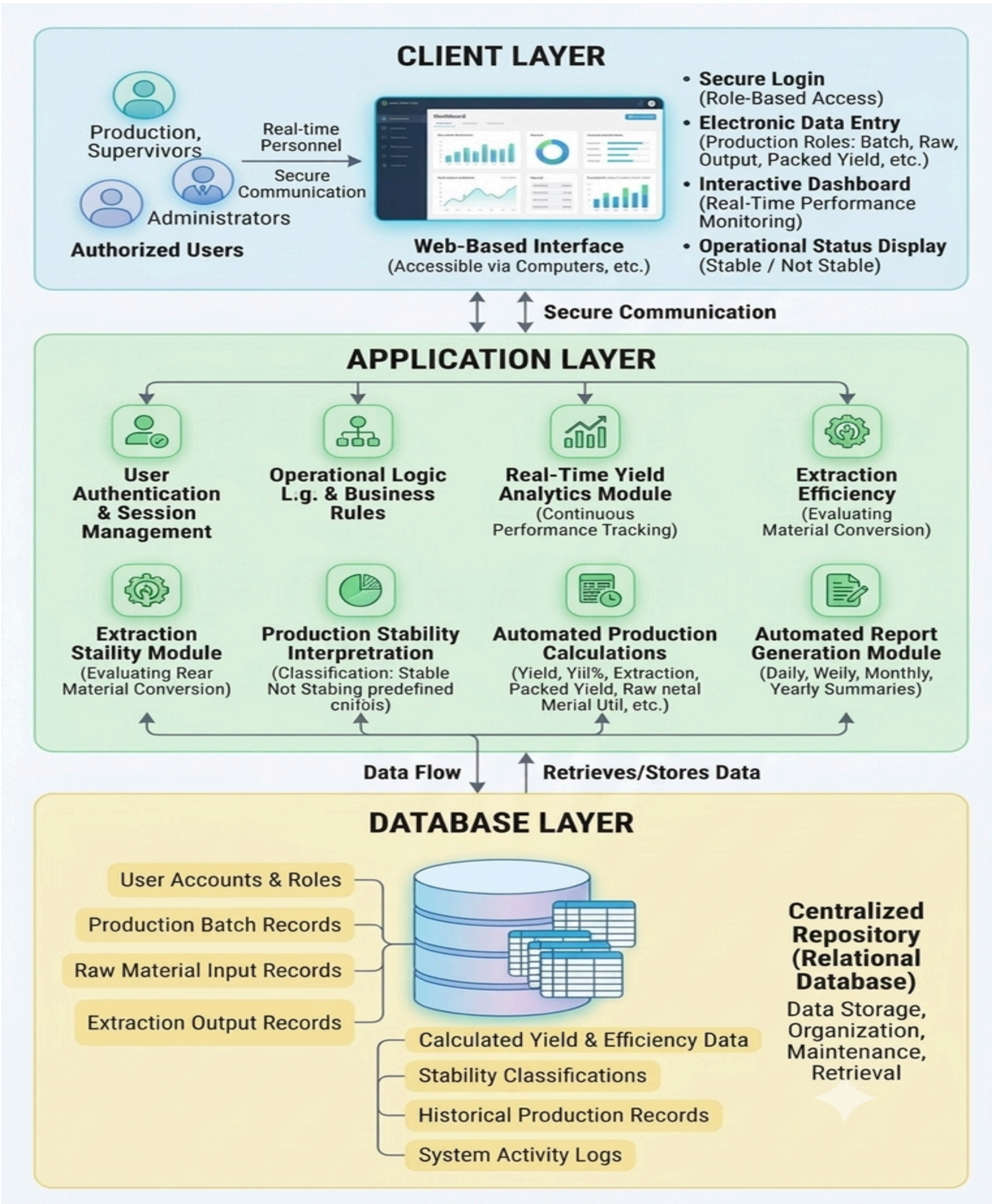
The interaction illustrated in Figure 4.2 demonstrates how the proposed system integrates production monitoring, analytics, database management, and reporting into a single web-based platform. Throughout the Agile development process, feedback obtained from administrators and intended users is continuously incorporated into succeeding iterations, ensuring that the developed application remains reliable, user-friendly, efficient, and aligned with the production monitoring requirements of AlDi Foods Inc.

4.4 System Architecture

The system architecture for AlDi Foods Inc. is separated into three core layers, forming a three-tier architecture:

- Client Layer (Presentation Layer): The web-based user interface where authorized users, including production personnel, supervisors, and administrators, securely log in, enter electronic production data, and view real-time dashboards and reports.
- Application Layer (Logic Layer / Central Processing Engine): The core engine that processes requests, validates submitted data according to business rules, performs automated calculations such as yield percentage and extraction efficiency, handles status classifications such as “Stable” and “Not Stable,” and manages automated report generation.
- Database Layer (Data Management Layer / Centralized Repository): The centralized relational database that securely stores, organizes, retrieves, and maintains all system records, including batch data, calculations, user roles, and production stability information.

Figure 4.3 illustrated the overall System Architecture of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.*



4.5 Hardware and Software Requirements

The successful implementation of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* requires appropriate hardware and software resources to ensure the efficient operation of the proposed web-based application. Since the system is responsible for recording production batch information, automatically computing production yield and extraction efficiency, generating graphical dashboards, and producing management reports, reliable computing resources are necessary to maintain system performance, data accuracy, and operational efficiency.

The hardware and software requirements are carefully selected based on the functional requirements identified during the system analysis and design phases. These resources provide the necessary environment for the installation, development, testing, deployment, and daily operation of the proposed application. The selected technologies are widely available, cost-effective, and compatible with the existing information technology infrastructure of AlDi Foods Inc., making the proposed system practical for implementation without requiring expensive specialized equipment.

The hardware specifications ensure that the application can efficiently process production data, perform automated computations, retrieve historical records, generate graphical visualizations, and support report generation with minimal processing delays. Likewise, the software requirements provide the development framework, programming environment, database management capabilities, and web technologies needed to build and operate the application successfully.

Since the proposed system utilizes a centralized database, all production information is securely stored and managed within a single repository. This centralized approach improves data consistency, minimizes data redundancy, and enables administrators to retrieve production records quickly whenever monitoring, analysis, or report generation is required. The recommended hardware specifications also provide sufficient processing power and storage capacity to accommodate increasing production records as the company continues to expand its operations.

The software technologies selected for the development of the system are based on open-source and industry-standard applications that are widely used in web development. These technologies support the implementation of the system's major functionalities, including secure administrator authentication, production batch management, automated yield computation, extraction efficiency monitoring, dashboard visualization, centralized database management, and automated report generation. The use of open-source software also minimizes implementation costs while ensuring that the system remains maintainable, scalable, and easy to update in the future.

Furthermore, the combination of the recommended hardware and software provides a stable and reliable environment for the continuous operation of the proposed application. The selected components ensure that the system can perform real-time production monitoring, efficiently process production information, maintain accurate historical records, and provide administrators with timely analytical reports that support operational planning and decision-making.

The following sections present the minimum and recommended hardware specifications as well as the software requirements necessary for the successful development, deployment, and implementation of the proposed *Digitizing Production: Real-Time Yield Analytics and Extraction*

Efficiency Monitoring System for AlDi Foods Inc. These requirements serve as the technical foundation that enables the application to operate efficiently while achieving the objectives of improving production monitoring, minimizing manual computation, enhancing data accuracy, and supporting data-driven decision-making within the company.

4.5.1 Hardware Requirements

The hardware requirements specify the minimum and recommended computer specifications necessary for the successful implementation and operation of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* These hardware components provide the computing resources needed to support the installation, execution, and maintenance of the proposed web-based application. Selecting appropriate hardware ensures that the system can perform efficiently while processing production information, executing automated computations, storing production records, and generating analytical reports without experiencing significant delays or performance issues.

Since the proposed system operates as a web-based application, the hardware must be capable of supporting server-side processing, database management, and real-time interaction between the administrator and the application. The recommended specifications are selected based on the expected workload of the system, including encoding production batch information, computing yield percentage and extraction efficiency, generating graphical dashboards through Chart.js, retrieving historical production records, and producing management reports.

Adequate processing power is essential because the application performs multiple operations simultaneously, including validating user inputs, processing production data, executing automated computations, communicating with the MySQL database, and displaying analytical results through graphical visualizations. A computer with a higher-performance processor enables the application to execute these operations more efficiently, resulting in faster response times and an improved user experience.

Memory (RAM) also plays an important role in ensuring smooth system performance. Sufficient memory allows the web browser, web server, database server, and other supporting applications to run simultaneously without affecting system responsiveness. As production records continue to increase over time, additional memory helps maintain stable system performance while retrieving historical production information and generating reports.

Storage capacity is another important hardware consideration because the application stores production batch information, yield records, extraction efficiency records, generated reports, and other related system files within the database. The use of Solid-State Drive (SSD) storage is recommended because it provides significantly faster data access, shorter application loading times, quicker database transactions, and improved overall system performance compared to traditional hard disk drives (HDD). Faster storage also enhances the retrieval of historical production records and improves the efficiency of report generation.

The monitor specification ensures that administrators can comfortably view the system's user interface, production forms, dashboards, charts, graphs, and generated reports. A higher display resolution provides clearer visualization of production analytics, allowing administrators to analyze production performance more effectively. Standard USB keyboard and mouse devices provide the necessary input tools for encoding production information, navigating the system, and performing administrative tasks efficiently.

Although the proposed system can operate within a local network environment, a reliable internet or network connection is recommended to support system accessibility, software updates, and future deployment in a networked production environment. A stable connection also facilitates faster communication between the web server and client devices when the application is accessed through multiple workstations.

Table 4.1 presents the minimum and recommended hardware specifications required for the implementation of the proposed system.

Table 4.1 Hardware Requirements of the Proposed System

Hardware Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 or AMD Ryzen 3	Intel Core i5 or AMD Ryzen 5 and above
Memory (RAM)	4 GB	8 GB or higher

Storage	256 GB SSD	512 GB SSD or higher
Monitor	1366 × 768 Resolution	Full HD (1920 × 1080)
Keyboard and Mouse	Standard USB Keyboard and Mouse	Standard USB Keyboard and Mouse
Internet Connection	10 Mbps	50 Mbps or higher

The minimum hardware specifications are sufficient to install and operate the proposed web-based application under normal production monitoring conditions. These specifications allow administrators to encode production information, perform automated yield and extraction efficiency computations, access production records, and generate reports with acceptable system performance.

However, the recommended hardware specifications provide improved processing capability, faster database operations, greater storage capacity, and enhanced responsiveness, particularly when handling larger volumes of production data and generating graphical dashboards and management reports. The additional computing resources also improve the overall reliability of the application during continuous daily operation and reduce the likelihood of performance bottlenecks.

Overall, the recommended hardware configuration provides a stable and efficient computing environment for the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* By utilizing these hardware resources, the proposed system can efficiently process production information, maintain accurate production records, generate real-time analytical results, and support effective production monitoring and decision-making. Furthermore, the selected hardware specifications provide sufficient capacity to accommodate future expansion of the system as additional production data and functional requirements are introduced.

4.5.2 Software Requirements

The proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* is developed using widely available, reliable, and open-source software technologies to ensure cost-effectiveness, compatibility, ease of maintenance, and long-term scalability. The selected software components provide the necessary development environment, web server, database management system, programming languages, and development tools required to build, deploy, and operate the proposed web-based application successfully.

The software requirements are carefully selected based on the functional and technical requirements identified during the system analysis and design phases of the study. Each software component performs a specific role in supporting the overall operation of the application, from developing the user interface and implementing the system's business logic to storing production information and generating graphical dashboards and management reports.

The use of open-source technologies significantly reduces implementation and maintenance costs for AlDi Foods Inc. while ensuring that the application remains flexible, reliable, and easy to upgrade. Since the proposed system is intended for small- and medium-sized manufacturing operations, the selected software provides a practical and sustainable solution without requiring expensive proprietary software licenses.

The operating environment of the application is based on Microsoft Windows 10 or Windows 11, which provides a stable platform for system development, deployment, and daily operation. The operating system supports all required development tools and server applications while providing compatibility with the selected programming languages and database management system.

For local development and deployment, the researchers utilize XAMPP, an integrated web server package that includes Apache, PHP, and MySQL. XAMPP simplifies the installation and configuration of the development environment by providing all essential server components within a single package. This environment enables the application to run locally during software development, testing, debugging, and demonstration before deployment in the production environment.

The Apache Web Server serves as the application's web server by processing client requests and delivering web pages to users through a web browser. Apache manages communication between the administrator and the application, allowing production information to be entered, processed, retrieved, and displayed efficiently. Its stability and widespread adoption make it suitable for supporting the daily operation of the proposed production monitoring system.

The core business logic of the application is developed using PHP (Hypertext Preprocessor). PHP is responsible for processing administrator requests, validating user inputs, communicating with the MySQL database, performing automated yield and extraction efficiency computations, and generating production reports. Through PHP, the application integrates all functional modules into a unified web-based platform that supports production monitoring and management decision-making.

The system interface is developed using HTML5, CSS3, and JavaScript, which work together to create a responsive, organized, and user-friendly web application. HTML5 provides the structural foundation of the web pages, CSS3 enhances the visual appearance and layout of the interface, while JavaScript adds interactivity and dynamic functionality. These technologies allow administrators to navigate the application efficiently, encode production information, view analytical dashboards, and generate reports with ease.

To improve data interpretation and production monitoring, the proposed system utilizes Chart.js for data visualization. This JavaScript library enables the application to display production information using charts, graphs, and graphical dashboards. Through visual representations of production yield and extraction efficiency, administrators can quickly identify production trends, monitor operational performance, compare production batches, and evaluate the effectiveness of the juice extraction process. The graphical dashboards provide management with meaningful insights that support timely and data-driven decision-making.

The proposed system uses MySQL as its Database Management System (DBMS). MySQL stores all production-related information, including administrator accounts, production batches, yield records, extraction records, and generated reports. The centralized database ensures that production information is stored securely, organized efficiently, and easily retrievable whenever monitoring, analysis, or report generation is required. The relational database structure also maintains data integrity through the use of primary keys and foreign key relationships, minimizing data redundancy while ensuring consistency across all production records.

During system development, Visual Studio Code serves as the primary source code editor. This lightweight yet powerful development environment enables the researchers to write, organize, debug, and maintain the application's source code efficiently. Its support for multiple programming languages, extensions, and integrated development features significantly improves coding productivity and software quality throughout the implementation phase.

The application is accessed using modern web browsers such as Google Chrome and Microsoft Edge, both of which fully support HTML5, CSS3, JavaScript, and Chart.js technologies. These browsers ensure proper rendering of the system interface, responsive navigation, accurate graphical visualization, and reliable execution of all application functions. Their compatibility with current web standards contributes to a consistent user experience during daily system operation.

To support collaborative software development and maintain version history, the researchers utilize Git as the version control system. Git enables the researchers to monitor code revisions, manage updates, restore previous versions when necessary, and maintain an organized development workflow throughout the implementation of the project. Version control also improves collaboration among the development team by ensuring that modifications to the source code are properly documented and managed.

Table 4.2 presents the software requirements used in the development and implementation of the proposed system.

Table 4.2 Software Requirements of the Proposed System

Software	**Purpose**
Windows 10/11	Operating System
XAMPP	Local Web Server Environment
Apache	Web Server
PHP	Server-side Programming Language
HTML5	Web Page Structure
CSS3	User Interface Design

JavaScript	Client-side Scripting
Chart.js	Data Visualization and Graph Generation
MySQL	Database Management System
Visual Studio Code	Source Code Editor
Google Chrome / Microsoft Edge	Web Browser
Git	Version Control (Development)

Software Requirements

Each of these software components contributes to the successful operation of the proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* The integration of these technologies enables the application to automate production recording, compute yield percentage and extraction efficiency, manage production records within a centralized database, generate graphical dashboards, and produce management reports efficiently.

Furthermore, the selected software technologies provide a stable, secure, and scalable environment that supports the long-term implementation of the proposed system. Their compatibility with one another ensures seamless communication between the web interface, application logic, and database, resulting in reliable system performance and efficient data processing. As production requirements continue to evolve, these technologies also provide a flexible foundation for future system enhancements, including additional analytical features, expanded reporting capabilities, multiple user roles, and integration with other production management functions.

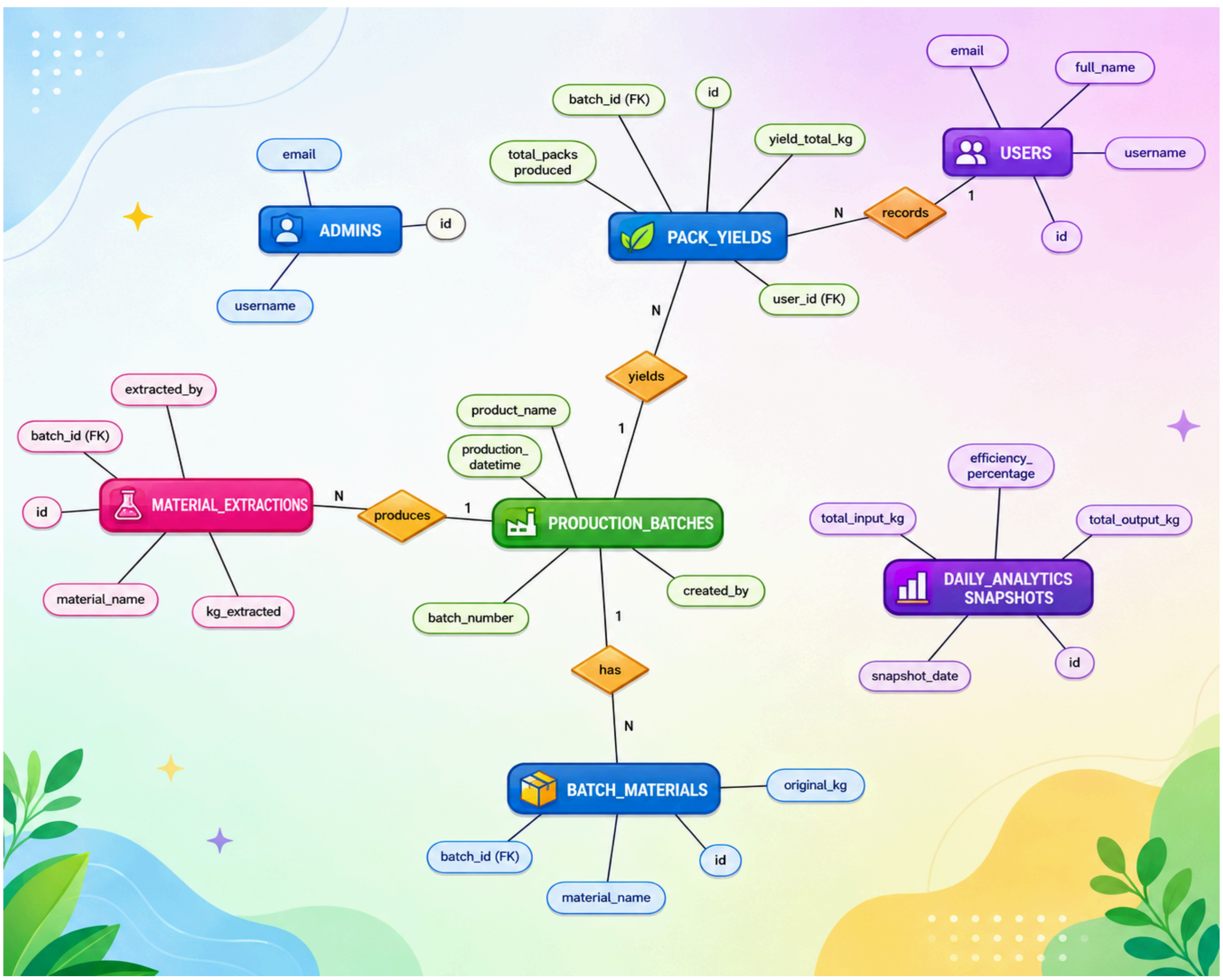
Overall, the software requirements presented in this section provide the technological foundation necessary for the successful development, deployment, and operation of the proposed web-based production monitoring system. By utilizing proven and widely adopted software technologies, the developed application delivers a reliable, user-friendly, and cost-effective solution that supports the production monitoring objectives of AlDi Foods Inc. while improving data accuracy, operational efficiency, and management decision-making.

4.6 System Design

The relational database model serves as the foundational structure for the proposed system by organizing operational information into distinct, normalized tables designed to minimize redundancy and maintain data integrity. At the core of the database is the USERS table, which manages user credentials and access permissions and connects directly to PRODUCTION_BATCHES to track batch creation. Each production batch acts as a central hub that links various operational processes, including raw material inputs (RAW_MATERIALS), output metrics from extraction operations (EXTRACTION_RECORDS), and final packaged output logs (PACKED_YIELDS).

To support real-time efficiency monitoring, the database structures analytical outputs into specialized tables derived directly from packed production data. Calculated performance indicators are systematically organized across YIELD_ANALYTICS, EFFICIENCY_RECORDS, and PRODUCTION_STABILITY to evaluate output ratios and assign operational status classifications such as *Stable* or *Not Stable*. These computational records feed into the REPORTS entity, enabling the automated generation of comprehensive daily, weekly, and monthly summaries without compromising system efficiency or data consistency.

Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.



The primary entities of the database include:

● Users

● Raw Materials

● Production Batches

● Extraction Records

● Packed Yields

● Yield Analytics

● Efficiency Records

● Production Stability

● Reports

Users Table

This table stores account credentials and access permissions for system users.

Field	Description
user_id	Primary Key
name	Complete Name
username	Login Username
password	Account Password

role	User Access Level

Production Batches Table

This table acts as the central hub tracking production batch information.

Field	Description
batch_id	Primary Key
user_id	Foreign Key (Created by User)
production_date	Date of Production
operator_name	Name of Assigned Operator
product_name	Name of Product
remarks	Additional Operational Notes

Raw Materials Table

This table records the raw materials utilized in each production batch.

Field	Description

raw_id	Primary Key
batch_id	Foreign Key
material_name	Name of Raw Material
quantity_input	Amount of Material Used
unit	Measurement Unit (e.g., kg, L)

Extraction Records Table

This table tracks the volume and details of extracted outputs from production.

Field	Description
extraction_id	Primary Key
batch_id	Foreign Key
extracted_output	Volume of Extracted Output
extraction_time	Time of Extraction Process

remarks	Extraction Notes

Packed Yields Table

This table stores the final packaged yield measurements derived from production batches.

Field	Description
packed_id	Primary Key
batch_id	Foreign Key
packed_output	Total Packaged Volume/Units
packed_date	Date Packaging Completed

Yield Analytics Table

This table stores calculated yield percentages to measure overall output performance.

Field	Description
yield_id	Primary Key
batch_id	Foreign Key

yield_percent	Calculated Yield Percentage

Efficiency Records Table

This table records resource-to-output conversion efficiency metrics.

Field	Description
efficiency_id	Primary Key
batch_id	Foreign Key
efficiency_%	Extraction Efficiency Metric (%)

Production Stability Table

This table evaluates batch performance stability against organizational criteria.

Field	Description
stability_id	Primary Key
batch_id	Foreign Key
status	Operational Status (e.g., Stable / Not Stable)

Reports Table

This table logs generated production summaries derived from analytical and stability metrics.

Field	Description
report_id	Primary Key
batch_id	Foreign Key
report_type	Type of Summary (e.g., Daily, Monthly)
generated_date	Date Report Created

The relationships among these tables allow the application to track raw material consumption, measure extraction performance, calculate packaged yields, evaluate production stability, and generate comprehensive management reports. Utilizing proper indexing and foreign key constraints helps maintain referential integrity while optimizing query performance across all operational and analytical workflows.

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusion and recommendations of the study entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* It summarizes the overall findings of the research, determines whether the objectives established in the study have been successfully achieved, and discusses the significance of the developed system in addressing the operational challenges experienced by AlDi Foods Inc. Furthermore, this chapter provides recommendations for future system enhancements, implementation strategies, and possible directions for future researchers who intend to continue or improve the proposed system.

The study is conducted in response to the company's existing manual production monitoring process, which relies heavily on handwritten records and manual computation of production data. Although the existing process enables the company to document production activities, it presents several limitations, including delays in recording production information, difficulty in monitoring production performance in real time, increased possibility of human errors during computation, and challenges in retrieving historical production records for analysis and reporting. These issues affect the organization's ability to monitor production efficiency promptly and make timely operational decisions.

To address these concerns, the researchers design and develop a web-based production monitoring system capable of digitizing production records and providing real-time yield analytics and extraction efficiency monitoring. The system centralizes production information into a single database where authorized users can record raw material usage, extraction output, and packed yield data. It also automates the computation of yield percentage and extraction efficiency while providing dashboard visualizations and periodic reports that support production monitoring and management decision-making.

Throughout the development process, the researchers follow the Agile Software Development Methodology to ensure that the system is continuously refined based on identified requirements and feedback gathered during each phase of development. The proposed system integrates several functional modules, including user authentication, production management, raw material recording, extraction monitoring, packed yield monitoring, production stability interpretation, dashboard visualization, historical record management, and automated report generation. These modules work together to provide an efficient and centralized digital platform for production monitoring.

The findings presented in this chapter are based on the successful completion of the system development process described in the previous chapters. The discussion highlights how the developed system addresses the problems identified in Chapter 1, incorporates concepts and best practices gathered from the related literature and studies discussed in Chapter 2, and implements the methodologies presented in Chapter 3. By comparing the objectives of the study with the actual capabilities of the developed system presented in Chapter 4, this chapter demonstrates the extent to which the proposed solution satisfies the operational requirements of AlDi Foods Inc.

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Finally, this chapter emphasizes the importance of adopting digital technologies in manufacturing environments, particularly for organizations that continue to rely on manual production monitoring procedures. The implementation of the proposed system demonstrates that integrating web-based information systems into daily production operations improves data accuracy, enhances production visibility, reduces manual workload, and supports more informed and timely decision-making. The conclusions and recommendations presented in this chapter provide valuable insights not only for AlDi Foods Inc. but also for future researchers and organizations seeking to modernize their production monitoring processes through digital transformation.

5.2 Summary of Findings

The study successfully achieves its primary objective of designing and developing a web-based production monitoring system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* The proposed system is developed to replace the company's existing manual production recording process with a centralized digital platform capable of providing real-time monitoring, automated computations, and production analytics.

Based on the requirements gathered through interviews, direct observations, and analysis of the company's current production workflow, the researchers identify several operational challenges affecting production monitoring. These challenges include the use of manual recording methods, delayed availability of production information, difficulty in monitoring production performance in real time, manual computation of yield percentages, and limited visibility of extraction efficiency, and the absence of a centralized repository for historical production records. These identified problems serve as the basis for designing the proposed system and guide the development of its core functionalities.

The completed system successfully digitizes the recording of production information by allowing authorized users to encode production batches, raw material quantities, extraction output, and packed yield through an organized web-based interface. Instead of relying on handwritten logs and manual computation, all production records are securely stored in a centralized MySQL database, making information more accessible, organized, and easier to retrieve whenever needed.

One of the major accomplishments of the proposed system is the automation of yield percentage computation. After users record the raw material input and production output, the system automatically calculates the corresponding yield percentage using predefined computational formulas. This automation significantly reduces the possibility of human error associated with manual calculations while improving the accuracy and consistency of production monitoring. The automated computation also enables production personnel and managers to obtain immediate production performance results without performing repetitive mathematical calculations.

The system also successfully performs extraction efficiency monitoring by comparing raw material input with extracted production output. This functionality allows production managers to evaluate the effectiveness of the extraction process for every production batch. Through continuous monitoring of extraction efficiency, management can identify production losses, monitor resource utilization, and evaluate whether production processes are operating within acceptable performance standards.

Another significant finding of the study is the successful implementation of production stability interpretation. Based on predefined production thresholds established during system development, the application automatically classifies production performance as either Stable or Not Stable. This feature enables production supervisors and managers to immediately recognize production inconsistencies and investigate potential causes before they affect succeeding production batches. The automated stability interpretation supports faster decision-making and contributes to improved operational control.

The dashboard module developed for the system provides real-time visualization of production performance through charts, graphs, summary cards, and production statistics. The dashboard consolidates production information into an easy-to-understand interface, allowing administrators and production managers to monitor key production indicators without manually reviewing individual production records. The availability of real-time graphical reports improves production visibility and enables management to evaluate operational performance more efficiently.

The developed system also supports historical production record management by storing all encoded production information within a centralized relational database. This enables authorized users to retrieve previous production records for analysis, comparison, auditing, and reporting purposes. Historical production information can be utilized to evaluate long-term production trends, identify recurring operational issues, and support future production planning.

Furthermore, the report generation module automatically produces production reports based on weekly, monthly, quarterly, and yearly production records. These reports summarize raw material usage, extraction output, yield percentage, extraction efficiency, and production stability results, providing management with comprehensive documentation of production performance. Automated report generation reduces the time required to prepare production summaries while improving the consistency and accuracy of organizational records.

The researchers also find that the integration of the different system modules—including user authentication, production management, raw material recording, extraction monitoring, packed yield monitoring, dashboard visualization, historical record management, and report generation—results in a unified production monitoring platform capable of supporting the day-to-day operational requirements of AlDi Foods Inc. The centralized nature of the system improves communication among users and ensures that production information is consistently updated and available for authorized personnel.

Overall, the findings demonstrate that the proposed system successfully fulfills the objectives established in Chapter 1. The developed application effectively digitizes production monitoring, automates production computations, provides real-time analytics, centralizes production information, and supports data-driven decision-making within AlDi Foods Inc. By replacing traditional manual recording methods with an integrated digital platform, the study contributes to improving production efficiency, reducing human error, and supporting the organization's ongoing digital transformation initiatives.

5.3 Conclusion

The primary objective of this study is to design and develop a web-based production monitoring system entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* The study is conducted in response to the operational challenges experienced by AlDi Foods Inc., particularly the company's reliance on manual production recording, delayed access to production information, manual computation of yield values, and the absence of a centralized platform for monitoring production performance. These limitations affect the efficiency of production monitoring and reduce the organization's ability to make timely and informed operational decisions.

Based on the findings presented in the previous chapters, the researchers conclude that the proposed system successfully achieves the objectives established at the beginning of the study. The developed application provides a centralized digital platform that replaces manual production recording with a more organized, accurate, and efficient method of managing production information. Through the implementation of computerized data recording and automated processing, the system improves the accessibility, consistency, and reliability of production records while minimizing errors commonly associated with handwritten documentation and manual computation.

One of the major accomplishments of the proposed system is the successful digitization of the production monitoring process. Instead of relying on traditional paper-based recording methods, authorized users are able to record production information through an organized web-based interface. Production batches, raw material quantities, extraction outputs, and packed yield records are securely stored in a centralized database where information can be retrieved efficiently whenever needed. This centralized approach not only improves record management but also reduces the possibility of data loss and duplication, which are common issues in manual record-keeping systems.

The developed system also successfully automates the computation of yield percentage and extraction efficiency. These calculations are performed automatically based on the production data entered by authorized users, eliminating the need for repetitive manual computations. Automation improves the accuracy of production calculations, reduces the likelihood of computational errors, and enables production personnel to focus more on operational activities rather than mathematical processing. The availability of immediate computation results also allows managers to evaluate production performance without unnecessary delays.

Another significant contribution of the study is the implementation of real-time dashboard analytics. The dashboard provides production managers and administrators with visual representations of production performance through charts, graphs, summary statistics, and production indicators. These visualizations simplify the interpretation of production information by presenting complex production data in an organized and understandable format. Real-time monitoring enables management to quickly identify operational issues, monitor production trends, and make informed decisions based on current production conditions rather than relying solely on historical records.

The system's production stability interpretation feature further enhances production monitoring by automatically classifying production performance as either Stable or Not Stable according to predefined production thresholds. This functionality provides immediate feedback regarding the consistency of production operations and assists supervisors in identifying potential production inefficiencies before they become significant operational problems. By providing timely production status information, the system contributes to better production control and continuous process improvement.

The automated report generation module also plays an important role in improving production management. Weekly, monthly, quarterly, and yearly reports can be generated directly from the centralized database without requiring extensive manual preparation. These reports provide comprehensive summaries of raw material consumption, production output, yield percentage, extraction efficiency, and production stability, enabling management to evaluate production performance over different reporting periods. The availability of organized reports also supports documentation requirements, performance evaluation, and strategic planning within the organization.

The successful integration of user authentication, production management, raw material recording, extraction monitoring, packed yield monitoring, dashboard visualization, historical data management, production stability interpretation, and report generation demonstrates that the proposed system functions as a complete production monitoring solution tailored to the operational needs of AlDi Foods Inc. The integration of these modules creates a unified platform that streamlines production monitoring activities while ensuring that production information remains accurate, secure, and easily accessible.

Furthermore, the findings of this study support the importance of digital transformation within the food manufacturing industry. As discussed in the related literature and studies presented in Chapter 2, many manufacturing organizations continue to adopt digital technologies to improve operational efficiency, reduce production waste, enhance data accuracy, and strengthen decision-making processes. The developed system demonstrates that even without implementing advanced industrial technologies such as IoT or machine learning, a well-designed web-based information system can significantly improve production monitoring through centralized data management, automated computations, and real-time visualization.

The use of the Agile Software Development Methodology also contributes significantly to the successful completion of the project. The iterative development process allows the researchers to continuously evaluate system functionalities, address identified issues during development, and ensure that the completed application meets the operational requirements gathered from AlDi Foods Inc. The structured development process results in a system that aligns with the organization's existing workflow while remaining flexible enough to accommodate future improvements and expansion.

Overall, the researchers conclude that the proposed *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.* system is a practical, reliable, and effective digital solution for modernizing production monitoring activities. The system successfully addresses the operational problems identified in the study by replacing manual production recording with an automated, centralized, and data-driven platform capable of supporting daily production operations. The implementation of the proposed system improves production visibility, enhances data accuracy, simplifies production analysis, and supports timely managerial decision-making.

Finally, the researchers conclude that the developed system contributes not only to the operational improvement of AlDi Foods Inc. but also to the broader objective of promoting digital transformation within the manufacturing sector. By demonstrating the successful application of information technology in production monitoring, the study provides evidence that digital solutions can effectively improve organizational efficiency while serving as a foundation for future technological advancements. The proposed system therefore represents a meaningful contribution to both the field of Information Technology and the continuing modernization of manufacturing processes in the Philippine food industry.

5.4 Recommendations

Based on the findings, conclusions, and overall evaluation of the proposed system, the researchers recommend several improvements and future directions to further enhance the effectiveness, usability, scalability, and long-term sustainability of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.* These recommendations are intended for AlDi Foods Inc., future researchers, and software developers who may continue the development and enhancement of the proposed web-based production monitoring system.

The researchers recommend that AlDi Foods Inc. fully implement the developed system in its daily production operations to maximize the benefits of digital production monitoring. The adoption of the system is expected to improve the accuracy of production recording, reduce manual computation, enhance production monitoring, and support faster and more informed managerial decision-making. The company is also encouraged to provide adequate training for administrators and production personnel to ensure the proper utilization and maintenance of the application.

The researchers also recommend the implementation of regular database backup and maintenance procedures to protect production records from accidental data loss and system failures. Routine software updates and preventive maintenance are likewise recommended to maintain the reliability, security, and efficiency of the application throughout its operational use.

For future system enhancement, the researchers recommend expanding the functionalities of the application by incorporating additional production management features such as inventory management, raw material tracking, supplier management, production scheduling, quality assurance monitoring, and employee performance monitoring. The integration of these modules would further improve the overall production management capabilities of the system.

The researchers further recommend the implementation of role-based access control by allowing multiple authorized users with different levels of system privileges. Future versions of the application can include separate user accounts for production supervisors, quality assurance personnel, warehouse staff, and management to improve collaboration while maintaining data security and accountability.

To improve accessibility and operational flexibility, the researchers recommend developing a mobile-responsive version of the application or a dedicated mobile application. This enhancement enables production personnel and management to monitor production activities, review reports, and access production information using smartphones or tablet devices.

The researchers also recommend the integration of advanced data analytics and predictive reporting features in future versions of the system. Additional analytical tools, such as production forecasting, trend analysis, and performance prediction, can assist management in making more strategic operational decisions and identifying potential production issues before they occur.

Although the current study intentionally excludes Industrial Internet of Things (IIoT) technologies and advanced automation to maintain affordability for small and medium-sized enterprises, future researchers are encouraged to investigate the possible integration of sensor-based data collection, automated production monitoring, or machine learning techniques if organizational resources and operational requirements permit such enhancements.

Future researchers are also encouraged to conduct further evaluation of the system using larger groups of users and longer implementation periods. Additional studies can assess the long-term effects of the system on production efficiency, operational performance, user satisfaction, and organizational productivity using different evaluation methodologies and performance indicators.

Lastly, the researchers recommend that future studies continue exploring innovative information technology solutions that support digital transformation within the manufacturing industry. Similar production monitoring systems can be developed for other food processing companies and manufacturing organizations to improve production management, enhance operational efficiency, reduce manual workloads, and promote data-driven decision-making through modern web-based technologies.

Overall, these recommendations are intended to support the continuous improvement, sustainability, and future expansion of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring System for AlDi Foods Inc.*, ensuring that the developed application remains relevant, adaptable, and capable of meeting the evolving operational requirements of the organization.

System Improvements

Although the developed system successfully achieves its primary objectives of digitizing production monitoring, automating yield computation, monitoring extraction efficiency, and generating production reports, there are still opportunities to improve its functionality and overall performance.

The researchers recommend incorporating an automated notification and alert feature into future versions of the application. This feature can notify administrators whenever production results fall below the expected yield percentage or extraction efficiency, or when production stability is classified as Not Stable. Such notifications allow production personnel to immediately review production records and take appropriate corrective actions.

Future versions of the system can also include a more comprehensive dashboard with additional visualization features. While the current system provides charts, graphs, and production summaries, future enhancements can include interactive filters, comparison of production batches by date, graphical trend analysis, and customizable dashboard layouts. These features provide management with clearer insights into production performance and support more effective decision-making.

Another recommended enhancement is the integration of barcode or QR code technology for production batch identification. This feature simplifies production batch recording, reduces manual encoding errors, improves the traceability of production records, and makes information retrieval more efficient. To strengthen system security, future versions should implement additional authentication and security features such as stronger password encryption, password recovery options, user activity logs, automatic session timeout, and role-based access control if additional user roles are introduced. These improvements help protect production information from unauthorized access while improving accountability among system users.

The researchers also recommend enhancing the report generation module by allowing reports to be exported in different formats such as PDF, Microsoft Excel, and CSV. Providing multiple export options makes it easier for management to archive production reports, prepare documentation, and perform further data analysis whenever necessary.

To ensure the safety of production records, future versions of the application should include automatic database backup and recovery functions. Regular backup procedures help protect valuable production information from accidental deletion, hardware failure, or unexpected system interruptions.

Finally, the researchers recommend improving the usability of the system by continuously refining the user interface based on user feedback. Enhancing navigation, simplifying data entry forms, and improving dashboard organization make the system easier to learn and use for production personnel and administrators.

Future Research Directions

This study focuses on developing a web-based production monitoring system for AlDi Foods Inc. that digitizes production recording, computes yield percentage and extraction efficiency, stores historical production records, and generates production reports. Although the proposed system successfully addresses the identified operational problems, the researchers recommend that future researchers can further improve the system by expanding its functionality while remaining aligned with the needs of manufacturing organizations.

The researchers recommend that future researchers can enhance the dashboard by incorporating more advanced statistical summaries, production comparisons, and additional graphical reports that provide deeper insights into production performance. These enhancements can help management evaluate long-term production trends and identify areas for operational improvement.

The researchers also recommend that future researchers can develop a mobile-responsive or dedicated mobile application version of the system to enable administrators and production supervisors to access production information using smartphones or tablets. This enhancement improves accessibility and allows authorized users to monitor production records even when they are away from the production office.

Another possible enhancement is the expansion of the system to include additional management modules such as inventory management, raw material inventory tracking, supplier information management, quality assurance monitoring, production scheduling, and employee management. Integrating these modules creates a more comprehensive manufacturing information system while maintaining the centralized database established in the current study.

The researchers also recommend that future studies can focus on improving report generation by incorporating more detailed production summaries, customized report templates, and comparative analyses of production performance over different reporting periods. Such improvements provide management with more comprehensive information for production planning and operational evaluation.

The researchers likewise encourage future researchers to conduct comparative studies involving other food manufacturing companies to determine whether the proposed production monitoring framework can be adapted and implemented in different production environments. These studies can help evaluate the flexibility and applicability of the developed system across various manufacturing processes.

Finally, the researchers recommend that future researchers evaluate the long-term impact of implementing digital production monitoring systems on operational efficiency, production consistency, data accuracy, reporting efficiency, and overall organizational productivity. Such studies provide additional evidence regarding the effectiveness of digital production monitoring systems in supporting continuous operational improvement.

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Implementation Strategies

For the successful implementation of the proposed system, the researchers recommend that AlDi Foods Inc. establish a structured deployment strategy that includes proper preparation, employee training, continuous monitoring, and regular system evaluation. Proper implementation helps maximize the benefits of the developed application while ensuring consistent and accurate production monitoring.

Before full implementation, the researchers recommend that production personnel and administrators undergo comprehensive user training covering system navigation, production data encoding, dashboard interpretation, report generation, and basic troubleshooting procedures. Adequate training improves user confidence, minimizes encoding errors, and promotes the proper utilization of the system.

The researchers also recommend that management establish standardized operating procedures (SOPs) that clearly define responsibilities for recording production information, validating encoded data, reviewing production reports, and maintaining database accuracy. Standardized procedures ensure consistency in production monitoring activities and improve the reliability of stored production information.

To maintain data integrity, the researchers recommend that regular database backups be performed according to an established schedule. Backup files are stored securely to minimize the risk of data loss caused by hardware failure, software malfunction, or other unforeseen circumstances.

The researchers further recommend that periodic system maintenance be conducted to ensure continuous system reliability and performance. This includes updating software components, correcting identified system errors, optimizing database performance, reviewing security settings, and monitoring overall system functionality.

The researchers also recommend that management regularly analyze the production reports and dashboard information generated by the system during production meetings and operational planning. Continuous evaluation of yield percentage, extraction efficiency, production stability, and production output would support informed decision-making and encourage continuous process improvement.

Finally, the researchers recommended that user feedback should be collected regularly after system implementation. Suggestions and observations from administrators and production personnel should be documented and considered in future system updates. Continuous improvement based on actual user experience would help ensure that the system remained effective, user-friendly, and responsive to the operational requirements of AlDi Foods Inc.

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5.5 Impact of the Study

The development of the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.* contributes significantly to the advancement of digital transformation in food manufacturing, particularly among small and medium-sized enterprises (SMEs). By replacing traditional manual production monitoring with a centralized digital platform, the study demonstrates how information technology improves operational efficiency, production visibility, and management decision-making.

For AlDi Foods Inc., the system serves as an effective production monitoring tool that digitizes the recording of raw material usage, extraction outputs, packed yield, and production performance. Through real-time yield analytics and extraction efficiency monitoring, management can immediately identify production trends, monitor operational performance, and detect potential production losses before they become significant. This enables supervisors and production managers to make timely, evidence-based decisions that contribute to improved productivity and reduced waste.

The study also minimizes the limitations associated with manual record-keeping. Traditional paper-based production logs are susceptible to human error, misplaced records, delayed reporting, and inconsistencies in data recording. By implementing a centralized database with automated calculations, the system improves data accuracy, consistency, and accessibility. Historical production records are securely stored and can be retrieved whenever needed for operational reviews, audits, and production planning.

Another significant contribution of the study is the enhancement of production transparency. Because production information updates in real time, authorized users can monitor production activities as they occur rather than waiting until the end of the production cycle. This immediate availability of information allows management to respond quickly to operational issues, helping maintain production stability and optimize resource utilization.

The system also promotes a culture of data-driven decision-making within the organization. Rather than relying solely on experience or manually compiled reports, production managers use real-time dashboards, graphical reports, and automated analytics to evaluate production performance objectively. Information such as yield percentage, extraction efficiency, production stability classification, and raw material utilization provides valuable insights that support strategic planning and continuous process improvement.

From an academic perspective, this study contributes to the field of Information Technology, particularly in the areas of systems analysis and design, database management, web application development, and business analytics. The project demonstrates how software engineering principles can be applied to solve practical operational problems in the manufacturing industry through the development of an integrated production monitoring system. It also illustrates the use of modern web technologies to create digital solutions that improve organizational efficiency.

The study likewise benefits future BSIT students and researchers by providing a practical reference for similar projects involving production monitoring, manufacturing information systems, dashboard development, and real-time analytics. The methodologies, system architecture, database design, Agile development approach, and evaluation procedures documented in this research serve as useful guides for future academic and industry-based system development projects.

Furthermore, the study contributes to the growing body of knowledge supporting the digital transformation of Philippine manufacturing SMEs. Many local businesses continue to rely on manual production monitoring because of financial limitations and limited access to advanced manufacturing technologies. The proposed system demonstrates that digital transformation can be achieved through practical, affordable, and scalable information systems without requiring expensive industrial automation or complex infrastructure. This makes the solution more accessible to organizations with limited technological resources.

Despite its contributions, the researchers recognize several limitations of the study. The system **is** developed as a prototype specifically tailored to the production workflow of AlDi Foods Inc., limiting its immediate applicability to other manufacturing environments without modification.

In addition, the study focuses only on production monitoring, yield analytics, extraction efficiency monitoring, and report generation. Other operational areas such as inventory management, purchasing, sales, accounting, supplier management, and predictive analytics are intentionally excluded from the project scope. Likewise, the prototype does not incorporate IoT devices or automated sensor technologies, as these are beyond the defined scope of the study.

Nevertheless, these limitations do not diminish the overall value of the developed system. Instead, they provide opportunities for future enhancements and research. The study successfully demonstrates that a carefully designed digital production monitoring platform can significantly improve production management, enhance operational visibility, reduce manual recording errors, and support informed managerial decision-making. Ultimately, the project serves as a practical example of how information technology can drive digital transformation and operational excellence within food manufacturing enterprises such as AlDi Foods Inc.

5.6 Summary

This chapter presents the overall conclusions and recommendations of the study entitled *"Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc."* It summarizes the major findings obtained during the design, development, implementation, testing, and evaluation of the proposed system. The chapter also discusses how the study achieves its research objectives and addresses the production monitoring challenges identified in Chapters 1, 2, 3, and 4.

The study successfully develops a web-based production monitoring system that digitizes the recording of production data, automates the computation of yield percentage and extraction efficiency, and provides real-time dashboard analytics for production management. By replacing manual recording practices with a centralized digital platform, the system improves data accuracy, enhances production visibility, minimizes human errors, and supports timely decision-making. The results of the system testing and user evaluation further demonstrate that the developed system is functional, reliable, accurate, efficient, and user-friendly for its intended users at AlDi Foods Inc.

The recommendations presented in this chapter emphasize the importance of continuous system enhancement through additional security features, expanded reporting capabilities, mobile accessibility, predictive analytics, and integration with other production management functions. The study also recommends future research involving emerging technologies such as Artificial Intelligence (AI), Machine Learning, and the Internet of Things (IoT) to further improve production monitoring and operational efficiency.

Overall, the study demonstrates that the application of information technology can significantly improve production management in food manufacturing by providing accurate, real-time production information that supports evidence-based decision-making. The developed system offers a practical and cost-effective solution for AlDi Foods Inc. and serves as a valuable reference for future researchers, software developers, and organizations seeking to implement digital production monitoring systems.

In conclusion, the *Digitizing Production: Real-Time Yield Analytics and Extraction Efficiency Monitoring for AlDi Foods Inc.* project successfully achieves its goal of designing and developing a digital production monitoring platform that aligns with the operational needs of the company. The study contributes to the growing adoption of digital transformation among manufacturing organizations by demonstrating how real-time analytics and centralized production monitoring can improve efficiency, reduce production losses, and support continuous operational improvement. It also provides a strong foundation for future developments aimed at creating more intelligent, scalable, and integrated manufacturing information systems.

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